

# INTRODUCTION TO ALGEBRAIC GEOMETRY

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Algebraic geometry is the study of solutions of polynomial equations. For example, if

$$p(x) = a_0 + a_1x + \dots + a_nx^n$$

with  $a_i \in \mathbb{R}$  and  $a_n \neq 0$  is a polynomial of degree  $n$ , then it is well known that  $p(x)$  has at most  $n$  real solutions. A real solution is any number  $a \in \mathbb{R}$  such that  $p(a) = 0$ . To see that there are at most  $n$  solutions, note that if  $a$  is a solution then  $(x - a)$  divides  $p(x)$  and hence  $p(x) = (x - a)p_1(x)$  where  $\deg p_1(x) = n - 1$ . Repeating this process, one sees that if  $a_1, \dots, a_r$  are distinct solutions, then  $(x - a_1)(x - a_2) \cdots (x - a_r)$  divides  $p(x)$ . But then  $n = \deg p(x) \geq r$ .

Of course it can happen that  $p(x)$  has less than  $\deg p(x)$  solutions. For example if  $p(x) = x^2 + bx + c$ , then  $p(x)$  may have 0, 1, 2 solutions depending on whether the value of  $b^2 - 4c$  is  $< 0$ ,  $0$  or  $> 0$ . On the other hand, if we look for complex solutions  $a \in \mathbb{C}$ , then  $p(x)$  has exactly  $\deg p(x)$  solutions when counted with multiplicity, and so we may write

$$p(x) = a_0(x - a_1) \cdots (x - a_n)$$

where  $a_i \in \mathbb{C}$ .

It is also natural to look for other kinds of solutions to polynomial equations. Consider for example the famous equation

$$x^n + y^n = z^n, \quad n \geq 2$$

then it is natural to look for solutions in  $\mathbb{Z}$ ,  $\mathbb{Q}$ ,  $\mathbb{R}$  or  $\mathbb{C}$ . Let's start with the case  $n = 2$ :  $x^2 + y^2 = z^2$ , then any solution  $x, y, z \in \mathbb{N}$  is a **Pythagorean triple**. For example  $(3, 4, 5)$ . Note that looking for solutions in  $\mathbb{N}$ ,  $\mathbb{Z}$  or  $\mathbb{Q}$  is essentially the same thing as looking for rational solutions to

$$x^2 + y^2 = 1$$

i.e. rational points on the unit circle. Given a solution  $a, b, c \in \mathbb{N}$  (or  $\in \mathbb{Q}$ ), then  $a/c, b/c$  is a rational solution of  $x^2 + y^2 = 1$ . Conversely given a rational solution  $a/c, b/c \in \mathbb{Q}$  of  $x^2 + y^2 = 1$ , then  $|a|, |b|, |c| \in \mathbb{N}$  is a solution to  $x^2 + y^2 = z^2$ . It is not hard to see that  $x^2 + y^2 = 1$  has infinitely many solutions  $\in \mathbb{Q}$  (cf. Exercise 0.4).

The behaviour of the solutions to the equation  $x^n + y^n = z^n$  when  $n \geq 3$  is dramatically different: there are no interesting solutions. More precisely, if  $a, b, c \in \mathbb{N}$  satisfy  $a^n + b^n = c^n$  for some  $n \geq 3$ , then  $(a, b, c) \in \{(0, 0, 0), (1, 0, 1), (0, 1, 1)\}$ . This result is the famous Fermat's last Theorem conjectured by the french mathematician Fermat in 1637 (Fermat famously wrote that he could prove this deceptively simple theorem but it would not fit in the margin of the book). The Theorem was finally proved more than 3 centuries later by Andrew Wiles in 1994 after intense effort by many mathematicians. This example illustrates that even though natural numbers  $\mathbb{N}$  are the simplest numbers, looking for solutions to polynomial equations that are natural numbers can be very complicated or even hopeless. However, by the Fundamental Theorem of Algebra, every polynomial  $p(x \in \mathbb{C}[x])$  admits a solution  $a \in \mathbb{C}$  and this suggests that the problem is more tractable when we look for solutions that are in the complex numbers  $\mathbb{C}$ . We will adopt this point of view throughout this course. We will often be more interested in qualitative results (eg. a degree 2 polynomial can have 0,

1, 2 real solutions and a polynomial of degree  $n$  has exactly  $n$  solutions when counted with multiplicities) rather than finding the exact solution set.

**Exercise 0.1.** Show that if  $p(x) \in \mathbb{C}[x]$  is a non-zero polynomial and  $a$  is a solution of  $p(x)$ , then  $(x - a)$  divides  $p(x)$ .

**Exercise 0.2.** Use the FTA (fundamental Theorem of Algebra) to show that if  $p(x) \in \mathbb{C}[x]$  is a non-zero polynomial, then  $p(x)$  has exactly  $\deg p(x)$  complex solutions when counted with multiplicity.

**Exercise 0.3.** Show that if  $a, b \in \mathbb{Z}$  is a solution of  $x^2 + y^2 = 1$ , then  $a^2 - b^2$ ,  $2ab$  is also a solution. Compute the first few solutions obtained from  $3/5$ ,  $4/5$ .

**Exercise 0.4.** Show that if  $a, b \in \mathbb{N}$ , then  $[(a^2 - b^2), 2ab, (a^2 + b^2)]$  solves  $x^2 + y^2 = z^2$ . Use this to show that the point  $(0, 1)$  is an accumulation point of solutions to  $x^2 + y^2 = 1$ . Finally deduce that the rational solutions to  $x^2 + y^2 = 1$  are dense in  $S^1 = \{(x, y) | x^2 + y^2 = 1\}$ .

**Exercise 0.5.** (Hard) State and prove the FTA (Fundamental Theorem of Algebra).

## 1. AFFINE VARIETIES AND THE ZARISKI TOPOLOGY

**Definition 1.1.** An **affine algebraic set**  $X \subset \mathbb{C}^N$  is the set of common zeroes of a set of polynomial equations  $\{p_i\}_{i \in I}$  where  $p_i \in \mathbb{C}[x_1, \dots, x_N]$  so that

$$X = \mathbb{V}(\{p_i\}_{i \in I}) = \{a = (a_1, \dots, a_N) \in \mathbb{C}^N | p_i(a) = 0 \text{ for all } i \in I\}.$$

In particular  $\mathbb{V}(p_i)$  denotes the set of points at which  $p_i$  vanishes. We have the following examples:

- $\emptyset = \mathbb{V}(1) \subset \mathbb{C}^N$ , and  $\mathbb{C}^N = \mathbb{V}(0)$ .
- $(a_1, \dots, a_N) \in \mathbb{C}^N$  is given by  $\mathbb{V}(x_1 - a_1, \dots, x_N - a_N)$ .
- $N = 1$ , then the only affine algebraic sets in  $\mathbb{C}$  are  $\emptyset$ ,  $\mathbb{C}$  and finite union of points  $a_1, \dots, a_r \subset \mathbb{C}$ .
- $N = 2$ , then the only affine algebraic sets in  $\mathbb{C}^2$  are  $\emptyset$ ,  $\mathbb{C}^2$  and finite unions of points  $(a, b) \in \mathbb{C}^2$  and curves given by zero sets of one polynomial equation (see Exercise 11.14). For example lines  $\mathbb{V}(ax + by + c)$ , quadrics (given by zero sets of polynomials of degree 2), cubics (given by zero sets of polynomials of degree 3) etc.
- Zero sets of linear equations are called **hyperplanes** and zero sets of one polynomial are called **hypersurfaces**.
- The sets  $\{a \in \mathbb{C} | 0 < |a| < 2\}$ ,  $\{1 - \frac{1}{n} | n \in \mathbb{N}\} \subset \mathbb{C}$ ,  $\mathbb{R} \subset \mathbb{C}$ ,  $\{(a, b) \in \mathbb{C}^2 | a\bar{a} + b\bar{b} \leq 1\}$  and  $\{(a, \exp(a)) | a \in \mathbb{C}\} \subset \mathbb{C}^2$  are all subsets of  $\mathbb{C}$  or  $\mathbb{C}^2$  that are not affine algebraic sets.

**Exercise 1.2.** Prove the last statement above.

**Exercise 1.3.** Suppose that  $X = \mathbb{V}(p_1, \dots, p_r) \subset \mathbb{C}^n$ . Let

$$I = (p_1, \dots, p_r) := \left\{ \sum_{i=1}^r p_i r_i \mid r_i \in \mathbb{C}[x_1, \dots, x_n] \right\}.$$

Show that  $X = \mathbb{V}(\{p | p \in I\})$ . We will denote  $\mathbb{V}(\{p | p \in I\})$  by  $\mathbb{V}(I)$ .

**Exercise 1.4.** Suppose that if  $\emptyset \neq I \subset J \subset \mathbb{C}[x_1, \dots, x_n]$ , then  $\mathbb{V}(I) \supset \mathbb{V}(J)$ .

**Definition 1.5.** Let  $I \subset \mathbb{C}[x_1, \dots, x_n]$  be a non empty subset, then  $I$  is an **ideal** if  $I$  is closed under addition and multiplication by arbitrary polynomials so that if  $i, j \in I$  and  $r \in \mathbb{C}[x_1, \dots, x_n]$ , then

$$i + j \in I, \text{ and } ri \in I.$$

**Exercise 1.6.** Show that if  $\{p_i\}_{i \in J}$  is a collection of polynomials  $p_i \in \mathbb{C}[x_1, \dots, x_n]$ , then the set  $I = (\{p_i\}_{i \in J})$  given by all finite linear combinations  $\sum r_i p_i$  where  $r_i \in \mathbb{C}[x_1, \dots, x_n]$ , is closed under addition and multiplication by elements of  $\mathbb{C}[x_1, \dots, x_n]$  (and hence  $I \subset \mathbb{C}[x_1, \dots, x_n]$  is an ideal; we say that  $I$  is **the ideal generated by**  $\{p_i\}_{i \in J}$ ).

Notice that the polynomial equations determining an affine set are not uniquely determined. For example  $\mathbb{V}(x) = \mathbb{V}(x^2) = \mathbb{V}(x^3) \dots \subset \mathbb{C}$  and  $\mathbb{V}(x, y) = \mathbb{V}(x + y, x - y) \subset \mathbb{C}^2$ . However given an affine algebraic set (or more generally any subset)  $X \subset \mathbb{C}^n$ , we may consider

$$\mathbb{I}(X) := \{p \in \mathbb{C}[x_1, \dots, x_n] \mid p(x) = 0 \ \forall x \in X\}.$$

For example  $\mathbb{I}(\mathbb{V}(x^2)) = (x) \subset \mathbb{C}[x]$ .

**Exercise 1.7.** Show that  $\mathbb{I}(\mathbb{V}(\{p\}_{p \in J})) \supset (\{p\}_{p \in J})$ .

**Exercise 1.8.** Show that if  $X \subset \mathbb{C}^n$  is a subset, then  $\mathbb{I}(X) \subset \mathbb{C}[x_1, \dots, x_n]$  is an ideal.

It is not hard to see that any affine set  $X \subset \mathbb{C}^n$  can be defined by finitely many polynomials.

**Theorem 1.9.** Let  $X \subset \mathbb{C}^n$  be an affine algebraic set, then there exists finitely many polynomials  $p_1, \dots, p_r \in \mathbb{C}[x_1, \dots, x_n]$  such that

$$X = \mathbb{V}(p_1, \dots, p_r).$$

*Proof.* Let  $X = \mathbb{V}(\{p\}_{p \in I}) \subset \mathbb{C}^n$  be an affine algebraic set. We may assume that  $I = \mathbb{I}(X)$ . We must show that  $X$  is defined by finitely many polynomial equations. By Exercise 1.8 we may assume that  $I$  is an ideal (i.e. it is closed under addition and multiplication by arbitrary polynomials). We will proceed by induction on  $n$ .

Suppose that  $n = 1$ , i.e.  $p_i \in \mathbb{C}[x]$ . Pick  $q \in I$  an element of minimal degree. We claim that  $(q) = I$ . To see this, pick any element  $p \in I$  and write  $p = qs + r$  where  $s, r \in \mathbb{C}[x]$  and  $\deg r < \deg q$  (long division). Since  $p, q \in I$ , then  $r = p - qs \in I$  ( $I$  is an ideal). Since  $\deg r < \deg q$ , we have  $r = 0$  proving the claim.

Suppose now that the claim holds in  $\mathbb{C}[x_1, \dots, x_{n-1}]$  and consider  $p_i \in \mathbb{C}[x_1, \dots, x_n] = \mathbb{C}[x_1, \dots, x_{n-1}][x_n]$  as a polynomial in the variable  $x_n$  with coefficients in  $\mathbb{C}[x_1, \dots, x_{n-1}]$ . Let  $q_i \in \mathbb{C}[x_1, \dots, x_{n-1}]$  be the leading coefficient so that

$$p_i = q_i x_n^{d_i} + (\text{terms of lower degree in } x_n).$$

In particular the degree of  $p_i$  with respect to  $x_n$  is  $d_i$ . Now pick  $p_1 \in I$  of minimal degree with respect to  $x_n$ , then inductively pick  $p_i \in I \setminus (p_1, \dots, p_{i-1})$  of minimal degree with respect to  $x_n$ . Let  $q_i \in \mathbb{C}[x_1, \dots, x_{n-1}]$  be the corresponding leading terms. Consider the sequence of ideals

$$(q_1) \subset (q_1, q_2) \subset (q_1, q_2, q_3) \subset \dots \subset Q = (q_i \mid i \in \mathbb{N}) \subset \mathbb{C}[x_1, \dots, x_{n-1}].$$

By our inductive assumption,  $Q$  is generated by finitely many elements and so  $Q = (q_1, \dots, q_k)$  for some  $k > 0$ . We claim that  $I = (p_1, \dots, p_k)$ . To see this, consider  $p_{k+1}$ . Then, the degree with respect to  $x_n$ , satisfies

$$d_{k+1} = \deg p_{k+1} \geq d_k = \deg p_k \geq d_{k-1} = \deg p_{k-1} \geq \dots \geq d_1 = \deg p_1.$$

But then  $q_{k+1} = \sum_{i=1}^k s_i q_i$  for some  $s_i \in \mathbb{C}[x_1, \dots, x_{n-1}]$  and so

$$p' := p_{k+1} - \sum_{i=1}^k s_i p_i x_n^{d_{k+1}-d_i} \in I$$

has degree  $< d_{k+1}$  (with respect to  $x_n$ ). But then  $p' \in (p_1, \dots, p_k)$ . But this easily implies that  $p_{k+1} = p' + \sum_{i=1}^k s_i p_i x_i^{d_{k+1}-d_i} \in (p_1, \dots, p_k)$  which is impossible.  $\square$

**Corollary 1.10.** *Let  $\{Z_i | i \in I\}$  be affine algebraic sets, then  $Z = \cap_{i \in I} Z_i$  is an affine set and in fact there exists a finite subset  $I_0 \subset I$  such that  $Z = \cap_{i \in I_0} Z_i$ .*

*Proof.* Assume for simplicity that  $Z_i = \mathbb{V}(p_i)$ , then  $Z = \mathbb{V}(\{p_i | i \in I\})$ . By Theorem 1.9, there is a finite subset  $I_0 \subset I$  such that  $Z = \mathbb{V}(\{p_i | i \in I_0\})$ . Thus  $Z = \cap_{i \in I_0} \mathbb{V}(p_i)$ .  $\square$

**Exercise 1.11.** *Show by example that the arbitrary union of algebraic sets is not in general an algebraic set.*

Next we will show that affine algebraic sets are the closed subset of the Zariski topology on  $\mathbb{C}^N$ . Recall that

**Definition 1.12.** *A topology on a space (or set)  $X$  is a collection of subsets (called open subsets)  $\Omega = \{U_i\}_{i \in I}$  such that  $U_i \subset X$  and*

- (1)  $\emptyset, X \in \Omega$ ,
- (2) *finite intersections of open subsets are open:*  $U_{i_1} \cap \dots \cap U_{i_r} \in \Omega$ , and
- (3) *arbitrary unions of open subsets are open:* for any  $J \subset I$  we have  $\cup_{j \in J} U_j \in \Omega$ .

*The closed subsets are the complements of open subsets.*

**Exercise 1.13.** *Show that  $\Omega$  is a topology on  $X$  if and only if  $\emptyset, X$  are closed, the union of two closed subsets is closed and arbitrary intersections of closed subsets are closed.*

**Exercise 1.14.** *The open subset of the (usual) Euclidean topology on  $\mathbb{R}^n$  are the arbitrary unions of open balls (an open ball of radius  $r > 0$  centered at  $a_1, \dots, a_n$  is just the set  $\{(b_1, \dots, b_n) \in \mathbb{R}^n | \sum (b_i - a_i)^2 < r^2\}$ ). Show that this defines a topology on  $\mathbb{R}^n$ .*

**Exercise 1.15.** *Show that the subsets of  $\mathbb{C}$  given by  $\emptyset, \mathbb{C}$  and the complements of finitely many points, define a topology on  $\mathbb{C}$ .*

**Exercise 1.16.** *Given any (non-empty) set  $X$ , show that one can consider the two following topologies:*

- (1)  $\Omega$  consists of all subsets of  $X$ ,
- (2)  $\Omega = \{\emptyset, X\}$ .

**Exercise 1.17.** *Show that  $\mathbb{V}(p) \cup \mathbb{V}(q) = \mathbb{V}(pq)$  as subsets of  $\mathbb{C}^n$  thus the union of two hypersurfaces is a hypersurface.*

**Exercise 1.18.** *Show that  $\mathbb{V}(p) \cap \mathbb{V}(q) = \mathbb{V}(p, q)$  thus the intersection of two hypersurfaces in  $\mathbb{C}^N$  is an affine algebraic set (but may not be a hypersurface).*

We have the following important fact.

**Theorem 1.19.** *Let  $\Omega$  be the set of all complements of all algebraic sets in  $\mathbb{C}^N$ , then  $\Omega$  defines a topology (the Zariski topology) on  $\mathbb{C}^N$ .*

*Proof.* Since  $\emptyset, \mathbb{C}^n$  are algebraic sets (as observed above), then their complements  $\mathbb{C}^n, \emptyset$  are open subsets.

Suppose that  $\mathbb{V}(p_1, \dots, p_r)$  and  $\mathbb{V}(q_1, \dots, q_s)$  are closed subsets of  $\mathbb{C}^n$  where  $p_i, q_j \in \mathbb{C}[x_1, \dots, x_n]$ , then

$$\mathbb{V}(p_1, \dots, p_r) \cup \mathbb{V}(q_1, \dots, q_s) = \mathbb{V}(p_i q_j | 1 \leq i \leq r, 1 \leq j \leq s).$$

To see the inclusion  $\subset$ , note that if  $a = (a_1, \dots, a_n) \in \mathbb{V}(p_1, \dots, p_r)$ , then  $p_i(a) = 0$  for  $1 \leq i \leq r$  and so  $(p_i q_j)(a) = p_i(a) q_j(a) = 0$  for  $1 \leq i \leq r, 1 \leq j \leq s$  so that  $a \in \mathbb{V}(p_i q_j | 1 \leq i \leq r, 1 \leq j \leq s)$ . Thus we have shown that  $\mathbb{V}(p_1, \dots, p_r) \subset \mathbb{V}(p_i q_j | 1 \leq i \leq r, 1 \leq j \leq s)$ . The inclusion  $\mathbb{V}(q_1, \dots, q_s) \subset \mathbb{V}(p_i q_j | 1 \leq i \leq r, 1 \leq j \leq s)$  is similar. To see the inclusion  $\supset$ , suppose that  $a \notin \mathbb{V}(p_1, \dots, p_r) \cup \mathbb{V}(q_1, \dots, q_s)$ , then there exists  $i, j$  such that  $p_i(a) \neq 0$  and  $q_j(a) \neq 0$ . Thus  $(p_i q_j)(a) = p_i(a) q_j(a) \neq 0$  so that  $a \notin \mathbb{V}(p_i q_j | 1 \leq i \leq r, 1 \leq j \leq s)$ .

Suppose that  $\{V_j\}_{j \in J}$  is an arbitrary collection of affine algebraic sets in  $\mathbb{C}^n$ , then each  $V_j$  is defined by a collection of polynomials  $p_j \subset \mathbb{C}[x_1, \dots, x_n]$ . But then

$$\bigcap_{j \in J} V_j = \mathbb{V}(\{p_j\}_{j \in J}).$$

Thus arbitrary intersection of closed subsets is closed.  $\square$

**Remark 1.20.** More generally, if  $X \subset \mathbb{C}^n$  is an algebraic set, then we define the Zariski topology by letting the closed subsets of  $X$  corresponding to the intersection of closed subsets of  $\mathbb{C}^n$  with  $X$  (so these closed sets are the sets defined by the vanishing of polynomial equations). It is immediate from Theorem 1.19 that this is in fact a topology.

**Exercise 1.21.** Show that if  $X \subset \mathbb{C}^n$  is an affine algebraic set (these are the closed sets in the Zariski topology), then it is closed in the Euclidean topology. Therefore the Euclidean topology has "more" closed (and open) subsets than the Zariski topology. we say that the Euclidean topology is **finer** than the Zariski topology.

**Definition 1.22.** An algebraic set  $X \subset \mathbb{C}^n$  is **reducible** if it can be written as the union of two algebraic sets  $X = X_1 \cup X_2$  such that  $X_1 \not\subset X_2$  and  $X_2 \not\subset X_1$ . If  $X$  is not reducible, then we say that  $X$  is **irreducible**. Irreducible affine algebraic sets will be called **affine varieties**.

We notice the following:

- (1)  $\mathbb{C}^n$  is irreducible.
- (2)  $\mathbb{V}(xy) \subset \mathbb{C}^2$  is reducible.
- (3) A hypersurface  $\mathbb{V}(p) \subset \mathbb{C}^n$  is irreducible if and only if  $p = q^n$  for some irreducible polynomial  $q \in \mathbb{C}[x_1, \dots, x_n]$ . In this case  $\mathbb{I}(\mathbb{V}(p)) = q$ .
- (4) If  $f : X \rightarrow Y$  is a surjective continuous map and  $Y$  is reducible, then  $X$  is reducible. (Recall that a map is **continuous** if inverse images of open subsets are open.)

**Definition 1.23.** Let  $X \subset \mathbb{C}^n$  be an algebraic variety, then the **dimension** of  $X$  is the maximum length of a chain of irreducible non-empty closed subsets  $X_0 \subset X_1 \subset \dots \subset X_n \subset X$ . In this case we say that  $\dim X = n$ .

We will see later that affine varieties are finite dimensional and  $\dim \mathbb{C}^n = n$ .

**Exercise 1.24.** Show that  $\mathbb{C}^n$  is irreducible.

**Exercise 1.25.** Show that  $\dim \mathbb{C}^n \geq n$ .

**Exercise 1.26.** Use the remarks after Definition 1.1 to show that  $\dim \mathbb{C}^2 = 2$ .

**Definition 1.27.** A topological space  $X$  is **Noetherian** if any decreasing sequence of closed subsets

$$X \supset X_1 \supset X_2 \supset \dots$$

is eventually constant (so that  $X_j = \cap_{i \in \mathbb{N}} X_i$  for all  $j \gg 0$ ).

We will see later that affine varieties are Noetherian topological spaces.

**Proposition 1.28.** let  $X$  be a Noetherian topological space, then any closed subset  $V \subset X$  is a finite union of irreducible closed subsets

$$V = V_1 \cup \dots \cup V_r.$$

If moreover, we require that there are no inclusions  $V_i \subset V_j$  for  $i \neq j$ , then the  $V_i$  are uniquely determined.

*Proof.* See [Hartshorne, 1.5]. □

**Lemma 1.29.** If  $X \subset \mathbb{C}^n$  is an algebraic set and  $f \in \mathbb{C}[x_1, \dots, x_n]$ , then  $X \setminus \mathbb{V}(f)$  is an algebraic set.

*Proof.* Let  $a = \mathbb{I}(X) \subset \mathbb{C}[x_1, \dots, x_n]$ , and consider the ideal

$$\tilde{a} = (a, x_0 f - 1) \subset \mathbb{C}[x_0, x_1, \dots, x_n]$$

and the corresponding affine variety  $X_f = \mathbb{V}(\tilde{a}) \subset \mathbb{C}^{n+1}$ . The projection  $\mathbb{C}^{n+1} \rightarrow \mathbb{C}^n$  defined by  $(x_0, x_1, \dots, x_n) \rightarrow (x_1, \dots, x_n)$  gives a bijection  $X_f \rightarrow X \setminus \mathbb{V}(f)$ . (In fact, if  $(a_0, \dots, a_n) \in \mathbb{V}(\tilde{a})$ , then clearly  $(a_1, \dots, a_n) \in \mathbb{V}(a)$  and since  $1 = a_0 f(a_1, \dots, a_n)$ , then  $f(a_1, \dots, a_n) \neq 0$  and conversely if  $(a_1, \dots, a_n) \in \mathbb{V}(a)$  and  $f(a_1, \dots, a_n) \neq 0$ , then let  $a_0 = 1/f(a_1, \dots, a_n)$  so that  $(a_0, \dots, a_n) \in \mathbb{V}(\tilde{a})$ ). In order to complete the proof, it suffices to check that the induced bijections  $X \setminus \mathbb{V}(f) \rightarrow X_f$  and  $X_f \rightarrow X \setminus \mathbb{V}(f)$  are continuous. □

**Remark 1.30.** We can think of the process of realizing the complement of a hypersurface in  $X$  as a hypersurface in an algebraic set of dimension one more. Another perspective is given by localization of rings (see Section 3).

**Exercise 1.31.** Show that the induced bijections  $X \setminus \mathbb{V}(f) \rightarrow X_f$  and  $X_f \rightarrow X \setminus \mathbb{V}(f)$  are continuous. (See Lemma 2.2.)

We say that the open subsets  $X_p$  where  $p \in \mathbb{C}[x_1, \dots, x_n]$  are **principal open subsets**. Next we show that the principal open subsets generate the Zariski topology.

**Lemma 1.32.** let  $U \subset X \subset \mathbb{C}^n$  be an open subset of an affine variety, then  $U$  is a finite union of principal open subsets  $U = \cup X_{p_i}$  where  $p \in \mathbb{C}[x_1, \dots, x_n]$ .

*Proof.* Let  $Z = X \setminus U$ , then since  $Z$  is closed, for any  $x \in U$ , there is a polynomial  $p_x \in \mathbb{I}(Z) \subset \mathbb{C}[x_1, \dots, x_n]$ , such that  $p_x(x) \neq 0$ . It is then clear that  $U = \cup_{x \in U} X_{p_x}$ . Note that  $Z = \cap X \setminus X_{p_x}$  is a closed subset and hence there exists a finite subset  $X_i = X_{p_i}$  such that  $Z = \cap X \setminus X_{p_i}$  (see Corolary 1.10) or equivalently  $U = \cup_{i=1}^r X_i$ . □

**Exercise 1.33.** Let  $V_1 = \mathbb{V}(x + 2y + 3)$ ,  $V_2 = \mathbb{V}(3x - 2y + 9) \subset \mathbb{C}^2$ . Compute  $V_1 \cup V_2$  and  $V_1 \cap V_2$ .

**Exercise 1.34.** Write  $\mathbb{C}^2 \setminus (0, 0)$  as a union of two principal open subsets of  $\mathbb{C}^2$ .

**Exercise 1.35.** Show that the Euclidean topology on  $\mathbb{R}^2$  is induced by the product of the Euclidean topologies on  $\mathbb{R}^1$  (every open subset for the Euclidean topology on  $\mathbb{R}^2$  is a union of products  $U_1 \times U_2$  where  $U_1, U_2$  are open subsets in  $\mathbb{R}$ ).

**Exercise 1.36.** Show that the Zariski topology on  $\mathbb{C}^2$  is not the product of the Zariski topologies on  $\mathbb{C}$ .

**Exercise 1.37.** Let  $C_n = \{(1, t, t^2, \dots, t^n) | t \in \mathbb{C}\} \subset \mathbb{C}^{n+1}$  be the **rational normal curve** of degree  $n$ . Show that  $C_n$  is defined by the  $2 \times 2$  minors of the matrix with rows  $(x_0, \dots, x_{n-1})$  and  $(x_1, \dots, x_n)$ .

**Exercise 1.38.** Use Corollary 1.10 to show that if  $X$  is an affine variety and  $\{U_i\}_{i \in I}$  is a collection of open subsets  $U_i \subset X$  such that  $\cup_{i \in I} U_i = X$ , then there is a finite subset  $I_0 \subset I$  such that  $X = \cup_{i \in I_0} U_i$ . In other words  $X$  is **compact**.

**Exercise 1.39.** Show that if  $X$  is an irreducible subset of a topological space  $W$ , then the **closure**  $\bar{X}$  of  $X$  in  $W$  (defined as the intersection of all closed subsets containing  $X$ ) is irreducible.

**Exercise 1.40.** (Hard) Show that a subset  $X \subset \mathbb{C}^n$  is compact in the Euclidean topology (meaning that if  $\{U_i\}_{i \in I}$  is an open cover of  $X$ , then there is a finite subset  $I_0 \subset I$  such that  $\{U_i\}_{i \in I_0}$  is an open cover of  $X$ ) if and only if  $X$  is bounded and any sequence of points  $x_i \in X$  admits a convergent subsequence  $x_{i_k}$  whose limit belongs to  $X$  i.e.  $\lim x_{i_k} \in X$ .

## 2. MORPHISMS OF ALGEBRAIC VARIETIES

We will now discuss maps between affine algebraic sets. The most natural maps to consider are the ones induced by polynomials: given  $P = (p_1, \dots, p_m)$  a collection of polynomials  $p_i \in \mathbb{C}[x_1, \dots, x_n]$  we define the **morphism** of affine spaces

$$F : \mathbb{C}^n \rightarrow \mathbb{C}^m, \quad a = (a_1, \dots, a_n) \rightarrow (p_1(a), \dots, p_m(a)).$$

Given affine varieties  $X \subset \mathbb{C}^n$  and  $Y \subset \mathbb{C}^m$  such that  $F(X) \subset Y$ , then we say that  $f = F|_X : X \rightarrow Y$  is a **morphism of affine varieties**.

**Exercise 2.1.** Show that the composition  $f \circ g$  of two morphisms of affine varieties  $g : X \rightarrow Y$  and  $f : Y \rightarrow Z$  is a morphism of affine varieties.

**Lemma 2.2.** Morphisms of algebraic varieties are continuous in the Zariski topology.

*Proof.* Recall that a map is **continuous** if the inverse image of every open subset is an open subset or equivalently the inverse image of every closed subset is a closed subset. Let  $f : X \rightarrow Y$  be a morphism of algebraic varieties given by a morphism of affine spaces  $F = (p_1, \dots, p_m) : \mathbb{C}^n \rightarrow \mathbb{C}^m$ . If  $V \subset Y$  is a closed subset, then  $V = \mathbb{V}(q_i | i \in I)$  is the zero of a collection of polynomials  $q_i \in \mathbb{C}[y_1, \dots, y_m]$ . The inverse image of  $V$  is

$$f^{-1}(V) = \mathbb{V}(q_i(p_1(x_1, \dots, x_n), \dots, p_m(x_1, \dots, x_n)) | i \in I)$$

which is clearly a closed subset. □

**Definition 2.3.** Let  $X \subset \mathbb{C}^n$  and  $Y \subset \mathbb{C}^m$ , then we say that  $X$  and  $Y$  are **isomorphic** if there are morphisms  $f : X \rightarrow Y$  and  $g : Y \rightarrow X$  such that  $f \circ g = \text{id}_Y$  and  $g \circ f = \text{id}_X$ .

**Exercise 2.4.** Show that  $\mathbb{V}(x^2 - y, x^3 - z) \subset \mathbb{C}^3$  is isomorphic to  $\mathbb{C}$ .



**Exercise 2.5.** Show that the projection of  $\mathbb{V}(y = x^2)$  on to the  $x$  axis is an isomorphism but the projection on to the  $y$  axis is not an isomorphism.

**Exercise 2.6.** (Hard) Show that  $\mathbb{V}(x^2 - y^3)$  is not isomorphic to  $\mathbb{C}$ , although they are homeomorphic. (Recall that two sets  $X, Y$  are **homeomorphic** if there is a bijection  $f : X \rightarrow Y$  such that  $f$  and  $f^{-1}$  are continuous.)

**Exercise 2.7.** Show that  $\mathbb{V}(xy)$  is not isomorphic to  $\mathbb{C}$ .

**Exercise 2.8.** Show that  $\mathbb{V}(xy - 1) \subset \mathbb{C}^2$  is homeomorphic to  $\mathbb{C}^*$  but not isomorphic to  $\mathbb{C}^*$  (as affine varieties; in fact  $\mathbb{C}^*$  is not an affine subvariety of  $\mathbb{C}$ . We will see later that they are isomorphic as quasi-projective varieties).

**Exercise 2.9.** Show that  $\mathbb{C}^2$  is not isomorphic to  $\mathbb{C}$ .

**Exercise 2.10.** Show that  $\dim$  is an isomorphism invariant.

**Exercise 2.11.** Give an example of a morphisms of algebraic varieties  $f : X \rightarrow Y$  such that  $f(X) \subset Y$  is not a closed subset.

### 3. RINGS AND IDEALS

In this section we will re-interpret the definitions and results of the previous sections in more algebraic terms. We begin by recalling the following algebraic definitions and results.

**Definition 3.1.** A **commutative ring with identity**  $(R, +, *)$  is a nonempty set  $R$  with two operations

$$+ : R \times R \rightarrow R \quad \text{and} \quad * : R \times R \rightarrow R$$

(addition and multiplication) such that  $(R, +)$  is an abelian group (i.e. for any  $a, b, c \in R$  we have  $(a + b) + c = a + (b + c)$ , there exists an element (the additive identity)  $0 \in R$  such that  $a + 0 = 0 + a = a$ , there exists additive inverses  $-a \in R$  such that  $a + (-a) = 0$ , and  $a + b = b + a$ ), multiplication is associative, commutative and has a multiplicative identity  $1 \neq 0$  ( $a * (b * c) = (a * b) * c$ ,  $a * b = b * a$  and  $1 * a = a * 1$ ) and the distributive law holds ( $a * (b + c) = a * b + a * c$ ).

Since the rings that we consider will almost always be commutative with identity, we will just refer to these as rings. A typical example of a ring is the polynomial ring in  $n$  variables  $\mathbb{C}[x_1, \dots, x_n]$ . Other examples are  $\mathbb{Z}, \mathbb{R}, \mathbb{Q}, \mathbb{C}$  and the corresponding polynomial rings.

**Exercise 3.2.** Show that the set of  $n \times n$  matrices with complex coefficients  $M_{n \times n}(\mathbb{C})$  is a non-commutative ring. Its identity is the identity matrix (i.e. the matrix  $I$  defined by the entries  $I_{i,j} = 0$  if  $i \neq j$  and  $I_{i,i} = 1$ ).

**Definition 3.3.** If  $I \subset R$  is a non-empty subset of a ring, then we say that  $I$  is an **ideal** if it is closed under addition, additive inverses (so that  $I \subset R$  is an additive subgroup) and  $ri \in I$  for all  $i \in I$  and  $r \in R$ .

**Exercise 3.4.** Given  $r \in R$ ,  $(r) := \{sr | s \in R\} \subset R$  is an ideal. These ideals are known as **principal ideals**.

**Exercise 3.5.** Let  $I, J$  be ideals. Show that  $I \cap J$  and  $I + J = \{i + j | i \in I, j \in J\}$  are ideals.

**Exercise 3.6.** Let  $I_1 \subset I_2 \subset \dots$  be an increasing sequence of ideals. Show that  $\cup_{i \geq 0} I_i$  is an ideal.

**Exercise 3.7.** Let  $I \subset R$  be a subset containing 0 and 1 such that if  $a, b \in I$ , then  $sa + tb \in I$  for any  $s, t \in R$ , then  $I$  is an ideal.

The importance of ideals is given by the fact that we can quotient rings by ideals to obtain new rings. This is accomplished as follows: Let  $I \subset R$  be an ideal, then we may define the equivalence relation  $r \sim r'$  if and only if  $r - r' \in I$ . It is easy to check that this is an equivalence relation with equivalence classes

$$R/I = \{r + I | r \in R\}.$$

We can then define addition and multiplication on  $R/I$  via

$$(r + I) + (s + I) = (r + s) + I, \quad (r + I) * (s + I) = (r * s) + I.$$

**Exercise 3.8.** Show that  $+$  and  $*$  defined above, are well defined (i.e. if  $r + I = r' + I$  and  $s + I = s' + I$ , then  $(r + I) + (s + I) = (r' + I) + (s' + I)$  and  $(r + I) * (s + I) = (r' + I) * (s' + I)$ ).

**Theorem 3.9.** Let  $I \subset R$  be an ideal, then  $R/I$  with addition and multiplication defined as above is a ring.

*Proof.* Exercise. □

**Exercise 3.10.** Let  $m \in \mathbb{N}$  be a positive integer. Show that  $\mathbb{Z}/(m)$  is a ring.

**Exercise 3.11.** Let  $R$  be a ring and  $\{r_i\}_{i \in I}$  a collection of elements of  $R$ , then  $(\{r_i\}_{i \in I}) = \{\sum r_{i_1} s_{i_1} + \dots + r_{i_k} s_{i_k} | s_i \in R\}$  is the smallest ideal containing  $\{r_i\}_{i \in I}$ .

**Exercise 3.12.** Show that if  $I$  is an ideal in  $\mathbb{C}[x]$ , then  $I = (p(x))$  for some  $p(x) \in \mathbb{C}[x]$ .

**Exercise 3.13.** Show that if  $I$  is an ideal in  $\mathbb{Z}$ , then  $I = (k)$  for some  $k \in \mathbb{Z}$ .

**Exercise 3.14.** Show that there are ideals in  $\mathbb{C}[x, y]$  such that  $I$  is not generated by one element.

**Exercise 3.15.** Show that there are ideals of  $\mathbb{Z}[x]$  that are not generated by one element.

Most rings that we will consider are **Noetherian**. This means that for any ideal  $I \subset R$ , there exist finitely many elements  $r_1, \dots, r_k \in R$  such that  $I = (r_1, \dots, r_k)$ . We have the following:

**Lemma 3.16.** Let  $R$  be a ring then the following are equivalent.

- (1)  $R$  is Noetherian.
- (2) If  $\{r_i\}_{i \in \mathbb{N}}$  is a sequence of elements of  $R$ , then there exists  $k > 0$  such that  $(\{r_i\}_{i \in \mathbb{N}}) = (r_1, \dots, r_k)$
- (3) If  $I_1 \subset I_2 \subset \dots$  is an increasing sequence of ideals, then  $I_k = I = \cup_{j > 0} I_j$  for all  $k \gg 0$  (or equivalently the sequence stabilizes so that there exists  $k_0$  such that  $I_k = I_{k_0}$  for all  $k \geq k_0$ ).

*Proof.* (2) implies (1): If (1) does not hold, then we can find a sequence of elements  $r_i \in I$  such that the inclusions

$$(r_1, \dots, r_k) \subset (r_1, \dots, r_{k+1})$$

are strict for all  $k > 0$ . By (2), this is impossible.

(1) implies (3): Suppose that  $R$  is Noetherian and let  $I = \cup_{j > 0} I_j$ . Let  $I = (r_1, \dots, r_k)$ , then each  $r_i$  belongs to some  $I_{j_i}$  and hence  $I = I_{j_1} \cup \dots \cup I_{j_k}$ .

(3) implies (2): Let  $I_i = (\{r_1, \dots, r_i\})$ , then  $I_i \subset I_{i+1} \dots$  and so by (3)  $I_k = \cup I_i$  so that  $(\{r_1, \dots, r_k\}) = (\{r_i\}_{i \in \mathbb{N}})$ . □

**Theorem 3.17.** *Let  $R$  be a ring (commutative with 1). If  $R$  is Noetherian, then so is  $R[x]$ .*

*Proof.* Exercise (follow the arguments in the proof of Theorem 1.9). □

**Definition 3.18.**  $\phi : R \rightarrow S$  is a **homomorphism of rings** if  $R, S$  are rings and  $\phi(a+b) = \phi(a) + \phi(b)$  and  $\phi(a*b) = \phi(a) * \phi(b)$ . The **kernel** of  $\phi$  is the subset  $\ker(\phi) = \{r \in R \mid \phi(r) = 0\}$ .  $\phi$  is **injective** if  $\ker(\phi) = 0$  and  $\phi$  is **surjective** if  $\phi(R) = S$ .

**Exercise 3.19.** Show that  $\ker(\phi) \subset R$  is an ideal and  $\phi(R) \subset S$  is a subring such that  $\phi(R) \cong R/\ker(\phi)$ .

**Exercise 3.20.** Show that if  $I \subset \phi(R)$  is an ideal, then  $\phi^{-1}(I) \subset R$  is an ideal.

**Exercise 3.21.** Show that if  $\phi : R \rightarrow S$  is surjective and  $I \subset R$  is an ideal, then  $\phi(I) \subset S$  is an ideal. Show by example that  $\phi$  being surjective is necessary.

**Exercise 3.22.** Show that the ideals of  $\phi(R)$  are in one to one correspondence with the ideals of  $R$  that contain  $K = \ker(\phi)$ .

**Exercise 3.23.** Show that if  $R$  is Noetherian, then so is  $\phi(R)$ .

**Exercise 3.24.** Show that if  $R'$  is a finitely generated ring over  $R$  (meaning that there is a surjective homomorphism  $R[x_1, \dots, x_n] \rightarrow R'$ ) and  $R$  is Noetherian, then so is  $R'$ .

**Definition 3.25.** An ideal  $P \subset R$  is **prime** if given  $x, y \in R$  such that  $x * y \in P$ , then  $x \in P$  or  $y \in P$ .

**Exercise 3.26.** Show that if  $m \geq 2$ , then  $(m) \subset \mathbb{Z}$  is prime if and only if  $m$  is a prime number.

**Exercise 3.27.** Show that if  $f \in \mathbb{C}[x_1, \dots, x_n]$ , then  $(f) \subset \mathbb{C}[x_1, \dots, x_n]$  is prime if and only if  $f$  is irreducible (i.e. it can not be written as the product of two non constant polynomials).

**Exercise 3.28.** If  $f : A \rightarrow B$  is a ring homomorphism and  $P \subset B$  is a prime ideal, then show that  $f^{-1}(P) \subset A$  is also a prime ideal.

**Definition 3.29.** An ideal  $\mathfrak{m} \subset R$  is **maximal** if given any other ideal  $\mathfrak{m} \subset I \subset R$  then either  $\mathfrak{m} = I$  or  $I = R$ .

**Exercise 3.30.** Show that the ideals  $(x_1 - a_1, \dots, x_n - a_n) \subset \mathbb{C}[x_1, \dots, x_n]$  are maximal (here  $a_i \in \mathbb{C}$ ).

**Exercise 3.31.** Show by example that if  $f : A \rightarrow B$  is a ring homomorphism and  $\mathfrak{m} \subset B$  is a maximal ideal, then  $f^{-1}(\mathfrak{m}) \subset A$  may not be maximal.

**Definition 3.32.** Let  $R$  be a ring (commutative with 1), then  $R$  is a **domain** if given  $r, s \in R$ , then  $r * s = 0$  implies  $r = 0$  or  $s = 0$  (in other words  $R$  has no **zero divisors**).

**Exercise 3.33.** Let  $I \subset R$  be an ideal, then  $I$  is prime if and only if  $R/I$  is a domain.

**Definition 3.34.** Let  $R$  be a ring (commutative with 1), then  $R$  is a **field** if given  $0 \neq r \in R$ , there exists  $s \in R$  such that  $r * s = 1$  (we will say that  $r$  is invertible and denote  $s = r^{-1}$ ).

**Exercise 3.35.** Show that multiplicative inverses are unique (if they exist)

**Exercise 3.36.** Show that any field is a domain.

**Exercise 3.37.** Let  $I \subset R$  be an ideal, then  $I$  is maximal if and only if  $R/I$  is a field. (I.e. show that any element  $r + I \neq 0 + I$  is invertible in  $R/I$ ; see §5 for the definition and properties of fields.)

**Definition 3.38.** Let  $I \subset R$  be an ideal, then the **radical** of  $I$  is

$$\sqrt{I} = \{f \in R \mid f^m \in I, \text{ for some } m > 0\}.$$

**Exercise 3.39.** Show that  $\sqrt{I} \subset R$  is an ideal.

**Exercise 3.40.** Show that  $R/I$  has no nilpotent elements (i.e. no elements  $0 \neq f \in R/I$  such that  $f^m = 0$  for some  $m > 0$ ) if and only if  $I$  is a radical ideal (i.e.  $\sqrt{I} = I$ ).

**Lemma 3.41.** If  $I \subset R$  is an ideal, then there is a 1-1 correspondence between the ideals of  $R/I$  and the ideals of  $R$  that contain  $I$ .

*Proof.* Exercise. □

**Definition 3.42.** A subset of a ring  $1 \in S \subset R$  is a **multiplicative subset** if for any  $s, t \in S$  we have  $st \in S$ . If  $R$  is a domain and  $0 \notin S$ , then we define

$$S^{-1}R := \{(a, s) \mid a \in R, s \in S\} / \sim$$

where  $\sim$  is the equivalence relation defined by  $(a, s) \sim (a', s')$  iff  $as' = a's$ .

**Exercise 3.43.** Show that  $(a, s) \sim (a', s')$  if and only if  $as' = a's$  is an equivalence relation. We denote the equivalence class of  $(a, s)$  by  $a/s$ .

**Lemma 3.44.** Define  $(a, s) + (a', s') = (as' + a's, ss')$  and  $(a, s)(a', s') = (aa', ss')$ , then  $S^{-1}R$  is a ring.

*Proof.* Exercise. □

Note that there is a natural inclusion of rings  $R \rightarrow S^{-1}R$ .

**Exercise 3.45.** Let  $I \subset S^{-1}R$  be an ideal, then  $I \cap R$  is also an ideal and if  $J \subset R$  is an ideal, then  $S^{-1}J \subset S^{-1}R$  is an ideal.

**Exercise 3.46.** Show that if  $I \subset S^{-1}R$  is an ideal, then  $I = (I \cap R)S^{-1}R$  and so the map  $I \rightarrow I \cap R$  is an injection from the set of all ideals in  $S^{-1}R$  to the set of ideals in  $R$ . Show that this injection takes prime ideals to prime ideals.

**Exercise 3.47.** Show that if  $I \subset R$  is an ideal, then  $I = J \cap R$  for some ideal  $J \subset S^{-1}R$  if and only if  $I = (IS^{-1}R) \cap R$ . Note that if  $I \cap S \neq \emptyset$ , then  $(IS^{-1}R) \cap R = R$  and so we have a bijection between the ideals of  $S^{-1}R$  and the ideals  $I \subset R$  such that  $I \cap S = \emptyset$ .

**Exercise 3.48.** Let  $R$  be a domain and  $S$  a multiplicative subset. Show that if  $R$  is Noetherian, then so is  $S^{-1}R$ .

**Exercise 3.49.** Show that if  $P \subset R$  is a prime ideal, then  $S = R \setminus P$  is a multiplicative set. We denote  $S^{-1}R = R_P$ .

**Exercise 3.50.** Show that if  $\mathfrak{m} \subset R$  is the unique maximal ideal, then every element of  $R \setminus \mathfrak{m}$  is invertible.

**Exercise 3.51.** Show that  $PR_P$  is the unique maximal ideal of  $R_P$ .

**Exercise 3.52.** Let  $S \subset T$  be multiplicative subsets of a domain  $R$ . Show that there is a natural inclusion  $S^{-1}R \subset T^{-1}R$  and an isomorphism  $T^{-1}(S^{-1}R) \cong T^{-1}R$ .

**Definition 3.53.** A ring  $R$  is **local** if it has a unique maximal ideal. In particular if  $P \in R$  is a prime ideal in a domain, then  $R_P$  is a local ring.

By 3.50, we know that a ring  $R$  with a maximal ideal  $\mathfrak{m}$  is local if and only if every element of  $R \setminus \mathfrak{m}$  is invertible.

**Exercise 3.54.** Let  $f \in R$  be a non zero element in a domain, then  $S = (f)$  is a multiplicative subset and we denote  $S^{-1}R$  by  $R_f$ . Show that the ideals of  $R_f$  are in one to one correspondence with ideals  $I \subset R$  such that  $I \cap (f) = \emptyset$ .

**Proposition 3.55.** Let  $R$  be a domain and  $Q(R)$  its field of fractions, then

$$R = \cap_P R_P = \cap_{\mathfrak{m}} R_{\mathfrak{m}}$$

where  $P \subset A$  are prime ideals (and  $\mathfrak{m} \subset A$  are prime ideals).

*Proof.* The inclusions  $R \subset \cap_P R_P \subset \cap_{\mathfrak{m}} R_{\mathfrak{m}}$  are immediate. Suppose that  $z \in \cap_{\mathfrak{m}} R_{\mathfrak{m}} \setminus R$ . Let  $I = \{r \in R \mid rz \in R\}$ , then  $I$  is a proper ideal of  $R$  (since  $1 \notin I$ ) and so  $I \subset \mathfrak{m}$  for some maximal ideal  $\mathfrak{m}$  of  $R$ . But then  $z \notin R_{\mathfrak{m}}$  since otherwise  $z \sim r/s$  where  $s \notin \mathfrak{m}$  and so  $sz = r$  i.e.  $s \in I \subset \mathfrak{m}$  which is the required contradiction.  $\square$

**Remark 3.56.** If  $R$  is a ring (not necessarily a domain), one can define the localization with respect to a multiplicative subset  $1 \in S \subset R$  by using the equivalence relation

$$(r, s) \sim (r', s') \quad \text{iff} \quad t(rs' - r's) = 0$$

for some  $t \in S$ . Addition and multiplication are defined in the same way.

**Exercise 3.57.** Show that this is an equivalence relation and multiplication and addition are well defined.

**Exercise 3.58.** Show that if  $\phi : R \rightarrow R'$  is a homomorphism of rings and  $S \subset R$  a multiplicative subset such that  $\phi(s) \in R'$  is a unit for all  $s \in S$ , then  $\phi$  extends uniquely to a homomorphism of rings  $S^{-1}R \rightarrow R'$ .

**Exercise 3.59.** Show that the natural map  $R \rightarrow S^{-1}R$  is not injective (this can happen only when  $R$  is not a domain).

## 4. MODULES

Modules over rings are a generalization of vector spaces over a field. The formal definition is as follows:

**Definition 4.1.** Let  $(M, +)$  be an abelian group and  $R$  a ring and  $R \times M \rightarrow M$  an operation denoted by  $(r, m) \rightarrow rm \in M$ , then  $M$  is a **(left)  $R$ -module** if the following properties hold.

- (1)  $1 \cdot m = m$  for all  $m \in M$ ,
- (2)  $(r + s)m = rm + sm$  for all  $r, s \in R, m \in M$ ,
- (3)  $(rs)m = r(sm)$  for all  $r, s \in R, m \in M$ , and
- (4)  $r(m + m') = rm + rm'$  for all  $r \in R, m, m' \in M$ .

**Exercise 4.2.** Show that if  $R$  is a field and  $M$  is an  $R$ -module, then  $M$  is an  $R$ -vector space.

**Exercise 4.3.** Show that every abelian group is a  $\mathbb{Z}$ -module.

**Exercise 4.4.** Show that if  $I \subset R$  is an ideal, then  $I$  is an  $R$ -module and  $R/I$  is an  $R$ -module.

**Exercise 4.5.** Show that if  $S \subset R$  is a multiplicative subset of a domain, then  $S^{-1}R$  is an  $R$ -module. More generally if  $M$  is an  $R$  module, define  $S^{-1}M$  similarly to  $S^{-1}R$  and show that  $S^{-1}M$  is both an  $R$  and an  $S^{-1}R$  module.

**Exercise 4.6.** Let  $R$  be a ring, show that  $R[x_1, \dots, x_n]$  is an  $R$ -module.

**Definition 4.7.** Let  $M$  be an  $R$ -module and  $N \subset M$  a subgroup, then  $N$  is a **sub  $R$ -module** if  $rn \in N$  for all  $n \in N$  and  $r \in R$ . Let  $M/N$  be the quotient of abelian groups, then  $M/N$  is naturally an  $R$ -module.

**Exercise 4.8.** If  $N \subset M$ ,  $L \subset M$  are submodules, then  $L \cap N \subset M$  is a submodule.

**Exercise 4.9.** If  $N \subset M$  is a submodule, then show that the submodules of  $M/N$  are in one to one correspondence with the submodules  $N \subset L \subset M$ .

**Definition 4.10.** Let  $M, N$  be  $R$  modules. A **homomorphism of  $R$ -modules**  $f : M \rightarrow N$  is a map such that

$$f(rm + sm') = rf(m) + sf(m') \quad \forall r, s \in R, m, m' \in M.$$

If  $f$  is bijective, we say that  $f$  is an **isomorphism of modules**. The **kernel** of  $f$  is the set  $\ker(f) = \{m \in M | f(m) = 0 \in N\}$ . It is a submodule of  $M$ . The **image** of  $f$  is the set  $\text{im}(f) = \{n \in N | n = f(m) \text{ for some } m \in M\}$ . It is a submodule of  $M$ .

**Exercise 4.11.** Show that the kernel and image of a homomorphism of  $R$ -modules is an  $R$ -module.

If  $N \subset M$  is a submodule, then  $M \rightarrow M/N$  is a homomorphism of modules. It is easy to see that if  $f : M \rightarrow N$  is a homomorphism of  $R$ -modules, then we have a short exact sequence of  $R$  modules

$$0 \rightarrow \ker(f) \rightarrow M \rightarrow \text{im}(f) \rightarrow 0.$$

As usual, this means that the first map is injective, the last one is surjective and  $\ker(M \rightarrow \text{im}(f)) = \text{im}(\ker(f) \rightarrow M)$ .

**Exercise 4.12.** Show that if  $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$  is a short exact sequence of  $R$ -modules, then  $M'' \cong M/M'$ .

**Lemma 4.13.** Let  $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$  and  $0 \rightarrow M'_1 \rightarrow M'_2 \rightarrow M'_3 \rightarrow 0$  be short exact sequences of  $R$  modules and  $f_i : M_i \rightarrow M'_i$  be compatible homomorphisms (so that the induced maps  $M_i \rightarrow M_{i+1} \rightarrow M'_{i+1}$  and  $M_i \rightarrow M'_i \rightarrow M'_{i+1}$  coincide). If  $f_1$  and  $f_3$  are isomorphisms, then so is  $f_2$ .

*Proof.* We must show that  $f_2$  is injective and surjective. If  $f_2(m) = 0$  for some  $m \in M_2$ , then the image in  $M'_3$  is 0, and since  $f_3$  is an isomorphism, the image of  $m$  in  $M_3$  is also 0 and by exactness of the first sequence,  $m$  is the image of an element  $m_1$  in  $M_1$ . Let  $m'_1 = f_1(m_1)$ , then the image of  $m'_1$  in  $M'_2$  is  $f_2(m) = 0$ . But since  $M'_1 \rightarrow M'_2$  is injective,  $m'_1 = 0$  and hence  $m_1 = 0$  but then  $m = 0$  and so  $f_2$  is injective.

The proof that  $f_2$  is surjective is similar and we leave it as an exercise. This line of reasoning is called diagram chasing (for obvious reasons).  $\square$

**Definition 4.14.** A **Noetherian module** is a module  $M$  such that for any subsequence of modules  $N_1 \subset N_2 \subset N_3 \subset \dots \subset M$  we have  $N_i = \cup_{j \in \mathbb{N}} N_j$  for all  $i \gg 0$ . We say that  $M$  satisfies the **ascending chain condition**.

**Exercise 4.15.** Show that a ring  $R$  is Noetherian if and only if it is a Noetherian  $R$ -module.

**Exercise 4.16.** Show that  $\mathbb{C}[x]$  is not a Noetherian  $\mathbb{C}$  module (but of course it is a Noetherian  $\mathbb{C}[x]$  module).

**Definition 4.17.** Let  $M$  be an  $R$ -module satisfying the **descending chain condition**, then we say that  $M$  is an **Artinian module**. Equivalently if  $M \supset N_1 \supset N_2 \supset \dots$ , then  $N_i = \cap_{j \in \mathbb{N}} N_j$  for all  $j \gg 0$ .

**Exercise 4.18.** Show that if  $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$  is a short exact sequence of  $R$ -modules such that  $M', M''$  are Noetherian, if and only if so is  $M$ .

**Definition 4.19.** An  $R$ -module  $M$  is **finitely generated** if there are elements  $m_1, \dots, m_n \in M$  such that any element  $m \in M$  can be expressed as a linear combination  $m = \sum r_i m_i$  with  $r_i \in R$ . In other words, the homomorphism  $\phi : R^{\oplus n} \rightarrow M$  given by  $\phi(r_1, \dots, r_n) \rightarrow \sum r_i m_i$  is surjective.

**Exercise 4.20.** Show that a submodule  $N \subset M$  of a finitely generated Noetherian module is also finitely generated.

**Exercise 4.21.** Show that  $M$  is Noetherian if and only if every submodule  $N \subset M$  is finitely generated.

**Exercise 4.22.** Give an example of a homomorphism  $\phi : M \rightarrow M$  of finitely generated  $R$  modules with  $\phi$  injective but not surjective. (Note that if  $\phi$  is surjective, then it is injective. See Exercise 6.10.)

**Exercise 4.23.** Give an example of a non Noetherian finitely generated module (hint let  $R$  be a polynomial ring with infinitely many variables).

**Theorem 4.24.** Let  $M$  be a finitely generated  $R$ -module. If  $R$  is a Noetherian ring, then  $M$  is a Noetherian  $R$ -module.

*Proof.* Let  $\phi : R^{\oplus n} \rightarrow M$  be the corresponding homomorphism. If  $N_1 \subset N_2 \subset N_3 \subset \dots \subset M$  is an increasing sequence of submodules, then so is  $f^{-1}(N_1) \subset f^{-1}(N_2) \subset f^{-1}(N_3) \subset \dots \subset R^{\oplus n}$ . It suffices to show that  $f^{-1}(N_i) = \cup_{j \in \mathbb{N}} f^{-1}(N_j)$  for any  $i \gg 0$ . Therefore we may assume that  $M = R^{\oplus n}$ .

We now reason by induction on  $n$ . If  $n = 1$ , the statement is clear. In general we have a homomorphism  $g : R^{\oplus n} \rightarrow R^{\oplus n-1}$ . Let  $N_j'' = g(N_j)$  and  $N_j' = \ker(N_j \rightarrow N_j'')$ . Therefore, we have a short exact sequence  $0 \rightarrow N_j' \rightarrow N_j \rightarrow N_j'' \rightarrow 0$ . By our inductive assumption  $N_i' = N' := \cup_{j \in \mathbb{N}} N_j'$  and  $N_i'' = N'' := \cup_{j \in \mathbb{N}} N_j''$  for all  $i \gg 0$ . But then we have short exact sequences  $0 \rightarrow N' \rightarrow N_i \rightarrow N'' \rightarrow 0$  for all  $i \gg 0$  and applying Lemma 4.13 to the inclusion  $N_i \subset N_{i+1}$ , one sees that  $N_i = N_{i+1}$  for all  $i \gg 0$  and the result follows.  $\square$

**Corollary 4.25.** If  $R$  is Noetherian, then  $M$  is Noetherian if and only if it is a finitely generated  $R$  module.

*Proof.* Exercise. □

**Theorem 4.26.** *Let  $M$  be a finitely generated  $\mathbb{Z}$  module, then  $M \cong \mathbb{Z}^{r_0} \oplus \mathbb{Z}/p_1^{r_1}\mathbb{Z} \oplus \dots \oplus \mathbb{Z}/p_t^{r_t}\mathbb{Z}$  where  $p_i$  are primes and  $r_i \in \mathbb{N}$ . (More generally the same result holds if  $R$  is a PID and  $M$  is a finitely generated  $R$  module.)*

*Proof.* Since  $M$  is finitely generated, there is a surjection  $f : \mathbb{Z}^{\oplus n} \rightarrow M$  with kernel  $K = \ker(f)$ . By Exercise 4.20,  $K$  is also finitely generated and so there is a surjection  $\mathbb{Z}^{\oplus m} \rightarrow K$ . Let  $g : \mathbb{Z}^{\oplus m} \rightarrow \mathbb{Z}^{\oplus n}$  be the induced homomorphism, then  $M$  is the cokernel of  $g$  (i.e.  $M \cong \mathbb{Z}^{\oplus n}/\text{im}(g)$ ). The map  $g$  is given by a  $n \times m$  matrix  $A$ . After performing row and column operations (which simply corresponds to choosing another basis for  $\mathbb{Z}^{\oplus n}$  and  $\mathbb{Z}^{\oplus m}$ ) we may assume that the only non-zero entries are diagonal entries  $a_{1,1}, \dots, a_{d,d}$  (one can show that  $a_{i,i}$  divides  $a_{i+1,i+1}$ ). It then follows that  $M \cong \mathbb{Z}^{n-d} \oplus \mathbb{Z}/a_{1,1}\mathbb{Z} \oplus \dots \oplus \mathbb{Z}/a_{d,d}\mathbb{Z}$ . The result now follows by applying unique factorization; i.e. if  $a = p_1^{r_1} \dots p_t^{r_t}$ , then  $\mathbb{Z}/a\mathbb{Z} \cong \mathbb{Z}/p_1^{r_1}\mathbb{Z} \oplus \dots \oplus \mathbb{Z}/p_t^{r_t}\mathbb{Z}$ . □

**Lemma 4.27.** *Let  $M$  be an  $R$  module, then  $m \in M$  is zero if and only if  $0 = m \in M_{\mathfrak{m}}$  for any  $\mathfrak{m} \subset R$  maximal ideal.*

*Proof.* If  $0 = m \in M$ , then clearly  $0 = m \in M_{\mathfrak{m}}$ . Suppose now that  $0 = m \in M_{\mathfrak{m}}$  for any  $\mathfrak{m} \subset R$  maximal ideal. Thus  $vm = 0$  for some  $v \in R \setminus \mathfrak{m}$ . Let  $I = \{r \in R | rm = 0 \in M\}$ , then  $I$  is an ideal and hence  $I \subset \mathfrak{m}$  for some maximal ideal  $\mathfrak{m} \subset R$ . this is the required contradiction. □

**Lemma 4.28.** *Let  $M$  be an  $R$  module, then  $M = 0$  if and only if  $M_{\mathfrak{m}} = 0$  for all maximal ideals  $\mathfrak{m} \subset R$ .*

*Proof.* Exercise. □

**Lemma 4.29.** *Let  $M, N$  be an  $R$  modules and  $\phi : M \rightarrow N$  a homomorphism of  $R$  modules, then  $\phi$  is injective (resp. surjective) if and only if the induced map  $\phi_{\mathfrak{m}} : M_{\mathfrak{m}} \rightarrow N_{\mathfrak{m}}$  is injective (resp. surjective) for all maximal ideals  $\mathfrak{m} \subset R$ .*

*Proof.* Exercise. □

## 5. $\mathbb{C}$ -ALGEBRAS

**Definition 5.1.** *A ring  $R$  is a  $\mathbb{C}$ -algebra if there is an inclusion of rings  $\mathbb{C} \hookrightarrow R$ . A homomorphism  $\phi : R \rightarrow S$  of  $\mathbb{C}$ -algebras is a  $\mathbb{C}$ -linear ring homomorphism i.e. a ring homomorphism such that  $\phi(\lambda r) = \lambda \phi(r)$  for all  $\lambda \in \mathbb{C}$ .*

**Exercise 5.2.** *Find a ring homomorphism (of  $\mathbb{C}$  algebras) that is not a  $\mathbb{C}$ -algebra homomorphism*

Note that a  $\mathbb{C}$ -algebra is automatically a  $\mathbb{C}$ -vector space. It is easy to see that if  $I \subset \mathbb{C}[x_1, \dots, x_n]$  is a proper ideal, then  $\mathbb{C}[x_1, \dots, x_n]/I$  is a  $\mathbb{C}$ -algebra. Given elements  $r_1, \dots, r_n \in R$  where  $R$  is a  $\mathbb{C}$ -algebra, we can always define a unique  $\mathbb{C}$  algebra homomorphism

$$\phi : \mathbb{C}[x_1, \dots, x_n] \rightarrow R$$

by letting  $\phi(p(x_1, \dots, x_n)) = p(r_1, \dots, r_n)$ .



**Definition 5.3.** We say that a  $\mathbb{C}$ -algebra  $R$  is **finitely generated** if there are elements  $r_1, \dots, r_n \in R$  such that the homomorphism

$$\mathbb{C}[x_1, \dots, x_n] \rightarrow R, \quad \phi(p(x_1, \dots, x_n)) = p(r_1, \dots, r_n)$$

is surjective (and hence  $R \cong \mathbb{C}[x_1, \dots, x_n]/\ker(\phi)$ ). If  $\ker(\phi) = 0$ , then  $R \cong \mathbb{C}[x_1, \dots, x_n]$  and we say that  $r_1, \dots, r_n$  are **algebraically independent** (over  $\mathbb{C}$ ).

**Remark 5.4.** In this case  $R$  is "polynomially" finitely generated over  $\mathbb{C}$  (i.e. finitely generated as a  $\mathbb{C}$ -algebra) but it is not finitely generated as a  $\mathbb{C}$  module.

**Definition 5.5.** Let  $B \subset A$  be an inclusion of  $\mathbb{C}$  algebras. we say that  $A$  is a **finite  $B$  algebra** if there are finitely many elements  $a_1, \dots, a_n$  such that for any  $a \in A$  there exists elements  $b_1, \dots, b_n \in B$  such that  $a = \sum_{i=1}^n b_i a_i$ . (Thus  $A$  is finitely generated as a  $B$  module.)

Note that finite dimensional vector spaces are finite  $\mathbb{C}$  algebras, but finitely generated  $\mathbb{C}$  algebras are typically infinite dimensional  $\mathbb{C}$  vector spaces (eg.  $\mathbb{C}[x]$ ).

**Lemma 5.6.** If  $A \subset B$  is a finite  $A$  algebra and  $B \subset C$  is a finite  $B$ -algebra, then  $A \subset C$  is a finite  $A$  algebra.

*Proof.* By definition, we may fix elements  $b_1, \dots, b_n$  generating  $B$  over  $A$  and  $c_1, \dots, c_m$  generating  $C$  over  $B$ . We claim that  $\{b_i c_j\}_{1 \leq i \leq n, 1 \leq j \leq m}$  generate  $C$  over  $A$ . To see this, pick any element  $c \in C$ , then we may write  $c = \sum c_j \beta_j$  where  $\beta_j \in B$ . Similarly each  $\beta_j$  can be written as  $\beta_j = \sum b_i \alpha_{i,j}$  for appropriate elements  $\alpha_{i,j} \in A$ . But then

$$c = \sum_j c_j \beta_j = \sum_j c_j \left( \sum_i b_i \alpha_{i,j} \right) = \sum_{i,j} \alpha_{i,j} b_i c_j.$$

□

Recall that a polynomial  $p \in A[x]$  is **monic** if its leading term is 1. We have the following.

**Lemma 5.7.** Let  $A \subset B$  be an inclusion of  $\mathbb{C}$ -algebras. An element  $b \in B$  satisfies a monic polynomial  $p \in A[x]$  if and only if  $A[b]$  is a finite  $A$  algebra.

*Proof.* Suppose that  $b$  satisfies a monic polynomial  $p(x) = x^d + a_1 x^{d-1} + \dots + a_{d-1} x + a_d$  with  $a_i \in A$ . Then  $A[b]$  is generated by  $1, b, b^2, \dots, b^{d-1}$ . In fact any element of  $A[b]$  may be written as  $f(b)$  where  $f \in A[x]$  is an arbitrary polynomial. Divide  $f(x)$  by  $p(x)$  to get  $f(x) = p(x)q(x) + r(x)$  where  $\deg r(x) \leq d-1$  (we can always do this when the leading coefficient of  $p(x)$  is invertible in  $B$ ). But then  $f(b) = p(b)q(b) + r(b) = r(b)$ .

Replacing  $B$  by  $A[b]$ , we may now suppose that  $B$  is finite over  $A$ . Let  $b_1, \dots, b_n$  be a generating set. For any  $x \in B$ , we write

$$xb_i = \sum_{j=1}^n a_{i,j} b_j.$$

This equation can be rewritten as

$$(xI_n - A) \cdot \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = 0,$$

where  $I_n$  is the  $n \times n$  identity matrix and  $A$  is the  $n \times n$  matrix with entries  $a_{i,j}$ . Recall from linear algebra that

$$\text{adj}(xI_n - A)(xI_n - A) = \det(xI_n - A)I_n$$

and so

$$0 = \text{adj}(xI_n - A)(xI_n - A) \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = \det(xI_n - A)I_n \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = \det(xI_n - A) \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$

But this implies that  $\det(xI_n - A) = 0$  i.e. that  $x$  satisfies a monic polynomial.  $\square$

**Exercise 5.8.** Show that  $\det(xI_n - A)$  is a monic polynomial.

**Exercise 5.9.** Show that  $\mathbb{Q}[\sqrt{2}]$  is a finite  $\mathbb{Q}$  algebra. Show that  $3\sqrt{2} + 1$  satisfies a monic polynomial.

**Definition 5.10.** An element  $a \in A$  is **algebraic** if  $\mathbb{C}[a]$  is a finite  $\mathbb{C}$  algebra (or equivalently if  $a$  satisfies a monic polynomial  $p \in \mathbb{C}[x]$ ). If  $a \in A$  is not algebraic, then we say that it is **transcendental**.

**Theorem 5.11** (Noether normalization lemma). Let  $A$  be a  $\mathbb{C}$  algebra generated by finitely many elements  $a_1, \dots, a_n \in A$ , then there exist transcendental elements  $x_1, \dots, x_m \in A$  such that  $\mathbb{C}[x_1, \dots, x_m] \subset A$  is a finite extension.

*Proof.* (Cf. [Reid, 3.13]) Consider the surjective homomorphism of  $\mathbb{C}$ -algebras  $\mathbb{C}[y_1, \dots, y_n] \rightarrow A = \mathbb{C}[a_1, \dots, a_n]$  with kernel  $I$ . If  $I = 0$  there is nothing to prove. Suppose therefore that there is an element  $0 \neq f \in I$ . If  $f$  is a monic polynomial of  $y_1$  with coefficients in  $\mathbb{C}[y_2, \dots, y_n]$ , then we conclude that  $A' = \mathbb{C}[a_2, \dots, a_n] \subset A$  is finite. Proceeding by induction on  $n$  we also have that there are transcendental elements  $x_1, \dots, x_m \in A' \subset A$  with  $m \leq n - 1$  such that  $\mathbb{C}[x_1, \dots, x_m] \subset A'$  is finite and by the above lemma we conclude that  $\mathbb{C}[x_1, \dots, x_m] \subset A$  is finite as required.

Therefore, we must show that if  $0 \neq I$ , then  $I$  contains a monic polynomial of  $y_1$ . To see this start with any element  $0 \neq f \in I$ . We may assume that  $\deg(f) = d > 0$  and we write  $f = f_d + g$  where  $\deg g \leq d - 1$ . We now consider elements  $a_2, \dots, a_n \in \mathbb{C}$  and new variables  $y'_2 = y_2 - a_2 y_1, \dots, y'_n = y_n - a_n y_1$  so that

$$f(y_1, y_2, \dots, y_n) = f(y_1, y_2 + a_2 y_1, \dots, y_n + a_n y_1) = f_d(1, a_2, \dots, a_n) y_1^d + h,$$

where  $h$  has degree  $\leq d - 1$  with respect to  $y_1$ . Since  $f_d(1, a_2, \dots, a_n) \neq 0$ , we may assume that  $f_d(1, a_2, \dots, a_n) \neq 0$  as required.  $\square$

**Exercise 5.12.** Apply Noether's normalization lemma to  $\mathbb{C}[x, y]/(x^2 - y^2 + 1)$ .

**Exercise 5.13.** Apply Noether's normalization lemma to  $\mathbb{C}[x, y, z]/(xy + z^2, x^2 y - xy^3 + z^4 - 1)$ .

## 6. NAKAYAMA'S LEMMA

Let  $R$  be a Noetherian ring and  $M$  a finitely generated  $R$ -module.

**Lemma 6.1.** If  $I \subset R$  is an ideal and  $\phi : M \rightarrow M$  is a homomorphism of  $R$  modules such that  $\phi(M) \subset IM$ , then there exists a monic polynomial  $p(x) = x^n + a_1 x^{n-1} + \dots + a_{n-1} x + a_n$  where  $a_i \in I^i$  and  $p(\phi) = 0$ .

*Proof.* Exercise. (See the proof of Lemma 5.7.)  $\square$

**Corollary 6.2.** *If  $IM = M$ , then there exists an element  $r \in 1 + I \subset R$  such that  $rM = 0$ .*

*Proof.* We apply Lemma 6.1 with  $\phi = \text{id}$ . Let  $p(x)$  be the corresponding polynomial, then let  $r = p(1) = 1 + \sum_{i \geq 1} a_i$ . Clearly  $r \in 1 + I$ . It follows easily that for any  $m \in M$ , we have  $0 = p(1)m = rm$ .  $\square$

**Corollary 6.3.** *If  $(R, \mathfrak{m})$  is a local ring and  $M$  is a finitely generated  $R$  module such that  $\mathfrak{m}M = M$ , then  $M = 0$ .*

*Proof.* Let  $r \in 1 + \mathfrak{m}$  be defined by Corollary 6.2, then  $r$  is invertible (see Exercise 3.50) and so  $M = (r^{-1}r)M = 0$ .  $\square$

**Exercise 6.4.** *Show that the finitely generated hypothesis is needed. (Hint:  $R = \mathbb{Z}_{(p)}$  and  $M = \mathbb{Q}$ .)*

**Corollary 6.5.** *If  $(R, \mathfrak{m})$  is a local ring and  $M$  is a finitely generated  $R$  module and  $N \subset M$  a sub module such that  $M = N + \mathfrak{m}M$ , then  $N = M$ .*

*Proof.* By Corollary 6.3 applied to  $M/N$ , since  $M/N = (N + \mathfrak{m}M)/N = \mathfrak{m}M/N$ , it follows that  $\mathfrak{m}M/N = 0$  and hence  $M = N$ .  $\square$

**Corollary 6.6.** *If  $(R, \mathfrak{m})$  is a local Noetherian ring and  $M$  is a finitely generated  $R$  module and  $m_1, \dots, m_n \in M$  are elements such that the  $R$  module  $M/\mathfrak{m}M$  is generated by the images of  $m_1, \dots, m_n$ , then  $M$  is generated by  $m_1, \dots, m_n$ .*

*Proof.* Apply Corollary 6.5 to the submodule  $N = \sum m_i R$ .  $\square$

**Corollary 6.7** (Nakayama's Lemma). *If  $(R, \mathfrak{m})$  is a local Noetherian ring and  $M$  is a finitely generated  $R$  module, then  $R/\mathfrak{m} = k$  is a field and  $M/\mathfrak{m}M$  is a  $k$ -vector space. There is a one to one correspondence between any basis  $v_1, \dots, v_n$  of  $M/\mathfrak{m}M$  and the minimal generating sets  $m_1, \dots, m_n$  of  $M$  (over  $R$ ). In particular two basis of  $M/\mathfrak{m}M$  (resp. two minimal generating sets of  $M$ ) are related by an invertible matrix with coefficients in  $k$  (resp.  $R$ ).*

*Proof.* Exercise.  $\square$

**Theorem 6.8.** [Going Up Theorem] *Let  $R \subset S$  be an integral extension of rings. If  $P \subset R$  is a prime ideal, then there is a prime ideal  $Q \subset S$  such that  $Q \cap R = P$ . If moreover  $Q_1$  is an ideal of  $S$  such that  $R \cap Q_1 \subset P$ , then we may assume that  $Q_1 \subset Q$ .*

*Proof.* Replacing  $S$  by  $S/Q_1$  and  $R$  by  $R/(Q_1 \cap R)$ , we may assume that  $Q_1 = 0$ . Localizing at  $P$ , we may assume that  $(R, P)$  is a local ring. Let  $Q \subset S$  be any maximal ideal containing  $PS$ , since clearly  $R \supset Q \cap R \supset P$ , it suffices to show that  $Q \neq S$  or equivalently that  $PS \neq S$ . If  $PS = S$ , then  $1 = \sum p_i s_i$  with  $p_i \in P$  and  $s_i \in S$ . Since this sum is finite,  $S' = \sum R s_i \subset S$  is a finitely generated  $R$ -module such that  $PS' = S'$ . By Nakayama's Lemma,  $S' = 0$  which is a contradiction.  $\square$

**Corollary 6.9.** *With the above notation, if  $P_1 \subset P_2 \subset \dots \subset P_n$  is a chain of prime ideals in  $R$  and  $Q_1 \subset Q_2 \subset \dots \subset Q_m$  is a chain of prime ideals in  $S$  with  $1 \leq m < n$  and  $P_i = Q_i \cap R$ , then there exists a  $Q_m \subset Q_{m+1} \subset \dots \subset Q_n$  a chain of prime ideals in  $S$  where  $P_i = Q_i \cap R$ .*

*Proof.* Exercise.  $\square$

**Exercise 6.10.** Let  $M$  be a finitely generated  $R$  module and  $\phi : M \rightarrow M$  a surjective homomorphism. Show that  $\phi$  is injective. (Hint: View  $M$  as an  $R[x]$  module, with  $x \cdot m = \phi(m)$ .)

## 7. FIELDS

**Definition 7.1.** A **field**  $F$  is a ring (commutative with 1) such that every non-zero element has a multiplicative inverse: if  $a \in F^* = F \setminus \{0\}$ , then there is an element  $b \in F$  such that  $ab = 1$ .

It is easy to see that multiplicative inverses are unique and hence we denote the multiplicative inverse of  $a$  by  $a^{-1}$ . Typical examples of fields are  $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ .

**Definition 7.2.** If  $F$  is a field, then a **subfield** of  $F$  is any subset  $\{0, 1\} \subset E \subset F$  such that  $E$  is closed under addition and multiplication, additive inverses and multiplicative inverses (of non-zero elements). It is easy to see that  $E$  with the operations induced by the addition and multiplication on  $F$  is also a field.

Note that if  $\{E_i\}_{i \in I}$  is a collection of subfields of a field  $F$  then  $\bigcap_{i \in I} E_i \subset F$  is a subfield. In particular if  $S \subset F$  is a subset, then  $(\bigcap_{F \supset E \supset S} E) \subset F$  is a field (where the intersection is over all subfields  $E \subset F$  containing  $S$ ).

**Definition 7.3.** Let  $E \subset F$  be an inclusion of fields, and  $\{f_i\}_{i \in I}$  be elements of  $F$ , then we let  $E(\{f_i\}_{i \in I})$  be the smallest field containing  $E$  and the  $f_i$ .

**Definition 7.4.** Let  $E \subset F$  be an inclusion of fields and  $f \in F$ , then we say that  $f$  is **algebraic** over  $E$  if  $\dim_E E(f)$  is finite. If  $f$  is not algebraic, then we say that  $f$  is **transcendental** over  $E$ .

**Lemma 7.5.** Let  $E \subset F$  be an inclusion of fields and  $f \in F$ , then the following are equivalent:

- (1)  $f$  is algebraic over  $E$ ,
- (2)  $E(f) = E[f]$  where  $E[f] = \{p(f) | p \in E[x]\}$ ,
- (3)  $f$  is the solution of a polynomial  $p \in E[x]$ .

*Proof.* (1) implies (3): Suppose that  $f$  is algebraic over  $E$ , then since  $\dim_E E(f)$  is finite, there exists  $d \in \mathbb{N}$  such that  $1, f, f^2, \dots, f^d$  are not linearly independent and so there exists  $e_0, \dots, e_d \in R$  such that  $\sum_{i=0}^d e_i f^i = 0$ . But then  $f$  is the solution of the polynomial  $p = \sum e_i x^i \in E[x]$ .

(3) implies (2): The inclusion  $\supset$  is clear and so it suffices to show that  $E[f]$  is a field, i.e. that every non-zero element of  $E[f]$  is invertible. Suppose that  $f$  is the solution of a polynomial  $p \in E[x]$ . We may assume that  $p$  is irreducible. Let  $0 \neq g \in E[f]$ , then  $g = q(f)$  for some polynomial  $q \in E[x]$ . Since  $0 \neq q(f)$ , one sees that  $p$  does not divide  $q$ , but since  $p \in E[x]$  is irreducible, then  $\gcd(p, q) = 1$  and so there exist polynomials  $a, b \in E[x]$  such that  $ap + bq = 1$ . but then

$$(gb)(f) = q(f)g(f) = 1 - a(f)p(f) = 1$$

and so  $g$  is invertible in  $E[f]$  and hence  $E[f]$  is a field.

(2) implies (1): Since  $E[f]$  is a field, there exists a polynomial  $q \in E[x]$  such that  $f \cdot q(f) = 1$ . Let  $p(x) = xq(x) - 1 \in E[x]$ , then  $p(f) = 0$  and so  $f$  is algebraic.  $\square$

**Corollary 7.6.** *If  $E \subset F$  is an inclusion of fields and  $f \in F$  is transcendental over  $E$ , then  $E(f) \cong E(x)$ ,  $E[f] \cong E[x]$  and  $\dim_E E(f)$  is infinite.*

*Proof.* Exercise. □

We next show how to associate a field to certain rings in a fashion similar to the construction of the rational numbers  $\mathbb{Q}$ .

**Definition 7.7.** *A ring  $R$  (commutative with 1) is a **domain** if it has no **zero divisors** i.e. there are no pairs of elements  $x, y \in R$  such that  $xy = 0$ .*

**Lemma 7.8.** *If  $I \subset \mathbb{C}[x_1, \dots, x_n]$  is an ideal, then  $\mathbb{C}[x_1, \dots, x_n]/I$  is a domain if and only if  $I$  is prime. Thus, if  $X$  is an affine variety, then  $\mathbb{C}[x_1, \dots, x_n]/\mathbb{I}(X)$  is a domain if and only if  $X$  is irreducible.*

*Proof.* If  $I$  is not prime, then there exist  $x, y \notin I$  such that  $xy \in I$ , but then  $x + I, y + I \neq I$  and  $(x + I)(y + I) = (xy + I) = I$  so that  $x + I, y + I \neq 0 \in R/I$  but  $(x + I)(y + I) = 0 \in R/I$ , i.e.  $R/I$  is not a domain. We leave the converse as an exercise. □

**Definition 7.9.** *Let  $R$  be a domain, then the **fraction field** of  $R$  is the set*

$$Q(R) = \{(f, g) | f \in R, g \in R^*\} / \sim, \quad (f, g) \sim (h, k), \text{ iff } fk = gh.$$

*Addition is defined by  $(f, g) + (h, k) = (fk + gh, gk)$  and multiplication is defined by  $(f, g)(h, k) = (fh, gk)$ . It will be convenient to denote the equivalence class of  $(f, g)$  simply by  $f/g$ . In other words  $Q(R) = S^{-1}R$  where  $S = R^*$ .*

**Definition 7.10.** *We say that an extension of fields  $F \subset E$  is a **finite field extension** if  $\dim_F E < +\infty$  (i.e. if the dimension of  $E$  as an  $F$  vector space is finite).*

Note that if  $e_1, \dots, e_r$  are algebraic over  $F$ , then  $F \subset F(e_1, \dots, e_r)$  is finite.

**Theorem 7.11** (Primitive element theorem). *Let  $F \subset E$  be a finite extension of a field of characteristic 0. Then there exists an element  $e \in E$  such that  $E = F(e)$ . In fact, if  $E = F(e_1, \dots, e_r)$ , then we may assume that  $e = \sum f_i e_i$  where  $f_i \in F$ .*

*Proof.* Unluckily the standard proof relies on Galois Theory. It is necessary to know that there are only finitely many intermediate field extensions  $F \subset F_i \subset E$ . If this is the case, then it suffices to pick an element  $\alpha \in E \setminus \cup F_i$ . □

**Remark 7.12.** *It is not necessary to assume that  $F$  have characteristic 0. It suffices to assume that  $F$  is infinite and  $E/F$  is separable. The primitive element theorem also holds for all finite fields (by a different proof).*

**Exercise 7.13.** *Show that  $Q(\mathbb{Z}) = \mathbb{Q}$ .*

**Exercise 7.14.** *Show that addition and multiplication are well defined in  $Q(R)$ ,  $1 = (1, 1)$  and  $0 = (0, 1)$  are the multiplicative and additive identities. The inverses are  $-(a, b) = (-a, b)$  and  $(a, b)^{-1} = (b, a)$ . Conclude that  $Q(R)$  is a field.*

**Exercise 7.15.** *Let  $F$  be a field, then either  $m \cdot 1 = 1 + \dots + 1 \neq 0$  for all  $m \in \mathbb{N}$  or there exists a prime number  $p$  such that  $p \cdot 1 = 0$ . In the first case we say that  $F$  has **characteristic 0** and in the second case we say that  $F$  has **characteristic  $p > 0$** .*

**Exercise 7.16.** *If  $F$  is a finite field (meaning that its cardinality  $|F|$  is finite), then  $|F| = p^e$  for some  $e \in \mathbb{N}$  and some prime number  $p > 0$ . It turns out that all fields of order  $p^e$  are isomorphic. We call such a field  $\mathbb{F}_{p^e}$ .*

**Exercise 7.17.** Show that  $\mathbb{Q}[\sqrt{2}]$  is a field and  $\dim_{\mathbb{Q}} \mathbb{Q}[\sqrt{2}] = 2$ .

**Exercise 7.18.** Let  $E \subset F$  be an inclusion of fields,  $p(x) \in E[x]$  be an irreducible polynomial of degree  $d$ , and  $f \in F$  a solution of  $p(x)$ , then show that  $E(f) \cong E[x]/(p)$  and  $\dim_E E(f) = d$ .

**Exercise 7.19.** Let  $\bar{\mathbb{Q}} = \{f \in \mathbb{C} \mid \dim_{\mathbb{Q}} \mathbb{Q}(f) < \infty\}$ . We say that  $\bar{\mathbb{Q}}$  is the **algebraic closure** of  $\mathbb{Q}$ . Show that  $\bar{\mathbb{Q}}$  is a field and  $\dim_{\mathbb{Q}} \bar{\mathbb{Q}} = \infty$ .

**Exercise 7.20.** Show that  $\mathbb{Q}(x, i)$  is not generated over  $\mathbb{Q}$  by a single element and that there are infinitely many intermediate field extensions  $\mathbb{Q} \subset F \subset \mathbb{Q}(x, i)$ .

**Exercise 7.21.** Find a primitive element for  $\mathbb{Q}(\sqrt{2}, \sqrt{3})/\mathbb{Q}$ .

**Exercise 7.22.** Show that  $E = \mathbb{F}_3[x]/(x^2 + 2x + 1)$  is a field and find a primitive element for  $E/\mathbb{F}_3$ .

**Exercise 7.23.** Show that  $E = \mathbb{F}_3[x]/(x^2 + x + 1)$  is not a field.

**Exercise 7.24.** Let  $p$  be a prime and  $q \in \mathbb{F}_p[x]$  a polynomial. Show that  $E = \mathbb{F}_p[x]/(q(x))$  is a field if and only if  $q(x)$  is irreducible.

**Exercise 7.25.** Let  $p$  be a prime number. Show that there is always a degree 2 irreducible polynomial in  $\mathbb{F}_p[x]$  and hence  $\mathbb{F}_p[x]/(q(x))$  is a field with  $q^2$  elements.

## 8. NULLSTELLNSATZ

**Theorem 8.1.** [Weak Nullstellensatz] If  $\mathfrak{m} \subset \mathbb{C}[x_1, \dots, x_n]$  is a maximal ideal, then  $\mathfrak{m} = (x - a_1, \dots, x - a_n)$  for  $a_1, \dots, a_n \in \mathbb{C}$ .

*Proof.* Note that  $(x - a_1, \dots, x - a_n)$  is a maximal ideal and  $\mathbb{C} \subset \mathbb{C}[x_1, \dots, x_n]/\mathfrak{m}$  is an inclusion of fields (cf. Exercise 3.37).

Suppose that the field  $\mathbb{C}[x_1, \dots, x_n]/\mathfrak{m}$  is an algebraic extension of  $\mathbb{C}$ . Since  $\mathbb{C}$  is algebraically closed, then  $\mathbb{C}[x_1, \dots, x_n]/\mathfrak{m} \cong \mathbb{C}$ . Let  $\phi : \mathbb{C}[x_1, \dots, x_n] \rightarrow \mathbb{C}$  be the induced homomorphism with kernel  $\mathfrak{m}$  and  $a_i = \phi(x_i)$ , then clearly  $\phi(x_i - a_i) = 0$  so that  $\mathfrak{m} \supset (x_1 - a_1, \dots, x_n - a_n)$ . Since  $(x_1 - a_1, \dots, x_n - a_n) \subset \mathbb{C}[x_1, \dots, x_n]$  is maximal (cf. 3.30), we have  $\mathfrak{m} = (x_1 - a_1, \dots, x_n - a_n)$  as required.

Therefore, we may assume that the field  $\mathbb{C}[x_1, \dots, x_n]/\mathfrak{m}$  is not an algebraic extension of  $\mathbb{C}$ . Thus there exists a transcendental element  $t \in \mathbb{C}[x_1, \dots, x_n]/\mathfrak{m}$  and hence

$$\mathbb{C}[x_1, \dots, x_n]/\mathfrak{m} \supset \mathbb{C}(t) \supset \mathbb{C}.$$

Note that the dimension of  $\mathbb{C}[x_1, \dots, x_n]/\mathfrak{m}$  (as a  $\mathbb{C}$  vector space) is clearly countable i.e. it does not contain an uncountable set of linearly independent elements. However it is easy to see that the set

$$\left\{ \frac{1}{t - a} \mid a \in \mathbb{C} \right\}$$

is uncountable and linearly independent. □

**Exercise 8.2.** Show that  $\left\{ \frac{1}{t - a} \mid a \in \mathbb{C} \right\}$  is linearly independent over  $\mathbb{C}$ .

**Exercise 8.3.** Let  $V$  be a vector space over  $\mathbb{C}$  with a countable set of generators. Show that any uncountable set of elements of  $V$  is linearly dependent.

**Corollary 8.4.** *If  $\mathbb{V}(p_1, \dots, p_r) = \emptyset$  for some polynomials  $(p_1, \dots, p_r) \in \mathbb{C}[x_1, \dots, x_n]$ , then  $(p_1, \dots, p_r) = \mathbb{C}[x_1, \dots, x_n]$  or equivalently there exist  $q_1, \dots, q_r \in \mathbb{C}[x_1, \dots, x_n]$  such that  $1 = \sum_{i=1}^r p_i q_i$ .*

*Proof.* Let  $I = (p_1, \dots, p_r)$  be the corresponding ideal. If  $I \neq \mathbb{C}[x_1, \dots, x_n]$  then  $I$  is contained in a maximal ideal  $I \subset \mathfrak{m}$ . By Theorem 8.1,  $\mathfrak{m} = (x - a_1, \dots, x - a_n)$  for  $a_1, \dots, a_n \in \mathbb{C}$ . But then each polynomial  $q \in I \subset \mathfrak{m}$  must vanish at  $(a_1, \dots, a_n) \in \mathbb{C}^n$ . This is a contradiction and so  $I = R$ . In particular  $1 \in I$  and so  $1 = \sum_{i=1}^r p_i q_i$  where  $q_1, \dots, q_r \in \mathbb{C}[x_1, \dots, x_n]$ .  $\square$

**Theorem 8.5.** [Nullstellensatz] *Let  $I \subset \mathbb{C}[x_1, \dots, x_n]$  be an ideal and  $V = \mathbb{V}(I) \subset \mathbb{C}^n$  the corresponding zero set. If  $f \in \mathbb{C}[x_1, \dots, x_n]$  vanishes along  $V$ , then  $f^m \in I$  for some  $m > 0$ . Equivalently, we have*

$$\mathbb{I}(\mathbb{V}(I)) = \sqrt{I}.$$

*Proof.* (Rabinowitsch's trick) Let  $I = (f_1, \dots, f_r)$ . The polynomials  $f_1, \dots, f_r, 1 - x_{n+1}f \in \mathbb{C}[x_1, \dots, x_{n+1}]$  have no common zeroes and so by Corollary 8.4,

$$1 = f_1 q_1 + \dots + f_r q_r + (1 - x_{n+1}f)q$$

where  $q_1, \dots, q_r, q \in \mathbb{C}[x_1, \dots, x_{n+1}]$ . Now substitute  $x_{n+1} = 1/f$  to get

$$1 = f_1(x_1, \dots, x_n)q_1(x_1, \dots, x_n, 1/f) + \dots + f_r(x_1, \dots, x_n)q_r(x_1, \dots, x_n, 1/f).$$

Let  $t$  be the biggest power of  $(1/f)$  appearing in the above expression, then  $q'_i := q_i \cdot f^t \in \mathbb{C}[x_1, \dots, x_n]$  and so

$$f^t = f_1 q'_1 + \dots + f_r q'_r \in I.$$

$\square$

**Corollary 8.6.** *Let  $I, J \subset \mathbb{C}[x_1, \dots, x_n]$  be two ideals. Then  $\mathbb{V}(I) = \mathbb{V}(J)$  if and only if  $\sqrt{I} = \sqrt{J}$ .*

*Proof.* Exercise.  $\square$

**Exercise 8.7.** *Show that  $p_1, \dots, p_k \in \mathbb{Q}[x_1, \dots, x_n]$  have a common root in  $\mathbb{C}^n$  if and only if  $(p_1, \dots, p_k) \neq \mathbb{Q}[x_1, \dots, x_n]$ .*

**Exercise 8.8.** *Use the Nullstellensatz to show that two distinct irreducible monic polynomials  $p, q \in \mathbb{C}[x_1, \dots, x_n]$  define different hypersurfaces in  $\mathbb{C}^n$ . Show that this fails in  $\mathbb{R}^n$ .*

**Exercise 8.9.** *Show that there are irreducible polynomials  $f \in \mathbb{R}[x, y]$  such that  $\mathbb{V}(f) \subset \mathbb{R}^2$  is reducible.*

## 9. THE COORDINATE RING OF AN AFFINE VARIETY

**Definition 9.1.** *The **coordinate ring** of an affine algebraic set  $X \subset \mathbb{C}^n$  is the ring  $A(X) = \mathbb{C}[x_1, \dots, x_n]/\mathbb{I}(X)$ . Note that this may be identified with the ring of functions on  $X$  induced by restricting polynomials from  $\mathbb{C}^n$  to  $X$ . In particular  $A(\mathbb{C}^n) = \mathbb{C}[x_1, \dots, x_n]$ .*

**Exercise 9.2.** *Show that the coordinate ring of an affine variety (resp. algebraic set) is a  $\mathbb{C}$  algebra that has no zero divisors (resp. no nilpotents). Equivalently  $A(X) = \mathbb{C}[x_1, \dots, x_n]/\mathbb{I}(X)$  where  $\mathbb{I}(X)$  is prime (resp.  $\mathbb{I}(X)$  is radical).*

**Exercise 9.3.** *Compute the coordinate rings of  $\mathbb{V}(y^2 - x^3)$  and  $\mathbb{V}(y^2 - x^2(x - 1))$  and show that they are not integrally closed.*

It is not hard to see that the coordinate rings actually determine the affine variety.

**Lemma 9.4.** *There is a bijection between finitely generated  $\mathbb{C}$  algebras without nilpotents (resp. zero divisors) and affine algebraic sets (resp. varieties).*

*Proof.* Let  $A$  be a finitely generated  $\mathbb{C}$  algebra without nilpotents, then we must show that there exists an affine algebraic set  $X \subset \mathbb{C}^n$  such that  $A = A(X)$ . Fix  $g_1, \dots, g_n$  generators of  $A$  (as a  $\mathbb{C}$ -algebra) and consider the surjective homomorphism of  $\mathbb{C}$ -algebras

$$\phi : \mathbb{C}[x_1, \dots, x_n] \rightarrow A, \quad p(x_1, \dots, x_n) \mapsto p(g_1, \dots, g_n).$$

If  $I = \ker(\phi)$ , then  $A \cong \mathbb{C}[x_1, \dots, x_n]/I$ . Since  $A$  has no nilpotents, then  $I$  is radical and so  $I = \mathbb{I}(\mathbb{V}(I))$ . Let  $X = \mathbb{V}(I)$ . We have shown that  $A = A(X)$ .  $\square$

Consider now  $F : X \rightarrow Y$  a morphism of affine varieties  $X \subset \mathbb{C}^n$  and  $Y \subset \mathbb{C}^m$  (see §2), then we have an induced homomorphism of  $\mathbb{C}$ -algebras

$$F^* : A(Y) \rightarrow A(X), \quad F^*(g) = g \circ F.$$

For example if  $F : \mathbb{C} \rightarrow \mathbb{C}^3$  is defined by  $F(t) = (t, t^2, t^3)$ , then  $F^*(x) = t$ ,  $F^*(y) = t^2$  and  $F^*(z) = t^3$  and so we have an induced homomorphism of  $\mathbb{C}$  algebras

$$F^* : \mathbb{C}[x, y, z] \rightarrow \mathbb{C}[t], \quad g(x, y, z) \mapsto g(t, t^2, t^3).$$

Notice also that  $F^*$  actually determines  $F$ ; in fact  $F^*(x) = t$ ,  $F^*(y) = t^2$  and  $F^*(z) = t^3$  implies that  $F(t) = (t, t^2, t^3)$ . One easily sees that:

**Lemma 9.5.** *There is a bijection between homomorphisms of finitely generated  $\mathbb{C}$ -algebras without nilpotents and morphisms of affine algebraic sets such that  $X \subset \mathbb{C}^n$ ,  $Y \subset \mathbb{C}^m$  are affine varieties, then*

$$\alpha : \text{Hom}(X, Y) \rightarrow \text{Hom}(A(Y), A(X))$$

*is defined as follows: for any morphism  $f : X \rightarrow Y$  let  $f^* \in \text{Hom}(A(Y), A(X))$  be the homomorphism of  $\mathbb{C}$ -algebras defined by  $f^*(g) = g \circ f$ .*

*Proof.* Let  $y_1, \dots, y_m$  be coordinates on  $\mathbb{C}^m$ . Given  $\Phi \in \text{Hom}(A(Y), A(X))$ , we define  $f : X \rightarrow \mathbb{C}^m$  by

$$f(p) = (\Phi(y_1)(p), \dots, \Phi(y_m)(p)).$$

Note that  $\Phi(y_j) \in A(X)$  and in order to check that we have a map  $f : X \rightarrow Y$ , we must check that if  $g \in \mathbb{I}(Y)$ , then  $g(\Phi(y_1)(p), \dots, \Phi(y_m)(p)) = 0$  for all  $p \in X$ . This is immediate since if  $\bar{y}_i = y_i|_Y$ , then for any  $g \in \mathbb{I}(Y)$  we have

$$0 = \Phi(g(\bar{y}_1, \dots, \bar{y}_m)) = g(\Phi(\bar{y}_1), \dots, \Phi(\bar{y}_m)) \in A(X).$$

We now check that  $f^* = \Phi$ . It suffices to check that this holds for the generators  $\bar{y}_1, \dots, \bar{y}_m$  of  $A(Y)$ . Again this is clear since

$$f^*(\bar{y}_i) = f_i = \Phi(\bar{y}_i).$$

The claim now follows easily.  $\square$

Let  $x \in X$  be a point on an affine variety, then  $A_x = \mathbb{C}$  since functions on points are clearly constant. From the lemma above, it follows that any point  $x \in X$  is determined by a non-zero homomorphism  $\phi_x : A(X) \rightarrow \mathbb{C}$ . We may think of this as being determined by evaluating functions  $f \in A(X)$  at  $x$ , i.e.  $\phi_x(f) = f(x)$ . Note that  $\phi_x$  is determined by  $\mathfrak{m}_x := \ker(\phi_x)$  where  $\mathfrak{m}_x$  is the ideal of functions vanishing at  $x$ . Since  $A(X)/\mathfrak{m}_x = \mathbb{C}$ , it follows that  $\mathfrak{m}_x$  is a maximal ideal. For example, the points of  $a \in \mathbb{C}^n$  are determined by  $n$ -tuples



$a = (a_1, \dots, a_n)$ ,  $\phi_a$  is given by  $p(x_1, \dots, x_n) \rightarrow p(a_1, \dots, a_n)$ ,  $\mathfrak{m}_a = (x_1 - a_1, \dots, x_n - a_n)$  and so

$$\phi_a : \mathbb{C}[x_1, \dots, x_n] \rightarrow \mathbb{C}[x_1, \dots, x_n]/\mathfrak{m}_a = (x_1 - a_1, \dots, x_n - a_n) \cong \mathbb{C}.$$

**Exercise 9.6.** Show that  $\mathbb{C}^2 \setminus \{(0, 0)\}$  is not affine.

**Exercise 9.7.** Suppose that  $X$  is an affine variety and  $f \in A(X)$  is not zero at every point  $x \in X$  i.e.  $f(x) \neq 0$ . then  $1/f \in A(X)$ .

**Definition 9.8.** Let  $X \subset \mathbb{C}^n$  be an irreducible affine variety, then the ring of rational functions on  $X$  is  $\mathbb{C}(X) = Q(A(X))$  the quotient field of  $A(X)$ .

**Definition 9.9.** Let  $X$  be an affine variety, then we let  $\mathbb{C}(X) = Q(A(X))$  be the **field of rational functions** on  $X$ . If  $X$  is a quasi-affine variety (i.e. an open subset of an affine variety), then we let  $\mathbb{C}(X) = \mathbb{C}(U)$  for some non-empty open affine subset of  $X$ .

**Exercise 9.10.** Check that the definition of  $\mathbb{C}(X)$  does not depend on the choice of the non-empty open subset  $U \subset X$ .

**Definition 9.11.** Let  $p \in X$  be a point on an affine variety, then we let

$$\mathcal{O}_p = \mathcal{O}_{X,p} = A(X)_{\mathfrak{m}_p}$$

be the **local ring** of  $p \in X$ . Here  $\mathfrak{m}_p$  denotes the maximal ideal of  $p \in X$  and  $A(X)_{\mathfrak{m}_p}$  is the localization of the coordinate ring of  $X$  at the maximal ideal corresponding point  $p \in X$ .

**Proposition 9.12.** Let  $p \in X$  be a point on an affine variety, then there are natural inclusions

$$A(X) \subset \mathcal{O}_{X,p} \subset Q(X).$$

Moreover we have the equality

$$A(X) = \bigcap_{P \in \text{Spec}(A(X))} A(X)_P = \bigcap_{\mathfrak{m} \in \text{Specmax}(A(X))} A(X)_{\mathfrak{m}}$$

where the **spectrum** of  $A(X)$  is  $\text{Spec}(A(X))$  is the set of all prime ideals in  $A(X)$  and the **maximal spectrum** of  $A(X)$  is  $\text{Specmax}(A(X))$  the set of all maximal ideals in  $A(X)$ .

*Proof.* The inclusions are clear and the equalities follow from Proposition 3.55.  $\square$

**Definition 9.13.** We say that a morphism of affine varieties  $f : X \rightarrow Y$  is **finite** if  $A(Y) \rightarrow A(X)$  is an integral extension of rings.

**Exercise 9.14.** Use the Going up Theorem 6.8 to show that if  $f : X \rightarrow Y$  is a finite morphism of affine varieties, then  $\dim X = \dim Y$ .

**Exercise 9.15.** Show that the inclusion  $\mathbb{C} \setminus \{0\} \hookrightarrow \mathbb{C}$  is not a finite morphism.

**Exercise 9.16.** Show that the projection  $\mathbb{C}^2 \rightarrow \mathbb{C}$  on to the first factor is not a finite morphism.

**Exercise 9.17.** Show that the inclusion  $X \subset W$  of an irreducible closed subset in an affine variety is a finite morphism.

**Exercise 9.18.** Show that the composition of two finite morphisms is a finite morphism.

**Exercise 9.19.** Apply Noether's Normalization Lemma to show that if  $X$  is an affine variety, then there exists a finite morphism  $X \rightarrow \mathbb{C}^n$ .

## 10. MORPHISMS OF QUASI-AFFINE VARIETIES

**Definition 10.1.** A **quasi-affine variety** is an open subset of an affine variety  $X \subset \mathbb{C}^n$ .

**Definition 10.2.** A **regular function** on a quasi-affine variety  $X \subset \mathbb{C}^n$  is a function  $f : X \rightarrow \mathbb{C}$  such that for any point  $p \in X$  there is an open subset  $p \in U \subset X$  such that  $f|_U = g/h$  where  $g, h \in \mathbb{C}[x_1, \dots, x_n]$  are polynomials and  $h(x) \neq 0$  for all  $x \in U$ .

The set of all regular functions on a quasi-affine variety is denoted by  $\mathcal{O}(X)$ . It is easy to see that  $\mathcal{O}(X)$  is a ring (commutative with 1).

**Exercise 10.3.** Show that a regular function is continuous in the Zariski topology.

**Exercise 10.4.** Show that if  $X \subset \mathbb{C}^n$  is an affine variety and  $I = \mathcal{I}(X)$ , then  $\mathcal{O}(X) = \mathbb{C}[x_1, \dots, x_n]/I$ .

**Definition 10.5.** A **morphism**  $f : X \rightarrow Y$  between two quasi-affine varieties  $X \subset \mathbb{C}^n$  and  $Y \subset \mathbb{C}^m$ , is a continuous map such that  $f = (f_1, \dots, f_m)$  where  $f_i \in \mathcal{O}(X)$ .

**Exercise 10.6.** Show that if  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$  are two morphisms of quasi-affine varieties, then  $g \circ f$  is a morphism of quasi-affine varieties.

**Proposition 10.7.** Let  $X$  be a quasi-affine variety and  $Y$  be an affine variety, then there is a natural equivalence

$$\alpha : \text{Hom}(X, Y) \rightarrow \text{Hom}(A(Y), \mathcal{O}(X)).$$

Here, an element  $f \in \text{Hom}(X, Y)$  is a morphism of quasi-affine varieties  $f : X \rightarrow Y$  and  $g \in \text{Hom}(A(Y), \mathcal{O}(X))$  is a homomorphism of  $\mathbb{C}$  algebras and  $\alpha(f) = f^*$  where  $f^*(p) = p \circ f \in \mathcal{O}(X)$  for any  $p \in A(Y)$ .

*Proof.* Similar to Lemma 9.5. □

## 11. DIMENSION

**Definition 11.1.** Let  $R$  be a ring, then the **Krull dimension** of  $R$  is the maximum length  $n$  of a chain of distinct prime ideals

$$P_0 \subset P_1 \subset \dots \subset P_n \subset R.$$

For example the dimension of  $\mathbb{C}[x_1, \dots, x_n]$  is  $n$ . The inequality  $\geq n$  follows easily from the inclusion of prime ideals

$$0 \subset (x_1) \subset (x_1, x_2) \subset \dots \subset \mathbb{C}[x_1, \dots, x_n].$$

Note that since prime ideals correspond to irreducible closed subsets, the above definition matches Definition 1.23, so that if  $R = A(X)_{\mathfrak{m}_x}$  is the local ring of  $X$  at  $x \in X$ , then  $\dim_x X$  is equal to the Krull dimension of  $(A(X)_{\mathfrak{m}_x}, \mathfrak{m}_x)$ .

**Exercise 11.2.** Show that if  $R$  is Noetherian, then the dimension of  $R$  always exists and is finite.

**Exercise 11.3.** Show that if  $R$  is a field, then the Krull dimension of  $R$  is 0.

We will denote the Krull dimension by  $\text{Krdim}(R)$ . We have the following important fact.

**Theorem 11.4.** Let  $(R, \mathfrak{m})$  be a local Noetherian domain, then

$$\text{Krdim}(R) = \max_{x \in \mathfrak{m} \setminus 0} \{\text{Krdim}(R/xR) + 1\}.$$

*Proof.* We begin by proving the inequality  $\geq$ . Suppose that  $x \in \mathfrak{m} \setminus 0$  and pick  $P_0 \subset P_1 \subset \dots \subset P_k \subset R/xR$  a chain of distinct prime ideals where  $k = \text{Krdim}(R/xR)$ . Let  $\tilde{P}_i$  be the inverse image of  $P_i$  in  $R$ . Then  $\tilde{P}_i$  are prime and the inclusions  $\tilde{P}_i \subset \tilde{P}_{i+1}$  are strict. Thus we have a sequence

$$(x) \subset \tilde{P}_0 \subset \tilde{P}_1 \subset \dots \subset \tilde{P}_k \subset R.$$

Since  $x \neq 0$ , the inclusion  $0 = \tilde{P}_{-1} \subset \tilde{P}_0$  is also strict and hence  $\text{Krdim}(R) \geq \text{Krdim}(R/xR) + 1$ .

We now prove the reverse inequality. Let  $\tilde{P}_{-1} \subset \tilde{P}_0 \subset \dots \subset \tilde{P}_k \subset R$  be a chain of distinct prime ideals where  $k+1 = \text{Krdim}(R)$ . Then there is an element  $x \in \tilde{P}_0 \cap (\mathfrak{m} \setminus 0)$  and so letting  $P_i = \tilde{P}_i/(x)$  for  $i \geq 0$ , we obtain a chain of distinct prime ideals  $P_0 \subset P_1 \subset \dots \subset P_k \subset R/xR$  and so  $\text{Krdim}(R) \leq \text{Krdim}(R/xR) + 1$ .  $\square$

**Definition 11.5.** If  $R$  is a Noetherian ring, then we let

$$\dim R = \sup_P \text{Krdim} R_P$$

where  $P \subset R$  is a prime ideal and  $R_P$  is the corresponding localized ring.

**Definition 11.6.** Let  $P \subset R$  be a prime ideal, then the **height** of  $P$  is the maximal length of distinct prime ideals  $P_0 \subset P_1 \subset \dots \subset P$ .

We next recall the following result.

**Theorem 11.7.** Let  $R$  be an integral domain which is a finitely generated  $\mathbb{C}$ -algebra. Then

- (1)  $\text{Krdim}(R) = \text{tr.deg.}_{\mathbb{C}} Q(R)$ , and
- (2) If  $P \subset R$  is prime, then  $\text{height}(P) + \text{Krdim}(R/P) = \text{Krdim}(R)$ .

In particular the length of all maximal chains of prime ideals is given by  $\dim R$ .

*Proof.* [Hartshorne, Theorem 1.8A]. Here is however an intuitive argument for (1). We will see later that if  $X$  is an affine variety,  $R = A(X)$  and the transcendence degree of  $Q(X)$  is  $d$ , then there is an open subset  $U \subset X$  which is isomorphic to an open subset of a hypersurface  $\mathbb{V}(f) \subset \mathbb{P}^{d+1}$ . But then by Exercise 11.14

$$\dim X = \dim U = \dim \mathbb{V}(f) = \dim \mathbb{P}^{d+1} - 1 = d.$$

$\square$

**Exercise 11.8.** Show that  $\dim \mathbb{C}^n = n$ .

**Exercise 11.9.** What is the dimension of  $\mathbb{C}[x, y, z]/x(y, z)$  localized at  $(x, y, z)$ ?

**Exercise 11.10.** Show that  $\dim R = \dim R/\sqrt{R}$ .

**Exercise 11.11.** Let  $R$  be a principal ideal ring (so that any ideal is generated by one element). Show that  $\dim R \leq 1$ .

**Exercise 11.12.** Show that if  $X$  is a quasi-affine variety, then  $\dim X = \dim \bar{X}$ . (See [Hartshorne, I.1.10].)

**Theorem 11.13** (Krull's Hauptidealsatz). Let  $f \in R$  be an element in a Noetherian ring which is neither a unit nor a zero divisor, then every minimal prime ideal  $P$  containing  $f$  has height 1.

*Proof.* We will assume that  $R = A(X)$  where  $X \subset \mathbb{C}^n$  is an affine variety. Let  $z_1 = f$ . We may assume that  $z_1, \dots, z_r \in \mathbb{C}[x_1, \dots, x_n]$  are algebraically independent in  $A(X)$  where  $r = \dim X$ . Let  $W = \mathbb{V}(P) \subset \mathbb{V}(f) = V$  so that  $W$  is an irreducible component of  $V$ . Pick  $g \notin P$  such  $g$  vanishes along  $V \setminus W$ . We will show that  $z_2, \dots, z_r$  are algebraically independent in  $A(W)$ . If this were not the case, then there is a nonzero polynomial  $F \in \mathbb{C}[t_2, \dots, t_r]$  such that  $F(z_2, \dots, z_r) = 0 \in A(W)$  and hence  $F(z_2, \dots, z_r)g = 0 \in A(X)$ . By the Nullstellensatz,  $z_1 = f$  divides  $(F(z_2, \dots, z_r)g)^n$  in  $A(X)$ , but then  $f$  divides  $g^n$  in  $A(X)$  and so  $g$  vanishes along  $V$  which is a contradiction.  $\square$

**Exercise 11.14.** Let  $Y \subset \mathbb{C}^n$  be an irreducible subset. Show that  $Y$  is the zero of an irreducible polynomial if and only if it has codimension 1 in  $\mathbb{C}^n$ .

**Exercise 11.15.** Show that if  $X \subset \mathbb{C}^n$  is defined by  $r$  equations, then  $\dim X \geq n - r$  (but strict inequality is possible).

## 12. PROJECTIVE VARIETIES

Let  $\bar{0} = (0, \dots, 0) \in \mathbb{C}^{n+1}$  and  $\mathbb{C}^* = \mathbb{C} \setminus 0$ . Consider the equivalence relation on  $\mathbb{C}^{n+1} \setminus \bar{0}$  defined by

$$(a_0, \dots, a_n) \sim (b_0, \dots, b_n) \text{ iff } \exists \lambda \in \mathbb{C}^* \text{ s.t. } a_0 = \lambda b_0, \dots, a_n = \lambda b_n.$$

In other words, if  $\bar{a} = (a_0, \dots, a_n)$  and  $\bar{b} = (b_0, \dots, b_n)$ , then  $\bar{a} \sim \bar{b}$  if and only if  $\bar{a} = \lambda \bar{b}$  for some  $\lambda \in \mathbb{C}^*$ . We denote the equivalence class of  $(a_0, \dots, a_n)$  by  $[a_0 : \dots : a_n]$ .

**Exercise 12.1.** Check that  $\sim$  defined above is an equivalence relation whose equivalence classes correspond to lines through the origin in  $\mathbb{C}^{n+1}$ .

**Definition 12.2.** We define  $n$ -dimensional **projective space** via

$$\mathbb{P}^n := (\mathbb{C}^{n+1} \setminus \bar{0}) / \sim.$$

Consider the natural inclusion  $\mathbb{C}^n \hookrightarrow \mathbb{P}^n$  given by

$$(a_1, \dots, a_n) \rightarrow [1 : a_1, \dots, a_n],$$

which gives a one to one correspondence between the set  $\mathbb{C}^n$  and the set  $\{x_0 \neq 0\} \subset \mathbb{P}^n$ . This correspondence can be inverted as follows: if  $[a_0 : \dots : a_n] \subset \mathbb{P}^n$  and  $a_0 \neq 0$ , then we let

$$[a_0 : \dots : a_n] \rightarrow (a_1/a_0, \dots, a_n/a_0).$$

Note that the subset  $\{x_0 = 0\} \subset \mathbb{P}^n$  can be identified with  $\mathbb{P}^{n-1}$  by the map  $\phi : \mathbb{P}^{n-1} \rightarrow \mathbb{P}^n$  defined by  $[x_1 : \dots : x_n] \rightarrow [0 : x_1 : \dots : x_n]$ . In this way we have

$$\mathbb{P}^n = \mathbb{C}^n \cup \mathbb{P}^{n-1} = \mathbb{C}^n \cup \mathbb{C}^{n-1} \cup \dots \cup \mathbb{C} \cup \{\text{pt.}\}.$$

We may think of  $\mathbb{P}^{n-1}$  as the set of points at infinity. For example if  $n = 2$  and  $ax + by + c = 0$  is a line  $l \subset \mathbb{C}^2$ , then the homogeneous equation  $aX + bY + cZ = 0$  defines a subset  $L \subset \mathbb{P}^2$  such that  $L \cap \mathbb{C}^2 = l$  (where  $\mathbb{C}^2$  corresponds to the subset  $Z \neq 0$ ). Notice that the intersection with the line at infinity is given by

$$L \cap \mathbb{P}^1 = \{aX + bY + cZ = 0\} \cap \{Z = 0\} = [b : -a : 0]$$

is determined by the slope of  $l$  and two lines  $l, l' \subset \mathbb{C}^2$  will meet at infinity if and only if they have the same slope. In this way, we have that two distinct lines in  $\mathbb{P}^2$  always meet at exactly one point.

**Lemma 12.3.**  $\mathbb{P}^n$  is an  $n$ -dimensional compact complex manifold.

*Proof.* We define local charts  $U_i \cong \mathbb{C}^n$  for  $0 \leq i \leq n$  by letting  $U_i = \{x_i \neq 0\} \subset \mathbb{P}^n$ . The natural inclusions  $\phi_i : U_i \hookrightarrow \mathbb{P}^n$  are given by

$$\phi_i(a_0, \dots, a_{i-1}, a_{i+1}, \dots, a_n) = [a_0 : \dots : a_{i-1} : 1 : a_{i+1} : \dots : a_n]$$

and therefore on  $U_{i,j} = U_i \cap U_j$  we have (holomorphic) transition functions  $\psi_{ij} = \phi_j^{-1} \circ \phi_i$  given by

$$\psi_{ij}(a_0, \dots, a_{i-1}, a_{i+1}, \dots, a_n) = (a_0/a_j, \dots, a_{i-1}/a_j, 1/a_j, a_{i+1}/a_j, \dots, a_{j-1}/a_j, a_{j+1}/a_j, \dots, a_n).$$

To see that  $\mathbb{P}^n$  is compact in the Euclidean topology, consider a sequence  $a^i \in \mathbb{P}^n$ . Writing  $a^i = [a_0^i : \dots : a_n^i]$ , after dividing by  $\|a^i\| = (\sum_{j=0}^n a_j^i \bar{a}_j^i)^{1/2}$ , we may assume that the  $a^i$  are represented by points in  $S^{2n+1} \subset \mathbb{C}^{n+1}$  which is compact. Therefore, after passing to a subsequence we may assume that the limit  $a = \lim_{j \rightarrow \infty} a^j \in S^{2n+1}$  exists. Since  $a \in S^{2n+1}$ , one of the coordinates of  $a = (a_0, \dots, a_n)$  is non-zero and hence  $[a_0 : \dots : a_n]$  represents a point in  $\mathbb{P}^n$ . Finally we must check that  $\lim a^i = [a_0 : \dots : a_n]$  in  $\mathbb{P}^n$ . Since the topology on  $\mathbb{P}^n$  is induced by the topology on the charts  $U_i$ , it suffices to check this on such a chart. Assume that  $a_0 \neq 0$ , then  $a_0^i \neq 0$  for all  $i \gg 0$  and so we may consider the corresponding points  $(1, a_1^i/a_0^i, \dots, a_n^i/a_0^i) \in U_0$ . Since the induced map  $f : S^{2n+1} \setminus \{x_0 = 0\} \rightarrow U_0$  defined by  $(x_0, \dots, x_n) \rightarrow (x_1/x_0, \dots, x_n/x_0)$  is continuous in a neighborhood of  $a$ , it follows that  $\lim f(a^i) = f(a)$  as required.  $\square$

We would like to define projective varieties in a similar fashion to what we did for affine varieties. Notice however that it does not make sense to evaluate a polynomial at a point of projective space. For example what should the value of  $x^2 - y^3 + z$  be at the point  $[1 : 1 : 0] \in \mathbb{P}^2$ ? You may want to say  $1^2 - 1^3 + 0 = 0$  however, recall that  $[1 : 1 : 0] = [2 : 2 : 0]$  and  $2^2 - 2^3 + 0 = -4 \neq 0$ . Notice however that if  $p \in \mathbb{C}[x_0, \dots, x_n]$  is a homogeneous polynomial of degree  $d$ , then

$$p(\lambda a_0, \dots, \lambda a_n) = \lambda^d p(a_0, \dots, a_n)$$

and so  $p(\lambda a_0, \dots, \lambda a_n) = 0$  if and only if  $p(a_0, \dots, a_n) = 0$  (since  $\lambda \neq 0$ ). Therefore, we can use homogeneous polynomials to define projective varieties.

Given an affine variety we can always define a corresponding projective variety. For example if  $X = \mathbb{V}(x^2 - y^3) \subset \mathbb{C}^2$ , then we can consider  $\bar{X} = \mathbb{V}(x^2 z - y^3) \subset \mathbb{P}^2$ . Notice that  $\bar{X} \cap \{z \neq 0\} = X$  and  $\bar{X} \cap \{z = 0\} = [1 : 0 : 0]$ . We have compactified the curve  $X$  by adding a point at infinity. In general if  $p \in \mathbb{C}[x_1, \dots, x_n]$  is any polynomial of degree  $d$ , we let

$$\tilde{p} = x_0^d p(x_1/x_0, \dots, x_n/x_0) \in \mathbb{C}[x_0, \dots, x_n]$$

be the corresponding homogeneous polynomial. So for example if  $p = x_1^2 - x_2^3$ , then

$$\tilde{p} = x_0^3 \left( \left( \frac{x_1}{x_0} \right)^2 - \left( \frac{x_2}{x_0} \right)^3 \right) = x_0 x_1^2 - x_2^3.$$

Notice that

$$\tilde{p}(1, x_1, \dots, x_n) = p(x_1, \dots, x_n)$$

and so we can recover the original polynomial  $p$  from its homogeneization  $\tilde{p}$ . Similarly, if  $X = \mathbb{V}(I) \subset \mathbb{C}^n$  for some ideal  $I = (f_1, \dots, f_r) \subset \mathbb{C}[x_1, \dots, x_n]$ , then we consider the ideal

$$\tilde{I} = (\tilde{f}_1, \dots, \tilde{f}_r) \subset \mathbb{C}[x_0, \dots, x_n].$$

As before, we can recover  $I$  from  $\tilde{I}$  and we have  $X = \tilde{X} \cap \{x_0 \neq 0\}$  where  $\tilde{X} = \mathbb{V}(\tilde{I}) \subset \mathbb{P}^n$ .

**Exercise 12.4.** Let  $X, Y$  be lines in  $\mathbb{C}^2$ . Show that the corresponding lines  $\tilde{X}, \tilde{Y} \subset \mathbb{P}^2$  always meet in exactly one point. If  $X, Y$  are not parallel, then  $\tilde{X} \cap \tilde{Y} = X \cap Y \subset \mathbb{C}^2 = \{x_0 \neq 0\}$  and if  $X, Y$  are parallel (but distinct), then  $\tilde{X} \cap \tilde{Y} = [0, 1, -m] \in \mathbb{P}^1 = \{x_0 = 0\}$  is the point at infinity corresponding to the slope  $m$  of the lines.

**Exercise 12.5.** Let  $X \subset \mathbb{P}^n$  be a projective variety. Show that  $X$  is compact in the Euclidean topology.

Given an affine variety, there are several advantages to considering the corresponding projective variety  $\tilde{X} \subset \mathbb{P}^n$ .

- (1)  $\tilde{X}$  is compact,
- (2) Intersection theory: As we have seen above, in  $\mathbb{P}^2$  two distinct lines always intersect at a point. We will see later that two distinct curves of degree  $d, d'$  intersect at  $dd'$  points in  $\mathbb{P}^2$  (when the points are counted with the correct multiplicity).

**Exercise 12.6.** Show that if  $C \subset \mathbb{P}^2$  is a curve of degree  $d > 0$  and  $L \subset \mathbb{P}^2$  is a line not contained in  $C$ , then  $L \cap C$  consists of  $d$  points when counted with multiplicity.

**Exercise 12.7.** Show that if  $C \subset \mathbb{P}^2$  is a curve of degree  $d > 0$  and  $Q \subset \mathbb{P}^2$  is a quadric and  $Q \cap C$  is finite, then  $Q \cap C$  consists of  $2d$  points when counted with multiplicity.

**Definition 12.8.** An ideal  $I \subset \mathbb{C}[x_0, \dots, x_n]$  is a **homogeneous ideal** if for any  $f \in I$ , then writing  $f = \sum_{i \geq 0} f_i$  where  $f_i$  is homogeneous of degree  $i$ , then  $f_i \in I$ . Notice that  $I$  is generated by finitely many homogeneous polynomials  $I = \{f_i\}_{i \in I}$  where  $f_i$  is homogeneous.

**Exercise 12.9.** Show that any homogeneous ideal  $I \subset \mathbb{C}[x_0, \dots, x_n]$  is generated by finitely many homogeneous polynomials.

**Definition 12.10.** A subset  $X \subset \mathbb{P}^n$  is a **projective variety** if it is defined by the vanishing of finitely many homogeneous polynomials or equivalently  $X = \mathbb{V}(I)$  where  $I \subset \mathbb{C}[x_0, \dots, x_n]$  is a homogeneous ideal.

Given any projective variety  $X \subset \mathbb{C}^n$  we let  $\mathbb{I}(\bar{X}) \subset \mathbb{C}[x_0, \dots, x_n]$  be the ideal corresponding to the closure of  $\bar{X}$  in  $\mathbb{P}^n$ . We note the following:

- (1)  $\mathbb{I}(\bar{X}) \subset \mathbb{C}[x_0, \dots, x_n]$  is a homogeneous ideal. If  $f \in \mathbb{I}(\bar{X})$ , then  $f$  vanishes at each point  $a = (1, a_1, \dots, a_n) \in X$ . Since  $\bar{X} \subset \mathbb{P}^n$  is a projective variety, for any  $\lambda \in \mathbb{C}^*$ ,  $f$  vanishes at each point  $\lambda \cdot a = (\lambda, \lambda a_1, \dots, \lambda a_n) \in \bar{X}$ , but then one sees that if  $f = \sum f_i$  where  $\deg(f_i) = i$ , then

$$0 = f(\lambda, \lambda a_1, \dots, \lambda a_n) = \sum f_i(\lambda, \lambda a_1, \dots, \lambda a_n) = \sum \lambda^i f_i(1, a_1, \dots, a_n)$$

holds for any  $\lambda \in \mathbb{C}^*$ . But then  $f_i(1, a_1, \dots, a_n) = 0$  for all  $i$  and so  $f_i \in \mathbb{I}(\bar{X})$  as required.

- (2)  $\mathbb{I}(\bar{X}) \subset \mathbb{C}[x_0, \dots, x_n]$  is a radical ideal, i.e.  $\sqrt{\mathbb{I}(\bar{X})} = \mathbb{I}(\bar{X})$ .
- (3) The set of homogeneous radical ideals are in one to one correspondence with the set of projective varieties with the exception of  $(x_0, \dots, x_n)$  that corresponds to the emptyset.

**Exercise 12.11.** Intersections of projective varieties are projective varieties and the union of two projective varieties is a projective variety.

**Definition 12.12.** The **Zariski topology** on a projective variety  $X \subset \mathbb{P}^n$  is defined by letting the closed subsets correspond to the zero sets of homogeneous ideals.

**Lemma 12.13.** Let  $\bar{X} \subset \mathbb{P}^n$  be a projective variety and  $X = \bar{X} \cap \{x_0 \neq 0\} \subset \mathbb{C}^n$  be the corresponding affine subvariety. Then the natural inclusion  $X \hookrightarrow \bar{X}$  is continuous in the respective Zariski topologies.

*Proof.* It suffices to show that if  $Z \subset \bar{X}$  is a closed subset then so is  $Z \cap X$ . Suppose that  $Z = \mathbb{V}(I)$  for some homogeneous ideal  $I = (f_1, \dots, f_r) \subset \mathbb{C}[x_0, \dots, x_n]$ , then  $Z \cap X = \mathbb{V}(I')$  where  $I' = (f_1(1, x_1, \dots, x_n), \dots, f_r(1, x_1, \dots, x_n)) \in \mathbb{C}[x_1, \dots, x_n]$  and so  $Z \cap X \subset \mathbb{C}^n$  is closed.  $\square$

**Exercise 12.14.** Show that projective varieties are compact in the Euclidean topology.

We now wish to define morphisms of projective varieties. The main difficulty is that points on projective varieties  $a \in X \subset \mathbb{P}^n$  are only defined up to multiplication by a non-zero scalar  $\lambda \in \mathbb{C}^*$ . However, consider the following example defined by

$$\phi : \mathbb{P}^1 \rightarrow \mathbb{P}^3, \quad \phi(t_0, t_1) = (t_0^3, t_0^2 t_1, t_0 t_1^2, t_1^3).$$

Notice that

$$\phi(\lambda t_0, \lambda t_1) = (\lambda^3 t_0^3, \lambda^3 t_0^2 t_1, \lambda^3 t_0 t_1^2, \lambda^3 t_1^3) = \lambda^3 (t_0^3, t_0^2 t_1, t_0 t_1^2, t_1^3)$$

and therefore  $\phi$  gives a well defined map  $\phi : \mathbb{P}^1 \rightarrow \mathbb{P}^3$ . One would be tempted to conclude that given any set of homogeneous polynomials of (the same) degree  $d$  say

$$p_1, \dots, p_r \in \mathbb{C}[x_0, \dots, x_n]_d$$

then we obtain a morphism

$$\phi : \mathbb{P}^n \rightarrow \mathbb{P}^r$$

however note that if  $a \in \mathbb{P}^n$  is contained in  $\mathbb{V}(p_1, \dots, p_r)$ , then  $\phi(a) = [0 : \dots : 0]$  is not defined! Therefore we only have a map

$$\phi : (\mathbb{P}^n \setminus \mathbb{V}(p_1, \dots, p_r)) \rightarrow \mathbb{P}^r.$$

**Definition 12.15.** A **quasi-projective variety** is an open subset of a projective variety  $X \subset \mathbb{P}^n$ .

Note that if  $X \subset \mathbb{P}^n$  is a projective variety and  $\mathbb{C}^n \cong U_i = \{x_i \neq 0\} \subset \mathbb{P}^n$ , then  $X_i := X \cap U_i$  is a quasi-projective variety which is an affine variety.

**Definition 12.16.** Let  $F : X \rightarrow Y$  be a map of quasi-projective varieties  $X \subset \mathbb{P}^n$  and  $Y \subset \mathbb{P}^m$ , then  $F$  is a **morphism** if for any  $x \in X$  there exists an open subset  $U \subset X$  such that

$$F|_U = [p_0 : \dots : p_m]$$

for some homogeneous polynomials of degree  $d$  in  $\mathbb{C}[x_0, \dots, x_n]_d$  such that  $\mathbb{V}(p_0, \dots, p_m) \cap U = \emptyset$ . We say that  $F$  is an **isomorphism** if there is a morphism  $G : Y \rightarrow X$  such that  $F \circ G = \text{id}_Y$  and  $G \circ F = \text{id}_X$ .

**Definition 12.17.** Let  $X$  be a quasi-projective variety. If  $X$  is isomorphic to an affine variety  $Y \subset \mathbb{C}^n \subset \mathbb{P}^n$ , then we say that  $X$  is an affine variety.

Here are some examples:

- (1)  $d$ -Uple embedding of  $\mathbb{P}^n$ . Let  $M_0, \dots, M_N$  be the monomials of degree  $d$  in  $x_0, \dots, x_n$ , so that  $N = \binom{n+d}{d} - 1$  and define

$$\rho_d : \mathbb{P}^n \rightarrow \mathbb{P}^N, \quad \rho_d(x) = [M_0(x) : \dots : M_N(x)].$$

So for example if  $n = 1, d = 3$ , then  $\rho_3(x, y) = [x^3 : x^2y : xy^2 : y^3]$  and if  $n = 2, d = 2$  then  $\rho_2(x, y, z) = [x^2 : y^2 : z^2 : xy : xz : yz]$ .

- (2) Segre embedding. Define  $\psi : \mathbb{P}^r \times \mathbb{P}^s \rightarrow \mathbb{P}^N$  where  $N = (r+1)(s+1) - 1$  by

$$\psi([x_0 : \dots : x_r] \times [y_0 : \dots : y_s]) = [x_0y_0 : \dots : x_0y_s : x_1y_0 : \dots : x_1y_s : \dots : x_ry_0 : \dots : x_ry_s].$$

- (3)  $\psi : \mathbb{P}^2 \setminus \{xyz = 0\} \rightarrow \mathbb{P}^2$  defined by  $\psi([x : y : z]) = ([yz : xz : xy])$ .

- (4) Mobius transformations: Let

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2, \mathbb{C})$$

be an invertible  $2 \times 2$  matrix, then we define the map

$$\phi_M : \mathbb{P}^1 \rightarrow \mathbb{P}^1, \quad [x : y] \rightarrow [ax + by : cx + dy].$$

Notice that since  $M$  is invertible, if  $(x : y) \neq (0, 0)$ , then  $(ax + by : cx + dy) \neq (0, 0)$  and so the map is well defined. It is also clear that  $\phi_M = \phi_{tM}$  for any  $t \in \mathbb{C}^*$  and one can check that  $\phi_M \neq \phi_N$  if  $M, N \in GL(2, \mathbb{C})$  are linearly independent. Notice also that  $\phi_M^{-1} = \phi_{M^{-1}}$ . Therefore the group  $PGL(2, \mathbb{C}) = GL(2, \mathbb{C})/\mathbb{C}^*$  acts on  $\mathbb{P}^1$ . Finally we remark that in affine coordinates (on  $\mathbb{C} = \mathbb{P}^1 \setminus \mathbb{V}(y)$ ) we obtain

$$[x : 1] \rightarrow \left[ \frac{ax + b}{cx + d} : 1 \right].$$

- (5)  $\mathbb{P}^2 \dashrightarrow \mathbb{P}^1 \times \mathbb{P}^1$  defined by  $[x : y : z] \dashrightarrow ([x : z], [y : z])$ .

**Exercise 12.18.** Show that the product of two affine varieties is an affine variety (in particular irreducible).

**Definition 12.19.** If  $X \subset \mathbb{P}^n$  is a projective variety, then  $S(X) = \mathbb{C}[x_0, \dots, x_n]/\mathbb{I}(X)$  is the **homogeneous coordinate ring** of  $X$ .

Since  $A(X)$  is Noetherian,  $S(X)$  is also Noetherian (see Exercise 3.23). Note that  $S(\mathbb{P}^n) = \mathbb{C}[x_0, \dots, x_n]$  and if  $U_i = \{x_i \neq 0\}$ , then  $A(U_i) \cong \mathbb{C}[x_0/x_i, \dots, x_{i-1}/x_i, x_{i+1}/x_i, \dots, x_n/x_i]$  which we think of as the set of degree 0 elements in the ring  $\mathbb{C}[x_0, \dots, x_n][1/x_i]$ .

Consider now the following example of an affine curve  $X = \mathbb{V}(x^2 - y^3) \subset \mathbb{C}^2$  corresponding to the projective curve  $\bar{X} = \mathbb{V}(zx^2 - y^3) \subset \mathbb{P}^2$ . Then  $S(\bar{X}) = \mathbb{C}[x, y, z]/(x^2z - y^3)$  and  $A(X) = \mathbb{C}[x, y]/(x^2 - y^3)$  which we identify with the set of degree 0 elements in  $\mathbb{C}[x, y, z, z^{-1}]/(x^2z - y^3)$  via the inclusion  $\phi : \mathbb{C}[x, y, z]/(x^2z - y^3) \rightarrow \mathbb{C}[x, y, z, z^{-1}]/(x^2z - y^3)$  defined by  $\phi(x) = x/z$ ,  $\phi(y) = y/z$  and hence  $\phi(p(x, y)) = p(x/z, y/z)$ . In particular  $\phi(x^2 - y^3) = (x/z)^2 - (y/z)^3 = z^{-3}(zx^2 - y^3) = 0 \in \mathbb{C}[x, y, z, z^{-1}]/(x^2z - y^3)$ .

**Lemma 12.20.** Let  $X \subset \mathbb{P}^n$  be a projective variety,  $X_i = X \cap U_i$ , then  $A(X_i)$  corresponds to the elements of degree 0 in  $S(X)[1/x_i]$  and  $S(X)[1/x_i] = A(X_i)[x_i, x_i^{-1}]$ .

*Proof.* Exercise. □

**Theorem 12.21.** Let  $X, Y \subset \mathbb{C}^n$  be affine varieties, then every irreducible component of  $X \cap Y$  has dimension  $\geq \dim X + \dim Y - n$ .



*Proof.* Suppose that  $Y = \mathbb{V}(f)$  is a hypersurface. If  $X \subset Y$ , then the claim is obvious, so suppose that  $X \not\subset Y$  and pick an irreducible component  $W$  of  $X \cap Y$ . If  $A(X)$  is the coordinate ring of  $X$ , then  $Y \cap X$  corresponds to  $A(X)/(f)$  and  $W$  to a minimal prime ideal  $P$  containing  $(f)$ . By Krull's Hauptidealsatz,  $P$  has height 1 and hence  $\dim W = \dim A(X)/P \geq \dim X - 1$  as required.

For the general case, we let  $\mathbb{C}^n \cong \Delta \subset \mathbb{C}^n \times \mathbb{C}^n$  be the diagonal and we observe that  $\Delta \cap (X \times Y) = X \cap Y$ . The result now follows by applying the hypersurface case  $n$  times, since  $\Delta = \mathbb{V}(x_1 - y_1, \dots, x_n - y_n)$ .  $\square$

**Theorem 12.22.** *Let  $X, Y \subset \mathbb{P}^n$  be projective varieties, then every irreducible component of  $X \cap Y$  has dimension  $\geq \dim X + \dim Y - n$  and if  $\dim X + \dim Y - n \geq 0$ , then  $X \cap Y \neq \emptyset$ .*

*Proof.* To see that  $X \cap Y$  has dimension  $\geq \dim X + \dim Y - n$  it suffices to observe that  $\mathbb{P}^n$  is covered by affine varieties  $U_i \cong \mathbb{C}^n$  and apply Theorem 12.21.

Suppose now that  $\dim X + \dim Y - n \geq 0$  and consider the corresponding cones  $C(X), C(Y) \subset \mathbb{C}^{n+1}$ . Since the intersection is non-empty (it contains  $\bar{0}$ ), it follows that there is an irreducible component  $W \subset C(X) \cap C(Y)$  of dimension

$$\geq \dim C(X) + \dim C(Y) - (n + 1) = \dim X + 1 + \dim Y + 1 - (n + 1) \geq 1.$$

But then  $W = C(Z)$  for some non empty subset  $Z \subset \mathbb{P}^n$  and hence  $X \cap Y \supset Z$  is not empty.  $\square$

**Exercise 12.23.** *Show that the homogeneous coordinate ring is not an invariant of  $X$  (eg. consider  $X = \mathbb{P}^1$  embedded in  $\mathbb{P}^3$  via  $[s : t] \rightarrow [s^4 : s^3t : st^3 : s^4]$  and  $X = \mathbb{P}^1$  embedded in  $\mathbb{P}^4$  via  $[s : t] \rightarrow [s^4 : s^3t : s^2t^2 : st^3 : s^4]$ ).*

**Exercise 12.24.** *Show that the rational normal curve in  $\mathbb{P}^3$  is not a complete intersection.*

**Exercise 12.25.** *Show that if  $H \in \mathbb{C}[x_0, \dots, x_n]$  is homogeneous of degree  $d > 0$ , then  $\mathbb{P}^n \setminus \mathbb{V}(H)$  is affine.*

### 13. REGULAR FUNCTIONS ON QUASI-PROJECTIVE VARIETIES

**Definition 13.1.** *Let  $X \subset \mathbb{P}^n$  be a quasi-projective variety and  $f : X \rightarrow \mathbb{C}$  a continuous function, then  $f$  is regular at  $x \in X$  if there is an open affine subset  $x \in U \subset X$  such that  $f|_U$  is regular.*

For the above definition to be useful, we will need the following observation:

**Lemma 13.2.** *Let  $x \in X \subset \mathbb{P}^n$  be a point on a quasi-projective variety, then there exists an open affine subset  $x \in U \subset X$ .*

*Proof.* Let  $x = [a_0 : \dots : a_n]$  and assume that  $a_i \neq 0$ , then  $x \in X_i = X \setminus \mathbb{V}(x_i) \subset X$  is a quasi-affine variety. By Lemma 1.32, there is an affine open subset  $x \in U \subset X_i$ .  $\square$

We make the following observations: Regular functions on  $X$  define a sheaf, so that

- (1) It is easy to see that if  $U \subset X$  is an open subset, then the set of regular functions on  $U$  given by

$$\mathcal{O}_X(U) = \{f : U \rightarrow \mathbb{C}, \text{ s.t. } f \text{ is regular at each point of } U\}$$

is a  $\mathbb{C}$  algebra.

- (2) If  $U \subset V \subset X$  are two open subsets and  $f \in \mathcal{O}_X(V)$ , then  $f|_U \in \mathcal{O}_X(U)$ . The reader should check that this defines a homomorphism of  $\mathbb{C}$ -algebras  $\mathcal{O}_X(V) \rightarrow \mathcal{O}_X(U)$ .

- (3) If  $U, V \subset X$  are two open subsets and  $f \in \mathcal{O}_X(V)$ ,  $g \in \mathcal{O}_X(U)$  agree on  $U \cap V$  (i.e.  $f|_{U \cap V} = g|_{U \cap V}$ ), then there exists a regular function  $h \in \mathcal{O}_X(U \cup V)$  such that  $h|_U = g$  and  $h|_V = f$ .

**Lemma 13.3.** *Let  $f : X \rightarrow Y$  be a map of quasi-projective varieties then  $f$  is a morphism if and only if for any  $p \in X$  there are affine neighborhoods  $p \in U \subset X$  and  $f(p) \in V \subset Y$  with  $f(U) \subset V$  such that  $f|_U$  is a regular map of affine varieties.*

*Proof.* Exercise. □

**Definition 13.4.** *A morphism of quasi-projective varieties  $f : X \rightarrow Y$  is a projective morphism if  $X$  is isomorphic to a closed subset of  $Y \times \mathbb{P}^n$  (which we identify with  $X$  so that we write  $X \subset Y \times \mathbb{P}^n$ ) and  $f$  is induced by the projection  $p : Y \times \mathbb{P}^n \rightarrow Y$  (i.e.  $f = p|_X$ ).*

Note that for any point  $p \in Y$ , we have that  $f^{-1}(p) \subset \mathbb{P}^n$  is a projective variety and hence is compact in the Euclidean topology. In fact, it follows that for any compact (in the Euclidean topology) subset  $V \subset Y$ , the inverse image  $f^{-1}(V)$  is also compact.

## 14. RATIONAL MAPS

**Definition 14.1.** *Let  $X \subset \mathbb{P}^n$  be a projective variety and  $p, q \in \mathbb{C}[x_0, \dots, x_n]$  be non-zero homogeneous polynomials of the same degree  $d$ . If  $p \notin \mathbb{I}(X)$ , then  $p/q$  defines a **rational function** on  $X \setminus \mathbb{V}(q)$ . Note that if  $[a_0 : \dots : a_n] \in \mathbb{P}^n$  and  $\lambda \in \mathbb{C}^*$ , then*

$$\frac{p(\lambda a_0 : \dots : \lambda a_n)}{q(\lambda a_0 : \dots : \lambda a_n)} = \frac{\lambda^d p(a_0 : \dots : a_n)}{\lambda^d q(a_0 : \dots : a_n)} = \frac{p(a_0 : \dots : a_n)}{q(a_0 : \dots : a_n)}$$

so that  $p/q$  is a well defined function on  $\mathbb{P}^n \setminus \mathbb{V}(q)$  and hence on  $X \setminus \mathbb{V}(q)$ .

**Exercise 14.2.** *Check that  $p/q$  and  $p'/q'$  define the same function on (a non-empty open subset of)  $X$  if and only if  $pq' - qp' \in \mathbb{I}(X)$ .*

**Definition 14.3.** *Let  $X \subset \mathbb{P}^n$  be a projective variety. We let*

$$\mathbb{C}(X) = \{p/q | p, q \in \mathbb{C}[x_0, \dots, x_n]_d, q \notin \mathbb{I}(X)\} / \sim$$

*be the field of rational functions of  $X$ . Here  $\sim$  is the equivalence relation given by  $p/q \sim p'/q'$  if  $pq' - qp' \in \mathbb{I}(X)$ .*

**Exercise 14.4.** *Check that  $\sim$  is an equivalence relation.*

**Exercise 14.5.** *Check that  $\mathbb{C}(X)$  is a field.*

**Exercise 14.6.** *Let  $X \subset \mathbb{C}^n$  be an affine variety and  $\bar{X} \subset \mathbb{P}^n$  be its projective closure. Show that  $\mathbb{C}(\bar{X})$  is the field of fractions of  $A(X)$ .*

**Definition 14.7.** *Let  $f$  be a function defined on an open subset  $U \subset X$  of a projective variety. For any point  $p \in U$ , we say that  $f$  is **regular** at  $p$  if there exists an open subset  $p \in V \subset U$  and a rational function  $p/q \in \mathbb{C}(X)$  such that  $V \cap \mathbb{V}(q) = \emptyset$  and  $f|_V = (p/q)|_V$ . We let  $\text{Reg}(f) \subset X$  be the set of all regular points of  $f$ . Note that this is an open subset.*

**Definition 14.8.** *Let  $X \subset \mathbb{C}^n$  (resp.  $X \subset \mathbb{P}^n$ ) be an affine (resp. projective) variety. A **rational map**  $\phi : X \dashrightarrow \mathbb{C}^m$  (resp.  $\phi : X \dashrightarrow \mathbb{P}^m$ ) is given by*

$$\phi : x \rightarrow (f_1(x), \dots, f_m(x)), \quad x \rightarrow [f_0(x) : \dots : f_m(x)]$$

where  $x \in X$  and  $f_i \in \mathbb{C}(X)$ . Note that  $\phi$  is not defined if  $f_i(x)$  is not defined (i.e. if  $x \notin \cap \text{Reg}(f_i)$ ) resp. if  $f_i(x)$  is not defined or if all  $f_0(x) = \dots = f_m(x) = 0$  (since  $[0 : \dots : 0]$  is not a point of  $\mathbb{P}^m$ ). This is why  $\phi$  is denoted with a broken arrow as it is only partially defined. We consider two rational maps  $\phi, \phi' : X \dashrightarrow \mathbb{P}^m$  to be equivalent if they agree on a common open subset  $U \subset X$ . In this case we have  $f_i(x) = g_i(x)f'_i(x)$  for all  $x \in U$  and so  $g_i(x) \in \mathbb{C}(U)$ . Given a rational map  $\phi : X \dashrightarrow \mathbb{P}^m$  we may then consider the **domain of definition**  $\text{Dom}(\phi) = \cup U_i$  where  $\phi|_{U_i} : U_i \rightarrow \mathbb{P}^m$  is defined by  $f_i \in \mathbb{C}(X)$  such that  $U_i \cap_{i=0}^m \mathbb{V}(f_i) = \emptyset$  and  $U_i \subset \cap \text{Reg}(f_i)$ .

**Exercise 14.9.** Show that  $\text{Dom}(\phi)$  is a non empty open subset of  $X$ .

**Remark 14.10.** Note that given  $\phi : X \dashrightarrow \mathbb{P}^m$  represented by  $x \rightarrow [f_0(x) : \dots : f_m(x)]$ , we may also consider the representative  $x \rightarrow [1 : f_1(x)/f_0(x) : \dots : f_m(x)/f_0(x)]$ . It is then easy to see that there is a one to one correspondence between the set of rational maps  $\phi : X \dashrightarrow \mathbb{C}^m$  and the set of rational maps  $\phi : X \dashrightarrow \mathbb{P}^m$ .

**Remark 14.11.** Note that if  $f_0, \dots, f_m \in \mathbb{C}[x_1, \dots, x_n]$  are homogeneous of the same degree say  $d$ , then we may define a rational map  $\phi : \mathbb{P}^n \rightarrow \mathbb{P}^m$  by  $[f_0/x_0^d : \dots : f_m/x_0^m]$  and this map is clearly equivalent to  $[f_0/x_i^d : \dots : f_m/x_i^m]$  for any  $0 \leq i \leq m$ .

We have the following examples:

- (1) **Rational normal curve of degree  $n$ :**  $\phi : \mathbb{P}^1 \dashrightarrow \mathbb{P}^n$  defined by  $[x : y] \rightarrow [x^n : x^{n-1}y : \dots : xy^{n-1} : y^n]$ .
- (2) **Projection from a linear subspace:** Let  $L \subset \mathbb{P}^n$  be a linear subspace of dimension  $r$  (i.e.  $L \cong \mathbb{P}^r$ ), then we may define the projection  $\phi : X \dashrightarrow \mathbb{P}^{n-r-1}$ . In appropriate homogeneous coordinates, we may assume that  $L = \mathbb{V}(x_{r+1}, \dots, x_n)$  and  $\phi(x) = [x_{r+1} : \dots : x_n]$ . It follows easily that  $\text{Dom}(\phi) = X \setminus L$ .
- (3)  $\phi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^2$  defined by  $[x : y : z] \dashrightarrow [1/x : 1/y : 1/z]$ . check that  $\text{Dom}(f) = \mathbb{P}^2 \setminus \{[1 : 0 : 0], [0 : 1 : 0], [0 : 0 : 1]\}$ .
- (4) **Projection of the cusp from the origin:** Let  $X = \mathbb{V}(zx^2 - y^3)$  and  $\phi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^1$  defined by  $[x : y : z] \rightarrow [x : y]$  be the projection from  $[0 : 0 : 1]$ , then  $\phi|_X : X \setminus \{[0 : 0 : 1]\} \rightarrow \mathbb{P}^1 \setminus [0 : 1]$  is a bijection. The rational map  $\psi : \mathbb{P}^1 \dashrightarrow X$  defined by  $[x : y] \rightarrow [x : y : y^3/x^2]$  is the inverse of  $\phi|_X$ . Thus  $X$  and  $\mathbb{P}^1$  are birational.
- (5) **Projection of a nodal cubic from the origin:** Let  $X = \mathbb{V}(y^2z - x^3 - x^2z)$  and  $\phi : \mathbb{P}^2 \dashrightarrow \mathbb{P}^1$  defined by  $[x : y : z] \rightarrow [x : y]$  be the projection from  $[0 : 0 : 1]$ , then  $\phi|_X : X \setminus \{[0 : 0 : 1]\} \rightarrow \mathbb{P}^1 \setminus \{[1 : 1], [1 : -1]\}$  is a bijection. The rational map  $\psi : \mathbb{P}^1 \dashrightarrow X$  defined by  $[x : y] \rightarrow [x : y : x^3/(y^2 - x^2)]$  is the inverse of  $\phi|_X$ . Thus  $X$  and  $\mathbb{P}^1$  are birational.

We will see later that neither the cusp nor the nodal cubic are isomorphic to  $\mathbb{P}^1$  (since they are singular, but  $\mathbb{P}^1$  is smooth).

**Definition 14.12.** Let  $\phi : X \dashrightarrow Y$  be a rational map of projective varieties. We say that  $\phi$  is **dominant** if  $f(\text{Dom}(\phi)) \subset Y$  is dense i.e.  $\overline{f(\text{Dom}(\phi))} = Y$ .

**Definition 14.13.** Let  $\phi : X \dashrightarrow Y$  be a dominant rational map of projective varieties then there is a naturally induced map of function fields  $\phi^* : \mathbb{C}(Y) \rightarrow \mathbb{C}(X)$  defined by  $\phi^*(g) = g \circ \phi$ . More explicitly, we may assume that  $X \subset \mathbb{P}^n$  (with coordinates  $x_0, \dots, x_n$ ) and  $Y \subset \mathbb{P}^m$  (with coordinates  $y_0, \dots, y_m$ ) so that if  $\phi$  is represented by  $[f_0 : \dots : f_m]$  for some rational functions  $f_i \in \mathbb{C}[x_0, \dots, x_n]$  whose regular locus intersects  $X$  and who do not all

vanish along  $X$ , then for any rational function  $g \in \mathbb{C}(Y)$ , we may consider  $g = g(y_0, \dots, y_m)$  and we let  $\phi^*(g) = g(f_0(x_0, \dots, x_n), \dots, (f_0(x_0, \dots, x_n)))$ . Of course one must check that this is all well defined. For example if  $g' \in \mathbb{C}[y_0, \dots, y_m]$  is another representative of  $g$  so that  $(g - g')|_Y = 0$ , then  $g = g' + h$  where  $h \in \mathbb{I}(Y)$ . But then since  $\phi^*(h) \in \mathbb{I}(X)$  it easily follows that  $\phi^*(g) - \phi^*(g') \in \mathbb{I}(X)$  so that  $\phi^*(g)$  and  $\phi^*(g')$  define the same element in  $\mathbb{C}(X)$ .

**Definition 14.14.** A rational map of projective varieties  $\phi : X \dashrightarrow Y$  is **birational** if there exists a rational map  $\psi : Y \dashrightarrow X$  such that  $\phi \circ \psi = \text{id}_Y$  and  $\psi \circ \phi = \text{id}_X$  (note that this means that  $\phi \circ \psi$  and  $\psi \circ \phi$  are the identities on some open subsets).

**Proposition 14.15.** Let  $\phi : X \dashrightarrow Y$  be a dominant rational map of projective varieties then the following are equivalent.

- (1)  $\phi$  is birational,
- (2)  $\phi^* : \mathbb{C}(Y) \rightarrow \mathbb{C}(X)$  is an isomorphism,
- (3) there exists open subsets  $X_0 \subset X$  and  $Y_0 \subset Y$  such that  $f|_{X_0} : X_0 \rightarrow Y_0$  is an isomorphism.

*Proof.* We will first show that (1) and (2) are equivalent. Assume that  $\phi$  is birational, then by assumption there is a rational map  $\psi : Y \dashrightarrow X$  such that  $\psi \circ \phi$  and  $\phi \circ \psi$  induce the identity on some open subsets of  $X$  and  $Y$  respectively. But then it is clear that  $\psi$  is also dominant and  $\phi^* \circ \psi^*$  and  $\psi^* \circ \phi^*$  are the identity on  $\mathbb{C}(X)$  and  $\mathbb{C}(Y)$ . Suppose now that  $\phi^* : \mathbb{C}(Y) \rightarrow \mathbb{C}(X)$  is an isomorphism. Since  $X \subset \mathbb{P}^n$  and  $Y \subset \mathbb{P}^m$  are projective, we may fix coordinates  $x_0, \dots, x_n$  and  $y_0, \dots, y_m$ . If  $f_i = \phi^*(y_i)$ , then  $\phi : X \rightarrow Y$  is represented by  $[f_0 : \dots : f_m]$ . Since  $\phi^*$  is an isomorphism we may pick  $g_0, \dots, g_m \in \mathbb{C}(Y)$  such that  $\phi^*(g_i) = x_i$  and we define  $\psi : Y \dashrightarrow X$  via  $[g_0 : \dots : g_m]$ . Note that

$$(\phi \circ \psi)^*(x_i) = \phi^*(\psi^*(x_i)) = \phi^*(g_i) = x_i$$

and so  $\phi \circ \psi$  is the identity on an open subset.

We next show that (1) and (3) are equivalent. Since (3) clearly implies (1), suppose that (1) holds i.e. there is a rational map  $\psi : Y \dashrightarrow X$  such that  $\psi \circ \phi$  and  $\phi \circ \psi$  induce the identity on  $X$  and  $Y$  respectively. Let  $X_0 = \text{Dom}(\phi)$  and  $Y_0 = \text{Dom}(\psi)$ ;  $\phi_0 = \phi|_{X_0}$  and  $\psi_0 = \psi|_{Y_0}$ ;  $X_1 = \phi_0^{-1}(Y_0)$  and  $Y_1 = \psi_0^{-1}(X_0)$ . Thus we have a map

$$\text{id}_{X_1} = (\psi \circ \phi)|_{X_1} : X_1 \rightarrow Y_0 \rightarrow X.$$

We claim that  $\phi_0(X_1) \subset Y_1$ . In other words we must show that if  $x \in X_1$ , then  $\psi_0(\phi_0(x)) \in X_0$ , but this is obvious since  $\psi_0(\phi_0(x)) = x \in X_1 \subset X_0$ . Clearly we also have  $\psi_0(Y_1) \subset X_1$ . But then it is clear that the induced maps  $X_1 \rightarrow Y_1$  and  $Y_1 \rightarrow X_1$  are isomorphisms. □

**Exercise 14.16.** Use the Primitive Element Theorem and Noether's normalization lemma, to show that any  $r$  dimensional variety is birational to a hypersurface in  $\mathbb{P}^{r+1}$ .

**Exercise 14.17.** Show that  $\mathbb{P}^2$  and  $\mathbb{P}^1 \times \mathbb{P}^1$  are birational.

**Theorem 14.18.** If  $X$  is a projective variety, then  $\mathcal{O}_X(X) = \mathbb{C}$ .

*Proof.* Let  $X_i = X \cap \{x_i \neq 0\} \subset \mathbb{C}^n$  be the corresponding affine varieties, then  $f|_{X_i} \in \mathcal{O}_X(X_i) \subset \mathbb{C}[x_0, \dots, x_n, x_i^{-1}]/\mathbb{I}(X_i)$  so  $f = g_i/x_i^{N_i}$  for some  $g_i \in \mathbb{C}[x_0, \dots, x_n]$  homogeneous of degree  $N_i$  and so  $f x_i^{N_i} \in S(X)_{N_i}$  (where  $S(X)_{N_i}$  denotes the subset corresponding to homogeneous polynomials of degree  $N_i$ ).

Let  $N = \sum N_i$ , then  $S(X)_N \cdot f \subset S(X)_N$ . To see this, it suffices to observe that if  $x_I$  is homogeneous of degree  $N$ , then  $x_i^{N_i}$  divides  $x_I$  for some  $i$  and so  $x_I \cdot f = x_I/x_i \cdot (x_i \cdot f) \in S(X)$  is homogeneous of degree  $N$ . But then,

$$S(X)_N \cdot x_0^{N_0} f^t \subset S(X)_N \cdot x_0^{N_0} f^{t-1} \subset \dots x_0^{N_0} \subset S(X)_N$$

and so  $S(X)[f] \subset x_0^{-N_0} S(X)$ . Since  $S(X)$  is a Noetherian ring and  $x_0^{-N_0} S(X)$  is a finitely generated  $S(X)$  module, it follows that  $S(X)[f]$  is finitely generated  $S(X)$  module (see Theorem 4.24). But then we may find a monic polynomial for  $f$  (see the proof of Lemma 5.7)

$$f^d + a_1 f^{d-1} + \dots + a_{d-1} f + a_d = 0$$

where  $a_i \in S(X)$ . Since  $f$  has degree 0, the above equation also holds if we replace each  $a_i$  by their homogeneous component of degree 0 and so we may assume that  $f$  satisfies a monic polynomial in  $\mathbb{C}[x]$ . Since  $\mathbb{C}$  is algebraically closed, then  $f \in \mathbb{C}$ .  $\square$

**Exercise 14.19.** Explain why there are  $\binom{n+d}{n}$  monomials of degree  $d$  in  $n$  variables.

**Exercise 14.20.** Let  $X = \mathbb{V}(xy - zw) \subset \mathbb{P}^3$ . Show that  $X$  is isomorphic to  $\mathbb{P}^1 \times \mathbb{P}^1$ .

**Exercise 14.21.** Show that  $\mathbb{P}^2$  is not isomorphic to  $\mathbb{C}^2$  nor to  $\mathbb{P}^1 \times \mathbb{P}^1$ .

**Exercise 14.22.** If an affine variety is isomorphic to a projective variety, then it is a point.

## 15. PROJECTIVE MORPHISMS ARE PROPER

Roughly speaking, a **projective morphism** is a morphism  $f : X \rightarrow Y$  of quasi projective varieties that factors through  $\mathbb{P}_Y^n := \mathbb{P}^n \times Y$ , more precisely  $f$  is induced by a closed immersion  $X \rightarrow \mathbb{P}_Y^n$  followed by the projection  $p : \mathbb{P}_Y^n \rightarrow Y$  (a closed immersion is a map  $f : X \rightarrow P$  which is a homeomorphism of topological spaces and induces a surjection  $f^\# \mathcal{O}_P \rightarrow \mathcal{O}_X$ . See [Hartshorne].)

**Theorem 15.1.** Let  $Y$  be a quasi-projective variety. Let  $X \subset \mathbb{P}^n \times Y$  be a closed subset and  $p : \mathbb{P}^n \times Y \rightarrow Y$  the projection on to the second factor, then  $p(X) \subset Y$  is a closed subset.

*Proof.* Since the claim is local on  $Y$ , we may assume that  $Y$  is affine and hence  $Y \subset \mathbb{C}^n$ . Notice that if  $p(X) \subset Y \subset \mathbb{C}^m$  is a closed subset of  $\mathbb{C}^m$ , then it is a closed subset of  $Y$ . Therefore we may assume that  $Y = \mathbb{C}^m$ .

Suppose now that  $X = \mathbb{V}(f_1(x, y), \dots, f_r(x, y))$  where  $x = [x_0 : \dots : x_n]$  are homogeneous coordinates on  $\mathbb{P}^n$  and  $y = (y_1, \dots, y_m)$  are coordinates on  $\mathbb{C}^m$  such that  $f_i(x, y)$  is homogeneous of degree  $d_i$  with respect to the coordinates  $[x_0 : \dots : x_n]$ . It suffices to show that if  $\bar{y} \in \mathbb{C}^m \setminus p(X)$ , then there is an open subset  $\bar{y} \in U \subset \mathbb{C}^m \setminus p(X)$ . Note that if  $\bar{y} \in \mathbb{C}^m \setminus p(X)$ , then  $\mathbb{V}(f_1(x, \bar{y}), \dots, f_r(x, \bar{y})) = \emptyset$  or equivalently there is a  $k > 0$  such that  $(f_1(x, \bar{y}), \dots, f_r(x, \bar{y})) \supset (x_0, \dots, x_n)^k$ . Consider now the ideal

$$I = (f_1(x, y), \dots, f_r(x, y)) \subset R = \mathbb{C}[x_0, \dots, x_n, y_1, \dots, y_m],$$

and Let  $R_k \subset \mathbb{C}[x_0, \dots, x_n, y_1, \dots, y_m]$  be the set of polynomials which are homogeneous of degree  $N$  with respect to  $x_0, \dots, x_n$ . By what we have seen above, we have

$$I \cap R/\mathfrak{m}_{\bar{y}} R \supset (x_0, \dots, x_n)^k \otimes R/\mathfrak{m}_{\bar{y}}.$$

It is easy to see that this implies  $I \cap R_k + \mathfrak{m}_{\bar{y}} R = R_k$  and hence that

$$\mathfrak{m}_{\bar{y}}(R_N/I \cap R_k) = R_k/(I \cap R_k).$$

Since  $R_N/(I \cap R_N)$  is a finitely generated  $k[y_1, \dots, y_m]$  module, it follows that  $(R_k/(I \cap R_k))_{\mathfrak{m}_{\bar{y}}}$  is a finitely generated  $\mathbb{C}[y_1, \dots, y_m]_{\mathfrak{m}_{\bar{y}}}$  module, and so by Nakayama's Lemma (Corollary 6.3) we have By Nakayama's lemma it follows that  $(R_k/(I \cap R_k))_{\mathfrak{m}_{\bar{y}}} = 0$ . But then there exists an  $f \in \mathbb{C}[y_1, \dots, y_m] \setminus \mathfrak{m}_{\bar{y}}$  such that  $f(R_k/(I \cap R_k)) = 0$  i.e.  $R_k = I \cap R_k$  over  $\mathbb{C}^m \setminus \mathbb{V}(f)$ . But then  $(f_1(x, \bar{y}), \dots, f_r(x, \bar{y})) \supset (x_0, \dots, x_n)^k$  for any  $\bar{y} \in \mathbb{C}^m \setminus \mathbb{V}(f)$ . This completes the proof.  $\square$

**Exercise 15.2.** Show that Theorem 15.1 fails if the morphism is not projective.

We use the above result to show the following result on the semicontinuity of the fiber dimension of a projective morphism.

**Theorem 15.3.** Let  $Y$  be a quasi-projective variety. Let  $X \subset \mathbb{P}^n \times Y$  be a closed subset and  $p : \mathbb{P}^n \times Y \rightarrow Y$  the projection on to the second factor, then

$$Y_k = \{y \in Y \mid \dim(p^{-1}(y)) \leq k\}$$

is open.

*Proof.* Since the question is local on  $Y$ , we may assume that  $Y$  is affine. If  $\dim(p^{-1}(y)) \leq k$  and  $H_1, \dots, H_{k+1}$  are general hyperplanes in  $|\mathcal{O}_{\mathbb{P}^n_Y}(1)|$ , then  $(p^{-1}(y)) \cap H_1 \cap \dots \cap H_{k+1} = \emptyset$ . Let  $Z = H_1 \cap \dots \cap H_{k+1}$  and  $W = p(Z) \subset Y$ . By Theorem 15.1,  $W$  is closed. Since  $p^{-1}(y') \cap H_1 \cap \dots \cap H_{k+1} = \emptyset$  for any  $y' \in Y \setminus W$ , it follows that  $\dim p^{-1}(y') \leq k$  for any  $y' \in Y \setminus W$  and the theorem is proven.  $\square$

**Exercise 15.4.** Show that if  $f : X \rightarrow Y$  is a surjective projective morphism,  $k = \dim X - \dim Y$ , then  $Y_{k-1} = \emptyset$ ,  $\overline{Y_k} = Y$  and  $\dim Y_{k+t} \leq \dim Y - t - 1$ .

## 16. NORMALITY

**Definition 16.1.** Let  $A \subset B$  be an inclusion of rings, then the **integral closure of  $A$  in  $B$**  is the union of all elements  $b \in B$  which are integral over  $A$  (i.e. they satisfy a monic polynomial with coefficients in  $A$ ). If  $A$  is a domain, then the **integral closure  $\bar{A}$  of  $A$**  is the integral closure of  $A$  in its quotient field  $Q(A)$ . Clearly  $A \subset \bar{A} \subset Q(A)$ . If  $A = \bar{A}$ , then we say that  $A$  is **integrally closed** or **normal**.

**Exercise 16.2.** Let  $A \subset B$  be an inclusion of rings. Show that the integral closure  $\bar{A} \subset B$  of  $A$  in  $B$  is a ring.

**Exercise 16.3.** Show that if  $A$  is factorial (i.e. it has a unique factorization) then  $A$  is integrally closed.

**Exercise 16.4.** Show that the integral closure  $\bar{A} \subset B$  is integrally closed in  $B$ .

**Exercise 16.5.** Show that the integral closure of  $\mathbb{Z} \subset \mathbb{Q}(i)$  is  $\mathbb{Z}[i]$ .

**Exercise 16.6.** Show that the integral closure of  $\mathbb{Z} \subset \mathbb{Q}(\sqrt{5})$  is  $\{(a + b\sqrt{5})/2 \mid 4 \mid (a^2 - 5b^2)\}$ .

**Exercise 16.7.** Let  $\xi$  be an  $n$ -th root of 1. Show that the integral closure of  $\mathbb{Z} \subset \mathbb{Q}(\xi)$  is  $\mathbb{Z}[\xi]$ .

**Exercise 16.8.** Suppose that  $A_i \subset B$  are normal in  $B$ , then show that  $A = \cap A_i \subset B$  is normal in  $B$ .

**Lemma 16.9.** *Let  $A \subset B$  be an inclusion of domains and  $S$  a multiplicatively closed subset of  $A$ . If  $\bar{A}$  is the integral closure of  $A$  in  $B$ , then  $S^{-1}\bar{A}$  is the integral closure of  $S^{-1}A$  in  $S^{-1}B$ .*

*Proof.* If  $b \in B$  is integral over  $A$  then it satisfies a monic polynomial  $x^n + a_1x^{n-1} + \dots + a_n \in A[x]$ . But then, for any  $s \in S$ , the element  $b/s \in S^{-1}\bar{A}$  satisfies the monic polynomial  $x^n + (a_1/s)x^{n-1} + \dots + (a_n/s^n) \in S^{-1}A[x]$  and so  $S^{-1}\bar{A}$  is contained in the integral closure of  $S^{-1}A$  in  $S^{-1}B$ .

Conversely, suppose that  $b/s$  is in the integral closure of  $S^{-1}A$  in  $S^{-1}B$ , then it satisfies a monic polynomial  $x^n + (a_1/s_1)x^{n-1} + \dots + (a_n/s_n) \in S^{-1}A[x]$ . But then  $bs_1 \cdots s_n$  satisfies the monic polynomial  $x^n + a_1s_1 \cdots s_ns/s_1x^{n-1} + \dots + a_n(s_1 \cdots s_ns)^n/s_n \in A[x]$ . Thus  $bs_1 \cdots s_n \in \bar{A}$  and hence  $b/s \in S^{-1}\bar{A}$ .  $\square$

**Definition 16.10.** *Let  $X$  be an affine variety, then  $X$  is **normal** if and only if  $A(X)$  is integrally closed.*

**Lemma 16.11.** *Let  $X$  be an affine variety, then  $X$  is normal if and only if the local rings  $\mathcal{O}_p$  are integrally closed rings for all points  $p \in X$ .*

*Proof.* Suppose that  $X$  is normal and hence  $A = A(X)$  is integrally closed in  $Q = Q(A)$ . Let  $\mathfrak{m} \subset A$  be the maximal ideal of  $p \in X$  then  $\mathcal{O}_p = S^{-1}A$  where  $S = A \setminus \mathfrak{m}$  is the corresponding multiplicative subset. Since  $X$  is normal  $A = \bar{A}$  is integrally closed. By Lemma 16.9,  $\mathcal{O}_p = S^{-1}A$  is integrally closed (in  $Q(S^{-1}A) = Q(A)$ ).

Conversely, suppose that the local rings  $\mathcal{O}_p$  are integrally closed rings for all points  $p \in X$ . Let  $b \in \bar{A} \subset Q$ , then as  $\bar{A} \subset \mathcal{O}_p \subset Q$  and  $\bar{\mathcal{O}}_p = \mathcal{O}_p$ , it follows that  $b \in \bigcap_{p \in X} \mathcal{O}_p = A$  (where for the last equality we used <sup>1</sup>).  $\square$

The advantage of the above lemma, is that the definition of normal can be made local and hence it applies to any variety.

**Definition 16.12.** *A quasi-projective variety  $X$  is **normal at a point**  $p \in X$  if  $\mathcal{O}_p$  is an integrally closed ring.  $X$  is **normal** if it is normal at every point  $p \in X$ .*

**Definition 16.13.** *Let  $X$  be an affine variety, then the normalization  $\nu : X^\nu \rightarrow X$  is defined by the inclusion of rings  $A(X) \rightarrow \overline{A(X)}$ . Note that  $X^\nu$  is normal and affine.*

**Lemma 16.14.** *Let  $Y$  be a normal affine variety and  $f : Y \rightarrow X$  be a dominant morphism of affine varieties, then  $f$  factors uniquely through the normalization  $X^\nu \rightarrow X$ .*

*Proof.* Let  $A = A(X)$  and  $B = A(Y)$ , then  $f : Y \rightarrow X$  is determined by the inclusion  $f^* : A \rightarrow B$  which induces an inclusion  $f^* : Q(A) \rightarrow Q(B)$ . It follows immediately that we have  $\bar{A} \subset \bar{B} = B$  and hence an induced map  $Y \rightarrow X^\nu \rightarrow X$ .  $\square$

**Lemma 16.15.** *Let  $X$  be an affine variety and  $\nu : X^\nu \rightarrow X$  its normalization. If  $U \subset X$  is an affine open subset, then  $\nu^{-1}(U) \rightarrow U$  is the normalization of  $U$ .*

*Proof.* By Lemma 16.14 there is a unique map  $U^\nu \rightarrow X^\nu$  factoring  $U^\nu \rightarrow U \rightarrow X$  and hence a map  $U^\nu \rightarrow \nu^{-1}(U)$ . On the other hand since  $\nu^{-1}(U)$  is normal, there is a unique map  $\nu^{-1}(U) \rightarrow U^\nu$ . By the uniqueness of these maps, it follows that  $\nu^{-1}(U) \cong U^\nu$ .  $\square$

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<sup>1</sup>ref

<sup>2</sup>ref

**Lemma 16.16.** *If  $X$  is a quasi-projective variety, then there exists a morphism  $\nu : X^\nu \rightarrow X$  such that for any affine subset  $U \subset X$ ,  $\nu^{-1}(U) \rightarrow U$  is the normalization of  $U$  (in particular  $\nu$  is uniquely determined by this property).*

*Proof.* Let  $X = \cup U_i$  where  $U_i$  is affine. Let  $\nu_i : U_i^\nu \rightarrow U_i$  be the corresponding normalizations, it suffices to show that there is a natural isomorphism between  $\nu_i^{-1}(U_{ij})$  and  $\nu_j^{-1}(U_{ij})$ . By Lemma 16.14  $\square$

Next we compute an interesting example. Let  $X = \mathbb{V}(y^2 - x^2(x+1)) \subset \mathbb{C}^2$ . we wish to compute the normalization  $X^\nu$ . Let  $t = y/x \in Q(A)$  where  $A = A(X) = \mathbb{C}[x, y]/(y^2 - x^2(x+1))$ . Then  $y/x \in \bar{A}$  since it satisfies the monic polynomial  $t^2 - (x+1) \in A[t]$ . Note that  $t^2 = (y/x)^2 = x+1$  and so  $x = t^2 - 1$  and  $y = tx = t(t^2 - 1)$  and so there is an inclusion  $A \subset \mathbb{C}[t] \subset Q(A)$  given by  $x \rightarrow t^2 - 1$  and  $y \rightarrow t(t^2 - 1)$ . Clearly  $\mathbb{C}[t]$  is integrally closed and hence  $\bar{A} = \mathbb{C}[y/x] \cong \mathbb{C}[t]$ .

**Exercise 16.17.** Let  $X = \mathbb{V}(xy) \subset \mathbb{C}^2$  compute the normalization  $\nu : X^\nu \rightarrow X$ .

**Exercise 16.18.** Let  $X = \mathbb{V}(x^2 - y^3) \subset \mathbb{C}^2$  compute the normalization  $\nu : X^\nu \rightarrow X$ .

**Exercise 16.19.** Let  $X = \mathbb{V}(x^2 - y^4) \subset \mathbb{C}^2$  compute the normalization  $\nu : X^\nu \rightarrow X$ .

Note that in all cases, we have that  $X^\nu$  is smooth. This is a general fact:

**Theorem 16.20.** *Let  $X$  be a one dimensional quasi projective variety (a curve), then  $X^\nu$  is smooth.*

**Corollary 16.21.** *Let  $X$  be a normal quasi-projective variety, then there is a closed subset  $Z \subset X$  of codimension  $\geq 2$  such that  $X \setminus Z$  is smooth.*

*Proof.* Exercise. This is not completely obvious. One approach is to mimic the proof of Theorem 16.20. See <http://www.math.utah.edu/~bertram/6030/Codim1.pdf> for details.  $\square$

Before proving Theorem 16.20, we make a brief detour.

**Definition 16.22.** A **discrete valuation** is a (non-zero) function  $\nu : K^* \rightarrow \mathbb{Z}$  where  $K$  is a field, such that

- (1)  $\nu(ab) = \nu(a) + \nu(b)$  for all  $a, b \in K^*$ , and
- (2)  $\nu(a+b) \geq \min\{\nu(a), \nu(b)\}$ .

We let

$$A_\nu = \{k \in K^* | \nu(k) \geq 0\} \cup \{0\}$$

be the corresponding DVR (discrete valuation ring).

**Exercise 16.23.** Show that if  $p$  is a prime number and  $\nu : \mathbb{Q}^* \rightarrow \mathbb{Z}$  is defined by  $\nu(a/b) = \nu(a) - \nu(b)$  where  $\nu(a)$  is the maximum power of  $p$  dividing  $a$ , then  $\nu : \mathbb{Q}^* \rightarrow \mathbb{Z}$  is a DVR and  $A_\nu u = \mathbb{Z}_{(p)} = S^{-1}\mathbb{Z}$  where  $S = \mathbb{Z} \setminus (p)$ .

**Exercise 16.24.** For any  $c \in \mathbb{C}$  and  $f/g \in \mathbb{C}(x)$  let  $\nu(f/g)$  be the order of the zero or pole of  $f/g$  at  $c$  (so that if  $c = 0$ ,  $\nu(x^k) = k$ ). Show that  $\nu$  is a discrete valuation and  $A_\nu = \mathbb{C}[x]_{(x-c)}$ .

**Exercise 16.25.** For any  $f/g \in \mathbb{C}(x)$  let  $\nu(f/g) = \deg(g) - \deg(f)$ . Show that  $\nu$  is a discrete valuation and  $A_\nu = \mathbb{C}[x^{-1}]_{x^{-1}}$ . What happens if we let  $\nu(f/g) = \deg(f) - \deg(g)$ .



**Exercise 16.26.** Show that if  $\nu$  is a discrete valuation, then  $\nu(K^*) = d\mathbb{Z}$  for some  $d > 0$ . If we let  $\nu'(k) = \nu(k)/d$ , then show that  $\nu'(K^*) = \mathbb{Z}$ ,  $\nu'$  is a discrete valuation and  $A_{\nu'} = A_{\nu}$ .

**Lemma 16.27.** Let  $\nu$  be a discrete valuation (such that  $\nu(K^*) = \mathbb{Z}$ ), then  $\mathfrak{m} := \{k \in K^* \mid \nu(k) > 0\}$  is a principal ideal in  $A_{\nu}$ . Thus  $\mathfrak{m} = (\pi)$  where  $\nu(\pi) = 1$ . We say that  $\pi$  is the **uniformizing parameter**.

*Proof.* Let  $a \in \mathfrak{m}$  and  $b \in A_{\nu}$ , then  $\nu(ab) = \nu(a) + \nu(b) > 0$  and so  $ab \in \mathfrak{m}$ . If  $a, a' \in \mathfrak{m}$ , then  $\nu(a + a') \geq \min\{\nu(a), \nu(a')\} > 0$ . Thus  $\mathfrak{m}$  is an ideal in  $A_{\nu}$ . To check that it is principal, pick  $\pi \in \mathfrak{m}$  of minimal degree. Clearly  $\nu(\pi) = 1$  (as  $\nu(K^*) = \mathbb{Z}$ ). For any  $m \in \mathfrak{m}$  we write  $m = \pi \cdot b$  where  $b = m/\pi$ . Note that  $\nu(m/\pi) = \nu(m) - 1 \geq 0$  and so  $m/\pi \in A_{\nu}$  as required.  $\square$

**Exercise 16.28.** Show that if  $I \subset A_{\nu}$  is any ideal, then  $I = \mathfrak{m}^k = (\pi^k)$  for some  $k \geq 0$ . It follows that any DVR is Noetherian and in fact a UFD. Note that  $\mathfrak{m}$  is the unique prime/maximal ideal in  $A_{\nu}$ .

**Lemma 16.29.** Let  $(A, \mathfrak{m})$  be a Noetherian local ring (so that  $A$  is a domain and  $\mathfrak{m}$  is the unique maximal ideal). If the Krull dimension of  $A$  is one (i.e.  $\mathfrak{m}$  is the unique prime ideal), then for any ideal  $0 \neq I \neq A$  we have  $\sqrt{I} = \mathfrak{m}$  and there is a unique  $n \in \mathbb{N}$  such that  $\mathfrak{m}^n \subset I$  but  $I \not\subset \mathfrak{m}^{n-1}$ .

*Proof.* We begin by observing that if  $a \in \mathfrak{m}$ , then  $A[a^{-1}] = K(A)$ . To this end, note that if  $J \subset A[a^{-1}]$  is a proper prime ideal, then  $J = P[a^{-1}]$  for some prime ideal  $P = J \cap A \subset A$ . By our assumption  $P = \mathfrak{m}$  and since  $a \in \mathfrak{m}$ ,  $1 = a/a \in J$  which is impossible.

We will now prove that  $\sqrt{I} = \mathfrak{m}$  for any ideal  $0 \neq I \subset A$ . It suffices to show  $\mathfrak{m} \subset \sqrt{I}$ . Pick  $m \in \mathfrak{m}$  and  $0 \neq a \in \sqrt{I} \setminus \mathfrak{m}$ , then (by what we have seen above)  $a^{-1} = b/m^n$  for some  $b \in A$  and  $n \geq 0$ . Note that in fact  $n > 0$  since  $a \notin \mathfrak{m}$ . But then  $m^n = ab$  and so  $m \in \sqrt{I}$ .

Finally, it suffices to show that  $\mathfrak{m}^n \subset I$  for some  $n$ . This is where we use the fact that  $A$  is Noetherian and so  $\mathfrak{m} = (m_1, \dots, m_k)$ . Since  $\sqrt{I} = \mathfrak{m}$ , there are integers  $n_i$  such that  $m_i^{n_i} \in I$ , but then if  $n = 1 - k + \sum n_i$ , it is easy to see that  $m^n \in I$  for any  $m \in \mathfrak{m}$ . Notice in fact that  $m = \sum m_i a_i$  and  $m^n = (a_1 m_1 + \dots + a_k m_k)^n$ . Expanding this out, we see that every term contains a power  $m_i^{n_i}$  for some  $1 \leq i \leq k$  and hence belongs to  $I$ .  $\square$

We have the following important result.

**Theorem 16.30.** Let  $(A, \mathfrak{m})$  be an integrally closed Noetherian local ring of dimension 1, then  $A$  is a DVR.

*Proof.* We begin by proving that  $\mathfrak{m}$  is principal i.e.  $\mathfrak{m} = (\pi)$  for some  $\pi \in A$ . Pick any  $0 \neq a \in \mathfrak{m}$ . By Lemma 16.29, there is an integer  $n$  such that  $\mathfrak{m}^n \subset (a)$  but  $\mathfrak{m}^{n-1} \not\subset (a)$ . Pick  $b \in \mathfrak{m}^{n-1} \setminus (a)$ , then  $b/a \in K(A) \setminus A$  and  $(b/a)\mathfrak{m} \subset A$  (this is because  $b\mathfrak{m} \subset \mathfrak{m}^n \subset (a)$ ). We claim that however  $(b/a)\mathfrak{m} \not\subset \mathfrak{m}$ . Grant this for the time being, then there is  $\pi \in \mathfrak{m}$  such that  $(a/b)\pi = u \notin \mathfrak{m}$  and hence  $u$  is a unit. But then for any  $m \in \mathfrak{m}$ , we have  $(a/b)m = c \in A$  (as  $(b/a)\mathfrak{m} \subset A$ ) and so  $m = c\pi u^{-1} \in (\pi)$  as required.

To see the claim, suppose that  $(b/a)\mathfrak{m} \subset \mathfrak{m}$ , then we have an inclusion

$$A\left[\frac{a}{b}\right] \hookrightarrow a \cdot A\left[\frac{a}{b}\right] \subset \mathfrak{m}\left[\frac{a}{b}\right] \subset \mathfrak{m}.$$

Since  $A$  is Noetherian, it then follows that  $A[\frac{a}{b}]$  is a finitely generated  $A$  module. But since  $A$  is integrally closed,  $\frac{a}{b} \in A$  which is the required contradiction.<sup>3</sup>

<sup>3</sup>add refs

Let  $0 \neq a \in A$ , if  $a$  is not a unit, then  $a \in \mathfrak{m} = (\pi)$  and we write  $a = \pi a_1$ . If  $a_1$  is not a unit, then we write  $a_1 = \pi a_2$  and hence  $a = \pi^2 a_2$ . We claim that after finitely many iterations, we may write  $a = \pi^n a_n$  where  $a_n$  is a unit. If this were not the case, then we would have a sequence of ideals  $(a) \subset (a_1) \subset (a_2) \subset \dots$ . Since  $A$  is Noetherian, this sequence is eventually constant so that  $(a_n) = (a_{n+1})$ . Since  $a_n = a_{n+1}\pi$  we have  $a_{n+1} = a_{n+1}\pi b$  for some  $b \in A$ . But then  $1 = \pi b$  as  $A$  is a domain) and so  $\pi$  is a unit, which is impossible. Writing  $a = \pi^n a_n$  where  $a_n$  is a unit, we let  $\nu(a) = n$ .

We must check that  $\nu$  is well defined: If  $a = \pi^n a_n = \pi^m b$  where  $m \geq n$ , then  $a_n = \pi^{m-n} b$ . Since  $\pi$  is not a unit,  $m - n = 0$ .

We leave the check that  $\nu$  is a discrete valuation as an exercise.  $\square$

**Exercise 16.31.** Check that  $\nu$  is a discrete valuation.

*Proof of Theorem 16.20.* Replacing  $X$  by  $X^\nu$  we may assume that  $X$  is normal. Since  $X$  is smooth on some open subset, replacing  $X$  by an affine open subset, we may assume that  $X$  is affine and  $x \in X$  is the unique singular point. Note that  $\mathbb{C}[x]$  is integrally closed and hence so is  $\mathbb{C}[x]_{\mathfrak{m}_x} C$ . But then, by Theorem 16.30,  $\mathbb{C}[x]_{\mathfrak{m}_x} C$  is a DVR. Let  $\pi$  be the uniformizing parameter so that  $(\pi) = \mathfrak{m}_x \mathbb{C}[x]_{\mathfrak{m}_x} C$ . After possibly replacing  $X$  by a smaller open subset containing  $x$ , we may assume that  $\pi \in \mathbb{C}[x]$  and  $\mathfrak{m}_x = (\pi)$ . But then  $\dim \mathfrak{m}_x / \mathfrak{m}_x^2 = 1 = \dim X$  and so  $x \in X$  is smooth.  $\square$

## 17. CONICS

We begin by investigating conics in  $\mathbb{C}^2$  i.e. subsets of  $\mathbb{C}^2$  defined by a polynomial of degree 2 say

$$q(x, y) = ax^2 + by^2 + 2cxy + 2dx + 2ey + f = 0.$$

It is often convenient to view this in matrix form

$$q(x, y) = \begin{pmatrix} x & y & 1 \end{pmatrix} \begin{pmatrix} a & c & d \\ c & b & e \\ d & e & f \end{pmatrix} \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} x & y & 1 \end{pmatrix} M \begin{pmatrix} x \\ y \\ 1 \end{pmatrix}.$$

Conics define familiar shapes in  $\mathbb{R}^2$  such as parabolas (eg.  $y = x^2$ ), ellipses (eg.  $x^2 + 3y^2 = 2$ ) and hyperbolas (eg.  $xy = 1$ ). Any two of these shapes can be identified via a change of coordinates, eg. the ellipse  $2(3x + y - 5)^2 + 3(x - 2y - 7)^2 = 3$  can be identified with the circle  $x^2 + y^2 = 3$  by letting  $x \rightarrow \sqrt{2}(3x + y - 5)$  and  $y \rightarrow \sqrt{3}(x - 2y - 7)$ . In other words we are allowed to act on  $(x, y)$  by an invertible linear transformation and to translate by arbitrary points in  $\mathbb{C}^2$ . Finally, we allow ourselves to rescale the constant term so that we can identify the circle  $x^2 + y^2 = 3$  with the unit circle  $x^2 + y^2 = 1$ .

It is then easy to see that after a linear transformation we may assume that the degree two term of  $q(x, y)$  is of the form

$$x^2 + y^2, \quad x^2, \quad 0$$

depending on the rank of the matrix  $\begin{pmatrix} a & c \\ c & b \end{pmatrix}$ . Suppose that we are in the rank 2 case so that we may write

$$q(x, y) = x^2 + y^2 + 2dx + 2ey + f = 0,$$

then, replacing  $x$  by  $(x - d)$  and  $y$  by  $(y - e)$  (and  $f$  by  $f - d^2 - e^2$ ) we obtain an expression of the form  $q(x, y) = x^2 + y^2 + f = 0$ . There are 2 cases to consider according to the rank of  $M$ .

If the rank of  $M$  is 2 then we have the equation  $x^2 + y^2 = 0$  and if the rank of  $M$  is 3, then after rescaling the constant term we have the equation  $x^2 + y^2 + 1 = 0$  (note that since we are over the complex numbers, it does not matter if  $f$  is positive or negative;  $x^2 + y^2 + 1 = 0$  is equivalent to  $x^2 + y^2 - 1 = 0$  via replacing  $(x, y)$  with  $(ix, iy)$ ). If the rank of  $M$  is 1, then after changing coordinates, we may assume that  $q(x, y) = x^2 + 2dx + 2ey + f = 0$ . Replacing  $x$  by  $x + d$  we may assume that  $q(x, y) = x^2 + 2ey + f$ . If  $e \neq 0$  (i.e. if the rank of  $M$  is 3), then replacing  $y$  by  $2ey + f$ , we may assume that  $q(x, y) = x^2 + y$  and otherwise if  $e = 0$  (so that the rank of  $M$  is 2 or 1) then we may assume that  $q(x, y) = x^2 + 1$  or  $q(x, y) = x^2$ . All in all we have the following cases:

- (1) Ellipse:  $x^2 + y^2 + 1 = 0$
- (2) Two lines meeting at a point:  $x^2 + y^2 = 0$  (note that this equation is equivalent to  $(x + iy)(x - iy) = 0$ ).
- (3) Parabola:  $x^2 + y = 0$
- (4) Two parallel lines:  $x^2 + 1 = 0$
- (5) A double line  $x^2 = 0$ .

**Remark 17.1.** *Note that over the real numbers, the classification is different since  $x^2 + y^2 = 1$  and  $x^2 + y^2 = -1$  have different zero sets. In this case the classification depends on the ranks of minors of  $M$  as well as their signature.*

The above classification is somewhat cumbersome because we must treat the coordinates  $x, y$  differently from the constant term (in particular we care about the rank of  $\begin{pmatrix} a & c \\ c & b \end{pmatrix}$  as well as the rank of  $M$ ). If we however consider the projective closure of these conics, i.e. conics in  $\mathbb{P}^2$ , then the situation is straightforward. A conic in  $\mathbb{P}^2$  is defined by a homogeneous polynomial of degree 2 say

$$q(x, y, z) = ax^2 + by^2 + 2cxy + 2dxz + 2eyz + fz^2 = 0.$$

Two conics are equivalent if they are obtained by an invertible base change i.e. by a matrix  $A \in GL(3, \mathbb{C})$ . If we replace  $(x, y, z)$  by coordinates  $(x, y, z)^t = A(x', y', z')^t$ , then we obtain a new equation

$$(x \ y \ z) M \begin{pmatrix} x \\ y \\ z \end{pmatrix} = (x' \ y' \ z') A^t M A \begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}$$

In other words, the base change replaces  $M$  by  $A^t M A$ . Since  $M$  is symmetric, we may pick a base change that diagonalizes  $M$  and so we may assume that  $q(x, y, z) = ax^2 + by^2 + fz^2$ . If  $a \neq 0$ , then replacing  $x$  by  $\sqrt{a}x$ , we may assume  $a = 1$  (and similarly for  $b$  and  $f$ ). Finally, since we may permute the variables, we only have three cases:

- (1) Rank  $M$  is 3:  $x^2 + y^2 + z^2 = 0$  an ellipse.
- (2) Rank  $M$  is 2:  $x^2 + y^2 = 0$  two lines.
- (3) Rank  $M$  is 1:  $x^2 = 0$  a double line.

Notice that with this description, we have lumped together the affine Ellipse and Parabola cases: a parabola is just an ellipse which is tangent to the line at infinity; for example  $y = x^2$  in homogeneous coordinates becomes  $zy - x^2 = 0$ . the line at infinity is defined by  $z = 0$ . The intersection point is  $[0 : 1 : 0]$  and in local coordinates ( $u = x/y, v = z/y$ ) on the chart  $y \neq 0$  we have  $v = u^2$  which is tangent to the line at infinity  $v = 0$ . We have also lumped

together two lines meeting at a point and two parallel lines (because two parallel lines are simply projective lines meeting at infinity).

**Exercise 17.2.** Study the classification of conics over  $\mathbb{P}_{\mathbb{R}}^2$ .

**Exercise 17.3.** What are the possible quadrics in  $\mathbb{P}^3$  (i.e. solutions of a homogeneous equation of degree 2). Show that any nonsingular quadric is isomorphic to  $\mathbb{V}(xy - zw) \subset \mathbb{P}^3$ .

**Exercise 17.4.** Show that any nonsingular quadric in  $\mathbb{P}^3$  is isomorphic to  $\mathbb{P}^1 \times \mathbb{P}^1$ .

## 18. SMOOTH VARIETIES

We begin with a few examples. Let  $y = x^2$  be a parabola, we wish to understand tangent lines to  $y = x^2$  at any given point  $(a, a^2)$ . From calculus, we know that the correct answer in parametric form is the line

$$t \rightarrow (a + t, a^2 + 2at) = (a + a^2) + t(1, 2a).$$

This reflects the fact that  $dy/dx = 2x$  and hence the slope at  $(a, a^2)$  is  $2a$ . If we evaluate  $x^2 - y$  along the tangent line, we get

$$(a + t)^2 - (a^2 + 2at) = t^2$$

so that the induced equation vanishes to order at least 2 at the origin. In fact this property (vanishing to order at least two at the origin) characterizes the tangent lines. Consider for simplicity the case  $a = 0$  i.e. the point  $(0, 0)$ . A line through the origin is given parametrically by  $t \rightarrow (a_1 t, a_2 t)$  where  $a_1, a_2$  are not both 0. We then restrict our equation  $x^2 - y$  to this line we obtain

$$(a_1 t)^2 - a_2 t$$

which vanishes to order 2 at the origin if and only if  $a_2 = 0$ . From a slightly different point of view, one can consider the gradient of  $f = x^2 - y$ , given by  $\nabla f = (2x, -1)$  so that the tangent line at  $(a, a^2)$  is given by

$$0 = \nabla f_{(a, a^2)} \cdot (x - a, y - a^2) = (2a, -1) \cdot (x - a, y - a^2) = 2a(x - a) - (y - a^2).$$

As a second example, consider the sphere  $x^2 + y^2 + z^2 = 1$  and the tangent plane at  $(0, 0, 1)$ . Since  $\nabla(f)(0, 0, 1) = (0, 0, 2)$ , the tangent plane is given by  $2(z - 1) = 0$ . The lines contained in this plane are of the form

$$(0, 0, 1) + t(a, b, 0).$$

Once again, restricting the equation  $x^2 + y^2 + z^2 = 1$  to the line  $(0, 0, 1) + t(a, b, 0)$  we obtain

$$t^2(a^2 + b^2) = 0$$

so that  $t = 0$  is a zero of order 2. On the other hand, restricting the equation  $x^2 + y^2 + z^2 = 1$  to the line  $(0, 0, 1) + t(a, b, c)$  where  $c \neq 0$  we obtain

$$t^2(a^2 + b^2 + c^2) + 2tc = 0$$

so that  $t = 0$  is a zero of order 1. This leads us to the following definition.

**Definition 18.1.** Let  $a \in X = \mathbb{V}(f_1, \dots, f_r) \subset \mathbb{C}^n$  be a point on an affine variety (or an algebraic set) and  $l(t) = a + tv$  for some  $0 \neq v \in \mathbb{C}^n$  be a line, then we say that a line  $l(t)$  is **tangent** to  $X$  at the point  $a \in X$  if  $l(0) = a$  and  $t = 0$  is a zero of multiplicity at least 2 for  $f_i(l(t))$  for all  $1 \leq i \leq r$ . If the minimum multiplicity of  $t = 0$  in  $f_i(l(t))$  is  $m + 1$ , then we say that  $l(t)$  is **tangent to order  $m$**  at  $a \in X$ . The **tangent space** to  $X$  at  $a$  is the

subspace  $T_a X \subset \mathbb{C}^n$  given by the union of the translates through the origin of all the lines tangent to  $X$  at  $a$ .

We consider next two more examples.

- (1)  $X = \mathbb{V}(x^2 - y^3)$  at  $(0, 0, 0)$ . If  $l = t(a, b)$ , then  $f(l(t)) = t^2(a^2 + b^3 t)$  has a zero of order 2 at  $t = 0$  for all  $a \neq 0$  and a zero of order 3 if  $a = 0$ . therefore  $T_{(0,0)} X = \mathbb{C}^2$ . Note that  $\nabla(x^2 - y^3)_{(0,0)} = (2x, -3y^2)_{(0,0)} = (0, 0)$ .
- (2)  $X = \mathbb{V}(x^2 + y^2 + y^3)$ . If  $l = t(a, b)$ , then  $f(l(t)) = t^2(a^2 + b^2 + b^3 t)$  which has a zero of order 2 at 0 if  $a \neq \pm ib$  and a zero of order 3 if  $a = \pm ib$ .

**Proposition 18.2.** *The tangent space  $T_a X$  does not depend on the choice of the basis of the ideal.*

*Proof.* Assume  $a = (0, \dots, 0)$ . Suppose that  $(f_1, \dots, f_r) = (g_1, \dots, g_s)$ . It suffices to show that if  $l(t) = tv$  is a line such that  $f_i(tv)$  is divisible by  $t^2$  for  $i = 1, \dots, r$ , then  $g_k(tv)$  is divisible by  $t^2$  for any  $1 \leq k \leq s$ . However note that  $g_k \in (f_1, \dots, f_r)$  and so  $g_k = \sum h_i f_i$  where  $h_1, \dots, h_r \in \mathbb{C}[x_1, \dots, x_n]$ . But then

$$g_k(tv) = \sum h_i(tv) f_i(tv)$$

is divisible by  $t^2$  since each  $f_i(tv)$  is divisible by  $t^2$ . □

**Proposition 18.3.** *Let  $X = \mathbb{V}(f_1, \dots, f_r)$  be an affine variety and  $p \in X$ , then a line  $l = p + tv$  is tangent to  $X$  at  $p$  if and only if*

$$l \subset \mathbb{V}((\nabla f_1)_p(x - p), \dots, (\nabla f_r)_p(x - p))$$

where  $(x - p) = (x_1 - p_1, \dots, x_r - p_r)$  and so  $(\nabla f_j)_p(x - p) = \sum \frac{\partial f_j}{\partial x_i}(p) \cdot (x_i - p_i)$ .

*Proof.* Assume that  $p = (0, \dots, 0)$ , then

$$f_i - \nabla f_i(p) \cdot x \in (x_1, \dots, x_n)^2$$

and so  $t^2$  divides  $f_i(tv)$  if and only if  $t^2$  divides  $\nabla f_i(p) \cdot tv = t(\nabla f_i(p) \cdot v)$  which is equivalent to  $\nabla f_i(p) \cdot v = 0$ . □

**Definition 18.4.** *Let  $p \in X$  be a point on an affine variety, then we say that  $p \in X$  is **smooth** if  $\dim T_p X = \dim_p X$ . If  $p \in X$  is not smooth, then  $\dim T_p X > \dim_p X$  and we say that  $p \in X$  is a **singular point**.*

**Remark 18.5.** *Note that we have to check that the definition is independent of the particular embedding of  $X \subset \mathbb{C}^N$  and we have not yet shown that for non smooth points we have the inequality  $\dim T_p X > \dim_p X$  (or equivalently that  $\dim T_p X \geq \dim_p X$  holds for any  $p \in X$ ).*

**Proposition 18.6.** *Let  $X = \mathbb{V}(f_1, \dots, f_r) \subset \mathbb{C}^n$  be a  $d$ -dimensional affine variety, then  $\text{Sing}(X)$  is a proper closed subvariety defined by the vanishing of the  $(n - d) \times (n - d)$  minors of the matrix*

$$[\nabla f_1, \dots, \nabla f_r].$$

*Proof.* For any  $p \in X$ , we have seen that the tangent space is defined by

$$T_p X = \mathbb{V}(\nabla f_1(p) \cdot (x - p), \dots, \nabla f_r(p) \cdot (x - p)).$$

Writing this in matrix form, we obtain that  $T_p X$  is determined by the equation

$$[\nabla f_1(p), \dots, \nabla f_r(p)]^t \cdot (x - p) = 0$$

where  $[\nabla f_1(p), \dots, \nabla f_r(p)]^t$  denotes the  $r \times n$  matrix with entries  $([\nabla f_1(p), \dots, \nabla f_r(p)]^t)_{i,j} = \frac{\partial f_i}{\partial x_j}(p)$  and  $x - p = (x_1 - p_1, \dots, x_r - p_r)^t$  is a column vector. By linear algebra, we have that

$$\dim(T_p X) = \dim \mathbb{C}^n - \text{rank}([\nabla f_1(p), \dots, \nabla f_r(p)])$$

and so  $\dim(T_p X) > \dim X$  if and only if  $\text{rank}([\nabla f_1(p), \dots, \nabla f_r(p)]) < n - d$ . By linear algebra, this last condition is equivalent to showing that all  $(n - d) \times (n - d)$  minors vanish.

We must now show that there is at least one non singular point on  $X$ . If  $X = V(f)$  is a hypersurface (i.e.  $X$  is defined by an irreducible polynomial  $f$ ), and every point  $p \in X$  is singular, then we would have that  $\partial f / \partial x_i$  vanishes at every point of  $X$  and so that  $f$  divides  $\partial f / \partial x_i$ . But this is impossible for degree reasons. To conclude the general case, it suffices to observe that every variety contains an open subset that is isomorphic to a hypersurface (see Proposition 21.3). □

**Exercise 18.7.** Show that if  $p \in X \cap Y$  where  $X, Y$  are varieties not containing each other, then  $p \in X \cup Y$  is a singular point.

## 19. INTRINSIC DEFINITION OF THE TANGENT SPACE

Let  $\bar{0} \in \mathbb{C}^n$  be the origin and  $M = (x_1, \dots, x_n) \subset \mathbb{C}[x_1, \dots, x_n]$  the corresponding maximal ideal, then we can identify the tangent space  $T_{\bar{0}}(\mathbb{C}^n)$  with  $M/M^2$ . To see this, consider the map

$$d : M \rightarrow (\mathbb{C}^n)^* := \text{linear functions on } \mathbb{C}^n$$

defined by  $d(f) = f_1$  where  $f = f_1 + f_2 + \dots$  and  $f_i \in \mathbb{C}[x_1, \dots, x_n]_i$  are homogeneous of degree  $i$ . Note that  $df = \sum \partial f / \partial x_i(0, \dots, 0) \cdot x_i = \nabla f(0, \dots, 0) \cdot x$ . It is also easy to see that  $d$  is surjective, since if  $L = \sum a_i x_i \in (\mathbb{C}^n)^*$ , then  $d(L) = L$ . Therefore we have shown that

$$d : M/M^2 \rightarrow (\mathbb{C}^n)^*$$

is an isomorphism. We claim that this isomorphism holds for any point of any affine variety.

**Theorem 19.1.** Let  $P \in X \subset \mathbb{C}^n$  be a point on an affine variety and  $\mathfrak{m}_P \subset A(X)$  be the maximal ideal of  $P \in X$ , then  $T_P(X) \cong (\mathfrak{m}_P / \mathfrak{m}_P^2)^*$ .

*Proof.* Recall that  $T_P(X) \subset \mathbb{C}^n$  is defined by the linear terms  $g_1$  of polynomials  $g \in \mathbb{I}(X)$ . (We may assume that the coordinates are chosen so that  $P = (0, \dots, 0)$ .) If  $(\mathbb{C}^n)^*$  is the dual space, then  $g_1 \in (\mathbb{C}^n)^*$  are the linear equations defining  $T_P(X)$  and  $T_P(X) = \mathbb{V}(\{g_1\}_{g \in \mathbb{I}(X)})$ . Notice that  $\mathfrak{m}_P = M / \mathbb{I}(X)$ . Consider the natural map

$$D = (i^* \circ d) : M \rightarrow (\mathbb{C}^n)^* \rightarrow (T_P(X))^*$$

where  $d : M \rightarrow (\mathbb{C}^n)^*$  was defined above and  $i^*$  is the dual map to  $i : T_P(X) \rightarrow \mathbb{C}^n$  (cf. Exercise 19.2). It is easy to see that  $D$  is surjective (cf. Exercise 19.3). We must therefore understand the kernel of  $D$  i.e. the set of functions  $f \in M$  such that  $f_1|_{T_P(X)} = 0$ . This condition is equivalent to saying that  $f_1 \in (\{g_1\}_{g \in \mathbb{I}(X)})$  or that  $f_1 = \sum a^i g_1^i$  for some  $g^i \in \mathbb{I}(X)$  and  $a^i \in \mathbb{C}$ . In other words  $f - \sum a^i g^i \in M^2$  for some  $g^i \in \mathbb{I}(X)$  i.e.  $f \in M^2 + \mathbb{I}(X)$ . But then

$$(T_P(X))^* = \frac{M}{M^2 + \mathbb{I}(X)} = \frac{\mathfrak{m}_P}{\mathfrak{m}_P^2}$$

where we have used the fact that  $\mathfrak{m}_P = M / \mathbb{I}(X)$  and Exercise 19.4. □

**Exercise 19.2.** Show that  $((\mathbb{C}^n)^*)^* = \mathbb{C}^n$  and show that if  $V$  is an infinite dimensional vector space, then  $(V^*)^* \neq V$ .

**Exercise 19.3.** Given a linear map of vector spaces  $f : V \rightarrow W$ , show that there is a naturally defined linear map of vector spaces  $f^* : W^* \rightarrow V^*$  and that if  $f^* = 0$ , then  $f = 0$ . If  $V, W$  are finite dimensional, then show that  $f^*$  is injective (resp surjective) if and only if  $f$  is surjective (resp. injective).

**Exercise 19.4.** Let  $M, I \subset R$  be ideals. Show that  $M/(M^2 + I) = (M/I)/(M/I)^2$ .

## 20. RESOLUTION OF SINGULARITIES

Recall that if  $X$  and  $Y$  are quasi-projective varieties, then  $X$  and  $Y$  are birational if there exists a rational map  $f : X \dashrightarrow Y$  and open subsets  $U \subset X$  and  $V \subset Y$  such that  $f|_U : U \rightarrow V$  is an isomorphism. Equivalently, we have seen that two quasi projective varieties are birational if and only if their function fields are isomorphic. Birational geometry is the study of quasi-projective varieties up to birational isomorphism. One of the most important results in birational geometry is Hironaka's resolution of singularities.

**Theorem 20.1.** Let  $X$  be any quasi projective variety, then there exists a projective morphism  $\mu : X' \rightarrow X$  such that  $X'$  is smooth and if  $X_0 = X \setminus \text{Sing}(X)$  and  $X'_0 = \mu^{-1}(X_0)$ , then  $\mu|_{X'_0} : X'_0 \rightarrow X_0$  is an isomorphism. In particular  $\mu$  is birational.

**Remark 20.2.** Hironaka's theorem is a very deep result whose proof is beyond the scope of these notes. We will however illustrate it with several examples and prove it in dimension 1.

It turns out that the resolution  $\mu : X' \rightarrow X$  can be constructed rather explicitly in terms of certain maps known as blow ups. We begin by considering the blow up of  $\mathbb{C}^n$  at the origin. Let

$$B = \{(x, y) \in \mathbb{C}^n \times \mathbb{P}^{n-1} | x \in y\} \subset \mathbb{C}^n \times \mathbb{P}^{n-1},$$

and the morphism  $\mu : B \rightarrow \mathbb{C}^n$  given by  $\mu(x, y) = x$ . Notice that we are identifying  $\mathbb{P}^{n-1}$  with the set of lines through the origin in  $\mathbb{C}^n$ . Note that  $B$  is a quasi-projective variety defined by equations  $x_i y_j = x_j y_i$  for all  $1 \leq i, j \leq n$ . It is clear that  $B$  is a quasi-projective variety and  $\mu$  is a projective morphism. If  $x \neq (0, \dots, 0)$  it follows that  $\mu^{-1}(x) = (x, [x])$  and  $E := \mu^{-1}((0, \dots, 0)) = (0, \dots, 0) \times \mathbb{P}^{n-1}$ . It follows easily that  $\mu : (B \setminus E) \rightarrow \mathbb{C}^n \setminus \{(0, \dots, 0)\}$  is an isomorphism. One should think of  $E$  as parametrizing the tangent directions to  $\mathbb{C}^n$  at the origin. We will now begin to investigate the effects of blowing up on affine varieties.

**Definition 20.3.** Let  $p \in X \subset \mathbb{C}^n$  be a point on an affine variety and  $\mu : B \rightarrow \mathbb{C}^n$  be the blow up of  $\mathbb{C}^n$  at  $p$ , then we define  $B_p(X)$  the **blow up of  $X$  at  $p$**  by taking the closure (in the Zariski topology) of  $\mu^{-1}(X \setminus \{p\})$ . In other words

$$B_p(X) := \overline{\mu^{-1}(X \setminus \{p\})}.$$

Note that  $\mu^{-1}(B_p \setminus \{p\})$  is isomorphic to  $B_p \setminus \{p\}$  and  $\mu^{-1}(p) = E \cap B_p(X)$ . We now illustrate the procedure with several examples.

- (1) **Lines:** Let  $X = \mathbb{V}(ax_1 + bx_2) \subset \mathbb{C}^2$  be a line (so that  $a, b \in \mathbb{C}$  are not both 0). Recall that  $X \subset \mathbb{C}^2 \times \mathbb{P}^1$  is defined by the equation  $x_1 y_2 = x_2 y_1$  where  $x_1, x_2$  are coordinates on  $\mathbb{C}^2$  and  $y_1, y_2$  are homogeneous coordinates on  $\mathbb{P}^1$ . Recall also that

we may cover  $\mathbb{P}^1$  by two affine charts  $U_2 = \{y_2 \neq 0\} \cong \mathbb{C}$  with coordinate  $t_2 = y_1/y_2$  and  $U_1 = \{y_1 \neq 0\} \cong \mathbb{C}$  with coordinate  $t_1 = y_2/y_1$ . Focusing on the first chart  $X_1 = X \cap (\mathbb{C}^2 \times U_1)$  is defined by  $x_1 t_1 = x_2$  and  $E_1 = E \cap \mathbb{C}^2 \times U_1$  is defined by the equation  $x_1 = 0$ . The equation of the line then becomes

$$ax_1 + bx_2 = x_1(a + bt_1).$$

This means that  $\mu^{-1}(X)$  has two irreducible components, namely

$$\mu^{-1}(X) = E \cup B_p(X)$$

where  $B_p(X) \cap \mathbb{C}^2 \times U_1$  is defined by  $\mathbb{V}(a + bt_1, x_1 t_1 - x_2)$ . Note that the intersection  $B_p(X) \cap E$  is given by letting  $x_2 = 0$  and hence is given by the point  $(0, 0, -a/b)$ . In other words this intersection recovers the slope of the original line.

- (2) **Cusp:** Let  $X = \mathbb{V}(x_1^2 - x_2^3) \subset \mathbb{C}^2$  and  $B \rightarrow \mathbb{C}^2$  be the blow up at the origin  $(0, 0)$ . We follow the above computation on  $\mathbb{C}^2 \times U_1$  where we have  $x_1 t_1 = x_2$  so that

$$x_1^2 - x_2^3 = x_1^2(t_1^2 - x_2).$$

This means that

$$\mu^{-1}(X) = 2E + B_p(X)$$

where  $B_p(X)$  intersects  $E$  at the point  $(0, 0, 0) \in \mathbb{C}^2 \times U_1$ . In local coordinates  $x_2, t_1$  we have that  $B_p(X) = \mathbb{V}(t_1^2 - x_2)$  is a parabola tangent to  $E$ . Note that we have used "2E" to record the fact that  $\mu^{-1}(X)$  vanishes to second order along  $E$ ; in set theoretic terms we of course have  $\mu^{-1}(X) = E + B_p(X)$ .

- (3) **Node:** Let  $X = \mathbb{V}(x_1^2 - x_2^3 - x_2^2) \subset \mathbb{C}^2$ . Proceeding as above we have

$$x_1^2 - x_2^3 - x_2^2 = x_2^2(t_1^2 - x_2 - 1)$$

so that

$$\mu^{-1}(X) = 2E + B_p(X)$$

where  $B_p(X)$  intersects  $E$  at the points  $(0, 0, \pm 1) \in \mathbb{C}^2 \times U_1$ . In local coordinates  $x_2, t_1 - 1$  (near  $(0, 1)$ ) we have that  $B_p(X) = \mathbb{V}(u(t_1 - 1) - x_2)$  where  $u = t_1 + 1$  is non-zero on a neighborhood of  $(0, 1)$  and so  $B_p(X)$  intersects  $E$  transversely at the points  $(0, 0, \pm 1)$ .

- (4) Consider a cone  $\mathbb{V}(x_1^2 + x_2^2 - x_3^2) \subset \mathbb{C}^3$  and blow up the origin  $p = (0, 0, 0)$ . We now have three affine charts  $\mathbb{P}^2 = U_1 \cup U_2 \cup U_3$  where  $U_i = \{y_i \neq 0\}$ . Computing on  $U_1$  we let  $t_2 = y_2/y_1$  and  $t_3 = y_3/y_1$  so that

$$B_p(\mathbb{C}^3) \cap \mathbb{C}^3 \times U_1 = \mathbb{V}(x_2 - t_2 x_1, x_3 - t_3 x_1)$$

and then  $\mu^{-1}(X)$  is defined by

$$x_1^2 - x_2^2 - x_3^2 = x_1^2(1 + t_2^2 + t_3^2)$$

so that  $\mu^{-1}(X) = 2E + B_p(X)$  where  $B_p(X)$  is smooth. A little reflection shows that  $B_p(X) \cap E = \mathbb{V}(x_1^2 + x_2^2 - x_3^2) \subset \mathbb{P}^2$  is a smooth circle and the map  $B_p(X) \rightarrow X$  is a resolution of singularities as in Hironaka's theorem.

- (5) This should not lead one to expect that one blow up always resolves a singularity. For example if  $X = \mathbb{V}(y^2 - x^2 y + y^4) \subset \mathbb{C}^2$ , then letting  $y = tx$  we obtain

$$y^2 - x^2 y + y^4 = x^2(t^2 - xt + t^4 x^2) = x^2 t(t - x + t^3 x^2)$$

and so one sees that  $B_p(X)$  is defined locally by  $\mathbb{V}(t(t - x + t^3 x^2))$  which is clearly singular at  $q = (0, 0)$  (in fact it corresponds to two curves meeting transversely). To



resolve the singularities of  $X$  it is therefore necessary to perform a second blow up  $B_q(B_p(X)) \rightarrow X$ .

- (6) It is easy to check that if  $X = \mathbb{V}(f) \subset \mathbb{C}^n$  is a hypersurface and  $B \rightarrow \mathbb{C}^n$  is the blow up of a point  $p \in X$ , then  $B_p(X) \cap E = \mathbb{V}(f_d) \subset \mathbb{P}^{n-1}$  is the projective hypersurface defined by the leading term of the Taylor expansion of  $f$  at  $p$ . For example if  $p = (0, \dots, 0)$  is the origin, the  $f - f_d \in (x_1, \dots, x_n)^{d+1}$  (i.e.  $f = f_d +$  higher order terms).

**Exercise 20.4.** Define the blow up of  $\mathbb{C}^{n+r}$  along a subspace  $\mathbb{C}^n$  as follows. Pick coordinates  $x_0, \dots, x_{n+r}$  such that  $\mathbb{C}^n = \mathbb{V}(x_{n+1}, \dots, x_{n+r})$ . Then consider the subset  $X \subset \mathbb{C}^{n+r} \times \mathbb{P}^{r-1}$  defined by the equations  $x_{n+i}y_j - x_{n+j}y_i$  for  $1 \leq i, j \leq r$ . Use this definition to define the blow up of a smooth  $n + r$ -dimensional variety along a smooth  $n$ -dimensional subvariety.

**Exercise 20.5.** Show that the blow up of  $\mathbb{P}^{n+r}$  along a linear subspace  $L \cong \mathbb{P}^n$  is isomorphic to the graph of the corresponding projection  $\mathbb{P}^{n+r} \dashrightarrow \mathbb{P}^r$ .

4

## 21. SMOOTH PROJECTIVE VARIETIES AND GENERIC SMOOTHNESS

**Theorem 21.1.** Let  $X$  be a smooth projective variety of dimension  $d$ , then  $X$  is isomorphic to a subvariety of  $\mathbb{P}^{2d+1}$ .

*Proof.* By assumption,  $X \subset \mathbb{P}^n$  for some  $n > 0$ . If  $n > 2d + 1$ , then the secant variety  $S(X) \subset \mathbb{P}^n$  given by the closure of the union of all lines through two points of  $X$  has dimension  $\leq 2d + 1$  and the tangent variety  $T(X) \subset \mathbb{P}^n$  given by the union of all tangent lines has dimension  $\leq 2d$  and hence we may pick a point  $p \in \mathbb{P}^n \setminus S(X) \cup T(X)$ . Consider the projection map  $\phi : \mathbb{P}^n \dashrightarrow \mathbb{P}^{n-1}$ . Since  $p \notin X$ , then  $\phi|_X : X \rightarrow \mathbb{P}^{n-1}$  is a morphism. Since  $p \notin S(X)$ ,  $\phi|_X$  is bijective and since  $p \notin T(X)$ , the differential of  $\phi|_X$  has maximal rank at each point  $x \in X$  and so  $X \rightarrow \phi(X)$  induces an isomorphism of tangent spaces.<sup>5</sup>  $\square$

**Exercise 21.2.** Show that the secant variety  $S(X) \subset \mathbb{P}^n$  is a variety of dimension  $\leq 2d + 1$  and the tangent variety  $T(X) \subset \mathbb{P}^n$  is a variety of dimension  $\leq 2d$ .

**Proposition 21.3.** Every projective variety is birational to a hypersurface.

*Proof.* Consider an embedding  $X \subset \mathbb{P}^n$ . Suppose that  $\dim X \leq n - 2$ . Let  $p \in \mathbb{P}^n \setminus X$ . Consider the projection map  $\phi : \mathbb{P}^n \dashrightarrow \mathbb{P}^{n-1}$ . If  $\phi|_X$  is not birational, then for any  $x \in X$ , the line  $\overline{px}$  through  $p$  and  $x$  intersects  $X$  at least in one other point, say  $y$ . Consider the map  $\pi : X \times X \times \mathbb{P}^1 \rightarrow S(X) \subset \mathbb{P}^n$ , then by what we observed above  $\dim \pi^{-1}(p) \geq d = \dim X$ . It follows that the set  $B \subset \mathbb{P}^n$  of points  $p$  such that the corresponding projection  $\phi|_X$  is not birational, has dimension  $\leq \dim(X \times X \times \mathbb{P}^1) - d = d + 1$ . Therefore we may pick a point  $p \in \mathbb{P}^n \setminus B$  and obtain a birational map  $\phi|_X : X \rightarrow X_1 \subset \mathbb{P}^{n-1}$ . Repeating this process  $n - d - 1$  times yields the proposition.  $\square$

**Theorem 21.4** (Bertini's Theorem). Let  $X \subset \mathbb{P}^n$  be a smooth projective variety and  $\hat{\mathbb{P}}^n$  be the dual projective variety parametrizing hyperplanes  $H \subset \mathbb{P}^n$ . Let  $W \subset \hat{\mathbb{P}}^n$  be the subset of planes such that  $H \cap X$  is smooth. Then  $W$  is a non-empty open subvariety of  $\hat{\mathbb{P}}^n$ .

<sup>4</sup>other examples. WU?

<sup>5</sup>Remark on why this is an isom.

*Proof.* Consider the subset  $B \subset X \times \hat{\mathbb{P}}^n$  consisting of pairs  $(x, H)$  such that  $x$  is a singular point of  $H \cap X$ . Since  $X$  is smooth, it is easy to see that for any  $x \in X$ ,  $H \cap x$  is singular if and only if  $T_x(X) \subset H$  and therefore the fibers of  $p : B \rightarrow X$  are projective spaces of dimension  $n - d - 1$  where  $d = \dim X$  (it is one condition to vanish at  $x$  and  $d$  conditions to contain  $T_x(X)$ ). Therefore, the dimension of  $B$  is  $n - d - 1 + d = n - 1$  and so if  $q : B \rightarrow \hat{\mathbb{P}}^n$  is the second projection, then  $q(B) \subset \hat{\mathbb{P}}^n$  is a proper subvariety and hence  $W = \hat{\mathbb{P}}^n \setminus q(B)$  is a non-empty open subvariety of  $\hat{\mathbb{P}}^n$ .  $\square$

**Exercise 21.5.** *Prove a version of Bertini's Theorem for varieties with isolated singularities.*

**Theorem 21.6** (Generic smoothness). *Let  $f : X \rightarrow Y$  be a morphism of complex projective varieties, then there exists a non-empty open subset  $X^0 \subset X$  such that  $f|_{X^0}$  is smooth. If moreover  $X$  is smooth, then there exists a non empty open subset  $Y^0 \subset Y$  such that for every  $y \in Y^0$ , the fiber  $X_y = f^{-1}(y)$  is smooth.*

*Proof.* Assume for simplicity that  $Y = \mathbb{C}$ . Let  $U \subset X$  be an affine open subset, then  $f$  is given by an element  $F \in \mathcal{O}_X(U)$  and since  $F$  is not constant, the differential  $dF : T_x(X) \rightarrow \mathbb{C} = T_y(Y)$  is surjective on an open subset.

We must now show that if  $X$  is smooth, then for all but finitely many  $y \in \mathbb{C}$  the differential  $T_x(X) \rightarrow \mathbb{C} = T_y(Y)$  is surjective. By what we have seen above, that there is an open subset  $X^0 \subset X$  where this holds. Let  $X_1$  be the complement of this open subset. For each irreducible component of  $X_1$ , there is an open subset  $X_1^0$  such that  $T_x(X_1^0) \rightarrow T_y(Y)$  is surjective for every  $x \in X_1^0$ . Since  $T_x(X_1^0) \subset T_x(X)$  it follows that  $T_x(X) \rightarrow T_y(Y)$  is onto. By Noetherian induction, there is an open subset  $Y^0 \subset Y$  such that  $T_x(X) \rightarrow T_y(Y)$  is onto for all  $x \in f^{-1}(Y^0)$ .  $\square$

## 22. CUBIC SURFACES

In this section we will study cubic surfaces i.e. smooth surfaces  $X \subset \mathbb{P}^3$  defined by a homogeneous equation of degree 3 sat  $f(x, y, z, w) \in \mathbb{C}[x, y, z, w]$ . We start with the following obvious remark.

**Lemma 22.1.** *Let  $H \subset \mathbb{P}^3$  be a hyperplane, then  $H$  intersects  $X$  in one of the following:*

- (1) *an irreducible cubic (degree 3) curve,*
- (2) *the union of a line and a smooth conic,*
- (3) *the union of three lines.*

*Proof.* Since  $X$  is smooth,  $H$  can not be contained in  $X$  (otherwise  $X = H \cup Y$  where  $Y$  is a conic and  $X$  is singular along  $H \cap Y$ , see Exercise 18.7). Therefore  $f|_H \neq 0$  is a homogeneous polynomial of degree 3. We must exclude the possibility that  $X \cap H$  contains a double line  $2L$ . We may choose local coordinates so that  $H = \mathbb{V}(t)$  and  $L = \mathbb{V}(t, z)$ . We then have that  $z^2$  divides  $f|_H(x, y, z) = f(x, y, z, 0)$  and so

$$f(x, y, z, w) = A(x, y, z)z^2 + B(x, y, z, w)w$$

where  $A, B$  are homogeneous of degree 1 and 2 respectively. Since

$$\nabla f = (A_x \cdot z^2 + B_x \cdot w, A_y \cdot z^2 + B_y \cdot w, A_z \cdot z^2 + A \cdot 2z + B_z \cdot w, B_w \cdot w + B),$$

then  $\nabla f = 0$  and  $f = 0$  if  $z = w = 0$  and  $B(x, y, 0, 0) = 0$ . (Note that  $B(x, y, 0, 0)$  has degree 2 and hence there are two such solutions counted with multiplicity.) Therefore  $X$  has a singular point which is impossible.  $\square$

**Lemma 22.2.** *If  $X$  contains three lines through a point, then the lines are coplanar.*

*Proof.* Suppose that  $x \in L_1, L_2, L_3 \subset X$ , then  $L_i = T_x(L_i) \subset T_x(X)$ .  $\square$

**Proposition 22.3.**  *$X$  contains at least one line  $L$ .*

*Proof.* There is a very elementary but rather lengthy proof of this fact in [Reid]. Here we follow the proof of [Beauville] which is less elementary but much shorter. The set of all cubic threefolds (including the singular ones) is just  $\mathbb{P}H^0(\mathcal{O}_{\mathbb{P}^3}(3)) \cong \mathbb{P}^{19}$ . We consider  $G$  the space of all lines in  $\mathbb{P}^3$ . Note that a line in  $\mathbb{P}^3$  is determined by a pair of distinct points  $X = [x_0 : x_1 : x_2 : x_3], Y = [y_0 : y_1 : y_2 : y_3] \in \mathbb{P}^3$ , but of course different points may determine the same line. It turns out that the line is uniquely determined by the point  $[\lambda_{12} : \lambda_{13} : \lambda_{14} : \lambda_{23} : \lambda_{24} : \lambda_{34}] \in \mathbb{P}^5$  where  $\lambda_{ij} = x_i y_j - x_j y_i$ . Note that  $X, Y$  are linearly independent and hence at least one  $2 \times 2$  minor is non-zero. One way to think of this is to use exterior algebra. The plane is determined by  $X \wedge Y$  (up to non-zero scalar). So we write  $X = \sum x_i e_i, Y = \sum y_i e_i$  and  $X \wedge Y = \sum x_i y_j e_i \wedge e_j$ . The coefficient of  $e_1 \wedge e_2$  is then  $x_1 y_2 - x_2 y_1 = \lambda_{12}$ . Note that the six minors satisfy the following relation  $\lambda_{12} \lambda_{34} - \lambda_{13} \lambda_{24} + \lambda_{14} \lambda_{23} = 0$  (this is an easy check). In fact  $G$  is defined by this equation (this is also an easy check, since  $\lambda_{12} \lambda_{34} - \lambda_{13} \lambda_{24} + \lambda_{14} \lambda_{23} = 0$  is irreducible, it suffices to see that  $\dim G = 4$ , but this follows easily since  $X = [1 : 0 : x_2 : x_3]$  and  $Y = [0 : 1 : y_2 : y_3]$  determine an open subset of  $G$  which is isomorphic to  $\mathbb{C}^4$ .) We now consider the incidence correspondence and projections

$$Z = \{(L, S) | L \subset S\} \subset G \times \mathbb{P}^{19}, \quad p : Z \rightarrow G, \quad q : Z \rightarrow \mathbb{P}^{19}.$$

Fix a line  $L$ , then in appropriate coordinates  $x, y, z, w$  on  $\mathbb{P}^3$ , we may assume that  $L = \mathbb{V}(z, w)$  and hence  $X$  contains  $L$  if and only if  $f \in (z, w)$  i.e. if and only if the coefficients of  $x^3, x^2 y, x y^2, y^3$  are all 0. This shows that the co-dimension of each fiber of  $p$  is 4 and hence  $\dim Z = \dim(G \times \mathbb{P}^{19}) - 4 = 19$ . We also claim that  $Z$  is non empty. To see this, simply exhibit a cubic containing a line! Consider now the projection  $q$ . If the proposition fails, then  $q(Z) \subset \mathbb{P}^{19}$  is a proper closed subset and thus each fiber of  $Z \rightarrow q(Z)$  is positive dimensional.<sup>6</sup> It suffices then to exhibit one cubic with finitely many lines. This requires a little preparation, and we will prove it later (see Lemma 22.10).  $\square$

**Exercise 22.4.** *Show that the open subset  $\lambda_{1,2} \neq 0$  is isomorphic to  $\mathbb{C}^4$ .*

**Exercise 22.5.** *Find the equation of a smooth cubic containing a line 9say  $\mathbb{V}(z, w)$ .*

**Proposition 22.6.** *Let  $L$  be a line on  $X$ , then there are 5 pairs of lines  $L_i, L'_i$  with  $1 \leq i \leq 5$  such that  $L, L_i, L'_i$  are coplanar for  $1 \leq i \leq 5$  and  $(L_i \cup L'_i) \cap (L_j \cup L'_j) = \emptyset$  for any  $1 \leq i < j \leq 5$ .*

*Proof.* Consider a plane  $P_t \supset L$ , then as we have seen  $X \cap P_t = L \cup C_t$  where  $C_t$  is a plane conic. We must show that there are exactly five planes for which  $C_t$  is a singular conic (and hence the union of two lines). The fact that  $(L_i \cup L'_i) \cap (L_j \cup L'_j) = \emptyset$  is then immediate from Lemma 22.2. Suppose in fact that (without loss of generality)  $L_1 \cap L_2 \neq \emptyset$ , then by construction  $L_1$  and  $L_2$  must meet at a point of  $L$ . But then  $L_1, L_2, L$  are coplanar and so  $L_2 = L'_1$  a contradiction.

We now choose coordinates such that  $L = \mathbb{V}(z, w)$  and so  $f \in \mathbb{I}(L) = (z, w)$ , and so

$$f = Ax^2 + 2Bxy + Cy^2 + 2Dx + 2Ey + F$$

---

<sup>6</sup>explain some details such as semicontinuity of fiber dimension.

where  $A, B, C, D, E, F$  are homogeneous of degrees 1, 2, 3 in  $\mathbb{C}[z, w]$ . As we have seen in Section 17, this conic is singular exactly when the

$$\Delta(z, w) = \det \begin{pmatrix} A & B & D \\ B & C & E \\ D & E & F \end{pmatrix} = ACF + 2BDE - CD^2 - B^2F - AE^2 = 0.$$

A plane  $P$  containing  $L$  is given by  $P_{[s:t]} = \mathbb{V}(sz - tw)$ . If  $s \neq 0$ , then we let  $\lambda = t/s$  and  $P_\lambda = \mathbb{V}(z - \lambda w)$ . Restricting to  $P_\lambda$  we obtain  $A|_P = A(z, w)|_P = A(\lambda w, w) = A(\lambda, 1)w$  and similarly for  $B|_P, C|_P, D|_P, E|_P$  and  $F|_P$ . Since  $f \in (z, w)$ , we have  $f|_P = w \cdot g$  where

$$g = A(\lambda, 1)x^2 + 2B(\lambda, 1)xy + C(\lambda, 1)y^2 + 2D(\lambda, 1)xw + 2E(\lambda, 1)yw + F(\lambda, 1)w^2.$$

Since  $\Delta(\lambda, 1)$  is a polynomial of degree 5 in  $\lambda$ , it follows that  $\Delta(\lambda, 1) = 0$  has five solutions, and we aim to show that these 5 solutions are distinct or equivalently that there are no multiple solution.

Pick any such root, we aim to show that this is not a repeated root. Without loss of generality we may assume that this root is  $\lambda = 0$  and we aim to find a contradiction. In this case  $P = \mathbb{V}(z)$  and  $P \cap X$  contains three lines (including the line  $L = \mathbb{V}(z, w)$ ). There are two cases depending on if the three lines contain a point or not. In this case we can assume that the coordinates have been chosen so that the remaining lines are given by the following equations:

- (1)  $L_1 = \mathbb{V}(z, x), L_2 = \mathbb{V}(z, x - z),$
- (2)  $L_1 = \mathbb{V}(z, x), L_2 = \mathbb{V}(z, y).$

Suppose that we are in case (2), then  $f|_P = xyw$  and so  $f = xyw + zq$  where  $q \in \mathbb{C}[x, y, z, w]$  is homogeneous of degree 2. But then  $z$  divides  $A, 2B - w, C, D, E, F$  so that  $B = w/2 + zb$  and  $A = za, C = zc, D = zd, E = ze$  and hence

$$\Delta(z, w) = z^2(acfz + 2(w + bz)de - cd^2z - (wb + zb^2)f - ae^2z) - w^2F/4.$$

Since  $X$  is smooth, we have that  $z^2$  does not divide  $F$  (for otherwise  $(0, 0, 1, 0)$  is a point of multiplicity  $\geq 2$  for  $f = Ax^2 + 2Bxy + Cy^2 + 2Dx + 2Ey + F$  i.e.  $X$  is singular at  $(0, 0, 1, 0)$ ). But then  $z^2$  does not divide  $\Delta(z, w)$  and so  $\lambda = 0$  is not a repeated root.

Case (1) is similar and we leave it as an exercise.  $\square$

**Exercise 22.7.** Prove case (1).

**Proposition 22.8.**  $X$  is birational to  $\mathbb{P}^2$ .

*Proof.* Let  $L, M$  be two disjoint lines (eg  $L_1$  and  $L_2$ ) and define rational maps

$$\phi : S \dashrightarrow L_1 \times L_2, \quad \psi : L_1 \times L_2 \dashrightarrow S$$

as follows: For any point  $P \in \mathbb{P}^3 \setminus (L_1 \cup L_2)$  there exists a unique line  $L_P$  containing  $P$  and intersecting  $L_1$  and  $L_2$  (this line is the intersection of the two planes containing  $p$  and  $L_i$  for  $i = 1, 2$ ). For any  $P \in S \setminus (L_1 \cup L_2)$ , we then define

$$\phi(P) = (L_P \cap L_1, L_P \cap L_2).$$

Suppose now that  $P_i \in L_i$ , then consider the line  $\overline{P_1, P_2}$  and note that if  $\overline{P_1, P_2}$  is not contained in  $X$ , then  $\overline{P_1, P_2} = P_1 \cup P_2 \cup Q$  and so we define

$$\psi(P_1, P_2) = Q.$$

It is easy to see that  $\phi$  and  $\psi$  are inverses (on appropriate open subsets) and hence  $X$  is birational to  $\mathbb{P}^1 \times \mathbb{P}^1$  (which in turn is clearly birational to  $\mathbb{P}^2$ ).  $\square$

**Remark 22.9.** *It is not hard to push these arguments a little further and to check that a smooth cubic surface contains exactly 27 lines (see [Reid] for the details.)*

Finally we must check that a smooth cubic threefold contains finitely many lines.

**Lemma 22.10.** *Let  $X$  be a smooth cubic threefold, then  $X$  contains only finitely many lines.*

*Proof.* Suppose that  $X$  contains a line (otherwise the claim is clear), then we may pick a plane  $P$  such that  $X \cap P = L \cup L_1 \cup L'_1$  is the union of 3 lines. If  $M$  is another line, then  $M$  intersects one of the lines  $L, L_1, L'_1$ . Without loss of generality we may assume that infinitely many lines intersect  $L$ . But this is a contradiction as we have seen that exactly 10 lines intersect  $L$ .  $\square$

## 23. RATIONAL AND UNIRATIONAL VARIETIES

**Definition 23.1.** *A projective variety  $X$  is rational if it is birational to  $\mathbb{P}^n$ . Equivalently  $X$  is rational if  $\mathbb{C}(X) \cong \mathbb{C}(x_1, \dots, x_n)$ . A variety  $X$  is unirational if there is a dominant morphism  $\mathbb{P}^n \dashrightarrow X$ .*

In general, it is very hard to tell if a variety is rational. For example it is clear that rational varieties are unirational but the converse is far from obvious. It is known that

**Theorem 23.2.** *If  $X$  is unirational and  $\dim X \leq 2$ , then  $X$  is rational.*

We can rephrase this result as follows: Let  $K \subset \mathbb{C}(x, y)$  be a field of transcendence degree 2, then  $K \cong \mathbb{C}(u, t)$ . Note that we are not claiming that  $K = \mathbb{C}(x, y)$ , as we could for example have the inclusion  $K = \mathbb{C}(x^2, y^2) \subset \mathbb{C}(x, y)$  which is not an equality. The proof of this result goes beyond the scope of these notes, however we mention the intuition behind the  $\dim X = 1$  case. In this case we must show that if  $\mathbb{P}^1 \rightarrow X$  is a dominant map, then  $X \cong \mathbb{P}^1$ . Let  $g(X)$  be the topological genus of the Riemann surface  $X$ , then we have to show that  $g(X) = 0$ . This follows from the well known **Hurwitz formula** which says

**Theorem 23.3.** *Let  $f : X \rightarrow Y$  be a dominant map of degree  $d$  of projective varieties of dimension 1, then*

$$2(g(X) - 2) = d(2g(Y) - 2) + \sum (r_i - 1),$$

where  $r_i$  denotes the ramification index.

Note that there exist open subsets  $Y_0 \subset Y$  and  $X_0 = f^{-1}(Y_0) \subset X$  such that  $f_0 = f|_{X_0} : X_0 \rightarrow Y_0$  is a smooth map (the rank of the differential is maximal at each point of  $X_0$ ). It follows that  $X_0 \rightarrow Y_0$  is an isomorphism locally in the analytic topology (this means that for any point  $x \in X_0$ , there are analytic neighborhoods  $x_0 \in B \subset X_0$  and  $y_0 = f(x_0) \in B' \subset Y_0$  such that  $f|_B : B \rightarrow B'$  is an isomorphism). If however  $x_0 \in X \setminus X_0$ , then we can choose analytic neighborhoods  $x_0 \in B \subset X$  and  $y_0 = f(x_0) \in B' \subset Y$  and analytic coordinates  $z$  on  $B$  and  $w$  on  $B'$  such that  $w = f(z) = z^r$  for some  $r \geq 1$ . We say that the **ramification index** at  $x_0$  is  $r$  (note that  $r = 1$  precisely when  $f$  is smooth, so if we accidentally include a smooth point in  $X \setminus X_0$ , then the above formula is not affected). The quantity  $2g - 2 = v - e + f$  can be computed from any triangulation where  $v$  is the number of vertices,  $e$  the number of edges and  $f$  the number of faces. If we choose a triangulation of  $Y$  such that each point of  $Y \setminus Y_0$  is a vertex, then the inverse image of this triangulation via  $f$  gives a triangulation with  $df$  faces,  $de$  edges and  $dv - \sum (r_i - 1)$  vertices. Therefore

$$2 - 2g(X) = df - de + dv - \sum (r_i - 1) = d(f - e + v) - \sum (r_i - 1) = d(2g(Y) - 2) - \sum (r_i - 1).$$

Applying the Hurwitz formula to a dominant morphism  $\mathbb{P}^1 \rightarrow X$  of degree  $d$  gives

$$-2 = 2g(\mathbb{P}^1) - 2 = d(2g(X) - 2) + \sum (r_i - 1).$$

Since  $\sum (r_i - 1) \geq 0$ , we must have  $2g(X) - 2 < 0$  and so  $g(X) = 0$ .

The proof of the two dimensional case is much harder. First of all it is necessary to generalize the concept of the genus of a curve (Riemann surface) to a complex surface (4-manifold). There are several natural options. First of all, we note that  $g(X) = \dim H^0(\Omega_X^1)$ . Here  $\Omega_X^1 = T_X^\vee$  is the cotangent bundle and  $H^0(\Omega_X^1)$  denotes the space of global holomorphic 1-forms. Recall that in local coordinates, a holomorphic 1-form may be written as  $f(z)dz$ , so if  $X = \cup U_i$  is an open cover and  $z_i$  is a local coordinate on  $U_i$ , then an element  $\omega \in H^0(\Omega_X^1)$  can be represented locally by

$$\{\omega_i = f_i dz_i \in \Gamma(\Omega_{U_i}^1)\}$$

which patch together on  $U_{i,j} = U_i \cap U_j$ . This means that if  $z_i = g_{i,j}(z_j)$  on  $U_{i,j}$ , then  $dz_i = \frac{\partial g_{i,j}}{\partial z_j} dz_j$ . For example, if  $X = \mathbb{P}^1$ , then there are two charts:  $U_1$  and  $U_2$  with local coordinates  $z$  and  $w$  respectively such that  $z = 1/w$  on  $U_{1,2} = U_1 \cap U_2$ . Suppose that  $f(z)dz$  and  $g(w)dw$  represent an element of  $H^0(\Omega_{\mathbb{P}^1}^1)$ , then  $dz = -\frac{1}{w^2}dw$  and so on  $U_{1,2}$  we must have  $f(z)dz = g(1/z)\frac{1}{z^2}dz$ . Since  $g(w)$  is a polynomial,  $g(1/z)\frac{1}{z^2}$  only involves powers  $z^a$  with  $a \leq -2$  while  $f(z)$  only involves positive powers of  $z$ . Thus  $f(z) = 0$  and  $g(w) = 0$ . This shows that  $H^0(\Omega_{\mathbb{P}^1}^1) = 0$  and hence  $\dim H^0(\Omega_{\mathbb{P}^1}^1) = g(\mathbb{P}^1)$ .

Returning to our goal of generalizing the genus to higher dimensional varieties, there are now two natural choices.

**Definition 23.4.** *Let  $X$  be a smooth projective variety, then the **irregularity** of  $X$  is*

$$q(X) = \dim H^0(\Omega_X^1).$$

A second natural choice involves the **canonical line bundle** which is just the top exterior power of the cotangent bundle. More precisely, if  $X$  is a smooth projective variety of dimension  $n$ , then let  $\omega_X = \wedge^n \Omega_X^1 = \Omega_X^n$  so that in local coordinates  $z_1, \dots, z_n$ , any section of  $\omega_X$  can be locally written as

$$f(z_1, \dots, z_n) dz_1 \wedge \dots \wedge dz_n,$$

where  $f$  is a regular function. If  $U_i \subset X$  is an open cover and  $\{z_1^i, \dots, z_n^i\}$  are local coordinates, then on  $U_{i,j}$  we can write  $z_k^i = g_k^{i,j}(z_1^j, \dots, z_n^j)$ . It is easy to see that the transition functions are now represented by the determinant of the matrix whose  $(k, l)$ -th entry is  $\partial g_k^{i,j} / \partial z_l^j$ . Similarly, we can consider the line bundles  $\omega_X^{\otimes m}$  for any  $m > 0$  whose sections can be locally written as

$$f(z_1, \dots, z_n) (dz_1 \wedge \dots \wedge dz_n)^{\otimes m},$$

and have transition functions  $\det(\partial g_k^{i,j} / \partial z_l^j)^m$ . We then define

**Definition 23.5.** *Let  $X$  be a smooth projective variety and  $m > 0$  a positive integer, then the  $m$ -th **plurigenera** of  $X$  is the number*

$$P_m(X) = \dim H^0(\omega_X^{\otimes m}).$$

Here  $H^0(\omega_X^{\otimes m})$  denotes the space of global sections of  $\omega_X^{\otimes m}$  described above, that patch together to give a global section.

We are now ready to state Castelnuovo's Criterion.

**Theorem 23.6.** *Let  $X$  be a projective surface with  $q(X) = P_2(X) = 0$ , then  $X$  is rational.*

The proof of this result is non-trivial and we do not recall it here. Note however that the irregularity and plurigenera are birational invariants. This means that if  $X \dashrightarrow Y$  is a birational map of smooth projective varieties, then  $q(X) = q(Y)$  and  $P_m(X) = P_m(Y)$  for all  $m > 0$ . In particular  $q(X) = P_2(X) = 0$  is a necessary condition for rationality.

Note that Theorem 23.3 in dimension 2 follows immediately from Castelnuovo's criterion. In fact, if  $f : \mathbb{P}^2 \dashrightarrow X$  is a dominant map of surfaces, then it is easy to see that  $P_2(X) = q(X) = 0$ . The reason is that if  $\omega \in H^0(\Omega_X^1)$  (or  $\omega \in H^0(\omega_X^{\otimes 2})$ ), then there is a naturally defined element  $f^*\omega \in H^0(\Omega_{\mathbb{P}^2}^1)$  (or  $f^*\omega \in H^0(\omega_X^{\otimes 2})$ ). If  $V \subset X$  and  $U = f^{-1}(V) \subset \mathbb{P}^2$  are open subsets such that  $f|_U : V \rightarrow U$  is smooth, then locally in the analytic topology  $U$  and  $V$  are isomorphic and  $\omega$  and  $f^*\omega$  are identified via this isomorphism. Since  $q(\mathbb{P}^2) = P_2(\mathbb{P}^2) = 0$  it follows that  $f^*\omega = 0$  and hence that  $q(X) = P_2(X) = 0$ . But then by Castelnuovo's Criterion  $X$  is birational to  $\mathbb{P}^2$  (we do not claim that  $f$  is a birational map).

It turns out that in general it is very hard to determine if an unirational variety is rational, however Clemens and Griffiths proved the following amazing result.

**Theorem 23.7.** *Let  $X \subset \mathbb{P}^4$  be a cubic threefold i.e. a smooth hypersurface of degree 3, then  $X$  is unirational but not rational. In particular  $\mathbb{C}(X) \subset \mathbb{C}(x_1, x_2, x_3)$  but  $\mathbb{C}(X)$  is not isomorphic to  $\mathbb{C}(t_1, t_2, t_3)$ .*

The proof of this result is beautiful but very non-trivial. In what follows we will only explain why a smooth cubic threefold is unirational.

**Proposition 23.8.** *Let  $X \subset \mathbb{P}^4$  be a cubic threefold, then  $X$  is unirational.*

*Proof.* **Claim 1.** We may find a line  $L \subset X$ .

To see this, simply consider a general hyperplane  $\mathbb{P}^3 \cong H \subset \mathbb{P}^4$ . Then  $H \cap X \subset H \cong \mathbb{P}^3$  is a smooth cubic surface (by Bertini's Theorem), and we have seen that a cubic surface contains a line (and in fact 27 lines). Just pick one of these.

Consider now the rational map  $\mathbb{P}^4 \dashrightarrow \mathbb{P}^2$  defined by projecting from  $L$ . This map is a morphism on  $\mathbb{P}^4 \setminus L$  and it becomes a morphism  $B_L\mathbb{P}^4 \rightarrow \mathbb{P}^2$  after blowing up  $\mathbb{P}^4$  along the line  $L$ . Note that the exceptional divisor is isomorphic to  $L \times \mathbb{P}^2$ .

**Claim 2.** Consider The blow up  $B_LX \subset B_L\mathbb{P}^4$ . Then, the fibers of the induced map  $B_LX \rightarrow \mathbb{P}^2$  are quadrics.

Since any plane  $\Pi_x$  through  $L$  (corresponding to a point  $x \in \mathbb{P}^2$ ) intersects  $X$  along  $L$  and a residual quadric curve say  $C_x$ , we have that the fibers of  $\phi$  are quadrics  $\phi^{-1}(x) \cong C_x$ , and the claim follows. Note that there is a non-empty open subset  $U \subset \mathbb{P}^2$  such that for any  $x \in U$ , we have  $C_x$  is smooth and hence  $C_x \cong \mathbb{P}^1$ .

**Claim 3.** Consider  $E = B_LX \cap L \times \mathbb{P}^2$  the exceptional divisor lying over  $L$ . Then the induced map  $E \rightarrow \mathbb{P}^2$  has generic degree 2 and the fibers of  $E \rightarrow \mathbb{P}^2$  are quadrics.

Since  $E = L \times \mathbb{P}^2 \cap B_LX$  and, the fiber of  $E$  over  $x \in \mathbb{P}^2$  is given by  $L \cap C_x$  and hence consists of two points. In other words  $E \dashrightarrow \mathbb{P}^2$  is generically of degree 2.

Consider now the fiber product  $Y = B_LX \times_{\mathbb{P}^2} E$ , then the fibers of  $B_LX \times_{\mathbb{P}^2} E \rightarrow E$  are just the rational curves  $C_x$  and  $Y \rightarrow E$  has a rational section. But then  $Y$  is birational to  $E \times \mathbb{P}^1$  and hence  $X$  is uniruled.  $\square$

**Exercise 23.9.** *Show that  $H^0(\Omega_{\mathbb{P}^n}^1) = 0$  and  $H^0(\omega_{\mathbb{P}^n}^{\otimes k}) = 0$ .*

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<sup>7</sup>to be defined

**Exercise 23.10.** Let  $X_0 \subset X$  be a big open subset of a smooth projective variety. Show that  $H^0(\Omega_X^1) \cong H^0(\Omega_{X_0}^1)$  and  $H^0(\omega_X^{\otimes m}) \cong H^0(\omega_{X_0}^{\otimes m})$  for all  $m > 0$ .

**Exercise 23.11.** Let  $f : X \rightarrow Y$  be a generically finite morphism. Show that there is a naturally defined homomorphism  $H^0(\Omega_Y^1) \rightarrow H^0(\Omega_X^1)$  and  $H^0(\omega_Y^{\otimes m}) \rightarrow H^0(\omega_X^{\otimes m})$  for any  $m > 0$ .

**Exercise 23.12.** Let  $f : X \rightarrow Y$  be a birational morphism, show that  $H^0(\omega_X^{\otimes k}) \cong H^0(\omega_Y^{\otimes k})$  and  $H^0(\Omega_X^1) \cong H^0(\Omega_Y^1)$ . Deduce that  $P_m(X)$  and  $q(X)$  are birational invariants. (Hint. Use the fact if  $f : X \rightarrow Y$  is a birational morphism, then there is a big open subset  $Y_0 \subset Y$  such that  $f : f^{-1}(Y_0) \rightarrow Y_0$  is an isomorphism).

**Exercise 23.13.** Let  $C_1, C_2$  be curves of genus  $g_1, g_2$ , compute  $P_m(C_1 \times C_2)$  and  $h^0(\Omega_{C_1 \times C_2}^1)$ .

## 24. SHEAVES

**Definition 24.1.** If  $X$  is a topological space, then a **presheaf** (resp. a **sheaf**) of abelian groups  $\mathcal{F}$  on  $X$  is given by assigning:

- (1) an abelian group  $\mathcal{F}(U)$  to any open subset  $U \subset X$ , and
- (2) for any inclusion of open subsets  $V \subset U \subset X$  a “restriction” homomorphism  $\rho_{UV} : \mathcal{F}(U) \rightarrow \mathcal{F}(V)$ ,

such that conditions (1-3) (resp. conditions (1-5)) below are satisfied by  $\mathcal{F}$  and  $\rho$ :

- (1)  $\mathcal{F}(\emptyset) = 0$ ,
- (2)  $\rho_{U,U} = \text{id}_U$ ,
- (3) if  $W \subset V \subset U \subset X$  are open subsets, then  $\rho_{UW} = \rho_{VW} \circ \rho_{UV}$ ,
- (4) if  $U \subset X$  is an open subset and  $\bigcup_i V_i$  is an open cover of  $U$  and  $s \in \mathcal{F}(U)$  satisfies  $\rho_{U,V_i}(s) = 0$  for all  $i$ , then  $s = 0$ , and
- (5) if  $U \subset X$  is an open subset and  $\bigcup_i V_i$  is an open cover of  $U$  and if  $s_i \in \mathcal{F}(V_i)$  satisfy  $\rho_{V_i V_{i,j}}(s_i) = \rho_{V_j V_{i,j}}(s_j)$  for all  $i, j$  then there is an element  $s \in \mathcal{F}(U)$  such that  $\rho_{U,V_i} = s_i$  for all  $i$ .

Note that as usual  $V_{i,j}$  denotes  $V_i \cap V_j$  and in what follows, we will denote  $\rho_{UV}$  simply by  $|_V$ . The above definitions seems at first very abstract but in practice there are many examples of a sheaf that you are familiar with.

- (1) The sheaf of regular functions  $\mathcal{O}_X$ . Let  $X$  be a quasi-projective variety, then  $\mathcal{O}_X(U)$  defines a sheaf of rings on  $X$ . (The hardest axiom to check is (5) which says that if  $f$  is a function which is regular when restricted to each open subset of an open cover, then  $f$  is in fact regular).
- (2) The sheaf of continuous functions on a topological space.
- (3) The sheaf of analytic functions on an analytic variety.
- (4) Given an abelian group  $A$  and a topological space  $X$ , one can define the **constant presheaf** by letting  $A(\emptyset) = 0$  and  $A(U) = A$  for any open subset  $\emptyset \neq U \subset X$ . It is easy to see that this is not a sheaf in general as (5) may fail.
- (5) Given an abelian group  $A$  and a topological space  $X$ , one can define the **constant sheaf** on  $X$  as follows: If  $U$  is an open subset, assuming that its connected components  $U_i$  are also open, then let  $A(U) = \prod_{i \in I} A(U_i) = \prod_{i \in I} A$ . In other words we let  $A(U) = A$  if  $U$  is an irreducible open subset and otherwise we let  $A(U)$  be a product of one copy of  $A$  for each irreducible component of  $U$ .



**Notation 24.2.** We call the elements of the abelian groups associated to  $X$  by the sheaf “sections” of  $\mathcal{F}$ , i.e. an element  $s \in \mathcal{F}(X)$  is called a **global section** of  $\mathcal{F}$  and an element  $s \in \mathcal{F}(U)$  is called a **local section** on the open set  $U$ . We often denote the set of global sections using the notation  $\Gamma(X, \mathcal{F}) := \mathcal{F}(X)$  and the set of local sections on  $U$  using the notation  $\Gamma(U, \mathcal{F}) := \mathcal{F}(U)$ .

By now the reader must expect that each time we introduce new objects we also introduce a notion of maps (and isomorphisms) between these objects. In this case we have

**Definition 24.3.** Let  $\mathcal{F}$  and  $\mathcal{G}$  be sheaves on a topological space  $X$ , then a **morphism of sheaves**  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  is a collection of homomorphisms  $\phi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)$  for any open subsets  $U \subset X$  which are compatible with inclusion of open subsets  $V \subset U \subset X$  (i.e.  $\rho_{UV} \circ \phi_U = \rho_{UV} \circ \phi_V$  where, by abuse of notation,  $\rho_{UV}$  denotes both restrictions  $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$  and  $\mathcal{G}(U) \rightarrow \mathcal{G}(V)$ ). An **isomorphism of sheaves** is a morphism of sheaves  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  such that there is a morphism of sheaves  $\psi : \mathcal{G} \rightarrow \mathcal{F}$  such that  $\phi \circ \psi$  and  $\psi \circ \phi$  are the identity morphism on  $\mathcal{F}$  and  $\mathcal{G}$ .

Just as in the case of groups or rings, we can talk about kernels, cokernels and images of a morphism.

**Definition 24.4.** Let  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  be a morphism of pre-sheaves on a topological space  $X$ , then we have:

- (1) **kernel presheaf:**  $\ker(\phi)(U) = \ker(\phi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U))$ ,
- (2) **cokernel presheaf:**  $\operatorname{coker}(\phi)(U) = \operatorname{coker}(\phi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)) = \mathcal{G}(U)/\phi_U(\mathcal{F}(U))$ ,
- (3) **image sheaf:**  $\operatorname{im}(\phi)(U) = \operatorname{im}(\phi_U : \mathcal{F}(U) \rightarrow \mathcal{G}(U)) = \phi_U(\mathcal{F}(U))$ .

**Exercise 24.5.** Show by example that image and cokernel presheaves may not be sheaves.

**Exercise 24.6.** Show that the kernel presheaf is always a sheaf.

It is much more convenient to work with sheaves. Luckily it is always possible to associate a sheaf to any presheaf in a natural way.

**Definition 24.7.** Let  $\mathcal{F}$  be a presheaf on a topological space  $X$ , then we define the sheaf  $\mathcal{F}^+$  on  $X$  in the following two steps: firstly, for any open subset  $U \subset X$  an element  $s \in \mathcal{F}^+(U)$  is given by an open cover  $U = \cup U_i$  and a collection of elements  $s_i \in \mathcal{F}(U_i)$  such that  $s_i|_{U_{ij}} = s_j|_{U_{ij}}$  for all  $i, j$ ; next we identify two elements of  $\mathcal{F}^+(U)$  if there is an open cover  $V_j$  of  $U$  such that  $s_i|_{V_j \cap U_i} = 0$  for all  $i, j$ .

We say that  $\mathcal{F}^+$  is the **sheafification** of  $\mathcal{F}$ .

**Exercise 24.8.** Show that the first step in the above definition ensures property (5) in the definition of a sheaf and the second step ensures property (4).

Note that the above definition is quite cumbersome (but natural). There is an alternative more efficient way to proceed. Intuitively, we are interested in the local behaviour of sheaves around a point  $P \in X$ . For example, we say that  $s \in \mathcal{F}(X)$  is zero locally near a point  $P$  if there is an open subset  $P \in U \subset X$  such that  $s|_U = 0$ . Therefore it is natural to consider the following definition.

**Definition 24.9.** Let  $\mathcal{F}$  be a presheaf on a topological space  $X$ , then for any point  $P \in X$  the **stalk** of  $\mathcal{F}$  at  $P$  is given by

$$\mathcal{F}_P = \{ \langle U, s \rangle \mid U \subset X \text{ is open, } s \in \mathcal{F}(U) \} / \sim$$

where  $\langle U, s \rangle \sim \langle U', s' \rangle$  if and only if there is an open subset  $P \in V \subset U \cap U'$  such that  $s|_V = t|_V$ . Given  $s \in \mathcal{F}(U)$  we then have elements  $s_P = \langle s, U \rangle \in \mathcal{F}_P$  for any  $P \in U$ . We will say that  $s_P$  is the **germ** of  $s$  at  $P$ .

With this definition we can give the following alternative definition for the sheaf  $\mathcal{F}^+$  associated to a presheaf  $\mathcal{F}$ .

**Definition 24.10.** For any open subset  $U \subset X$  let  $\mathcal{F}'(U)$  be the set of functions from  $s : U \rightarrow \cup_{P \in U} \mathcal{F}_P$  such that

- (1) for any  $P \in U$ ,  $s(P) \in \mathcal{F}_P$ , and
- (2) for any  $P \in U$  there exists an open subset  $P \subset V \subset U$  and an element  $t \in \mathcal{F}(V)$  such that  $s(Q) = t_Q$  for all  $Q \in V$ .

**Exercise 24.11.** Show that  $\mathcal{F}^+ = \mathcal{F}'$ .

**Exercise 24.12.** Show that if  $\mathcal{F}$  is a presheaf on  $X$ , then  $\mathcal{F}_P^+ = \mathcal{F}_P$  for any  $P \in X$ .

**Exercise 24.13.** Show that there is a natural morphism  $\phi : \mathcal{F} \rightarrow \mathcal{F}^+$  satisfying the following universal property: For any morphism of pre-sheaves  $\psi : \mathcal{F} \rightarrow \mathcal{G}$ , if  $\mathcal{G}$  is a sheaf, then there is a unique morphism of sheaves  $\psi^+ : \mathcal{F}^+ \rightarrow \mathcal{G}$  such that  $\psi^+ \circ \phi = \psi$ .

**Exercise 24.14.** If  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  is a morphism of sheaves then show that there is a natural induced map  $\phi_p : \mathcal{F}_P \rightarrow \mathcal{G}_P$ .

**Exercise 24.15.** If  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  is a morphism of sheaves then  $\phi$  is an isomorphism if and only if the induced map of stalks  $\phi_p : \mathcal{F}_P \rightarrow \mathcal{G}_P$  is an isomorphism for all  $P \in X$ .

**Definition 24.16.** If  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  is a morphism of sheaves, then we define

- (1) **kernel sheaf:**  $\ker(\phi)$  is just the kernel presheaf of  $\phi$  (as observed above this is automatically a sheaf).
- (2) **image sheaf:** If we denote the corresponding image presheaf by  $\text{im}'(\phi) \subset \mathcal{G}$ , then we define the image sheaf by  $\text{im}(\phi) = \text{im}'(\phi)^+ \subset \mathcal{G}^+ = \mathcal{G}$ . (In other words the image sheaf is the sheaf naturally associated to the image presheaf.)
- (3) **cokernel sheaf:** If we denote the corresponding cokernel presheaf by  $\text{coker}'(\phi)$ , then we define the cokernel sheaf by  $\text{coker}(\phi) = \text{coker}'(\phi)^+$ . (In other words the cokernel sheaf is the sheaf naturally associated to the cokernel presheaf.)

**Definition 24.17.** If  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  is a morphism of sheaves, then we say that  $\phi$  is **injective** if  $\ker(\phi) = 0$ ,  $\phi$  is **surjective** if  $\text{im}(\phi) = \mathcal{G}$ .

**Exercise 24.18.** Check that  $\phi : \mathcal{F} \rightarrow \mathcal{G}$  is injective (resp. surj.) if and only if  $\phi_P$  is injective (resp. surjective) for all  $P \in U$ .

**Definition 24.19.** Consider a sequence of morphism of sheaves  $\phi^i : \mathcal{F}^i \rightarrow \mathcal{F}^{i+1}$ . We say that this sequence is exact if  $\ker(\phi^i) = \text{im}(\phi^{i-1})$ . We will be particularly interested in short exact sequences i.e. in exact sequences of the form

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0.$$

This is equivalent to requiring that  $\mathcal{F}' \rightarrow \mathcal{F}$  is injective,  $\mathcal{F} \rightarrow \mathcal{F}''$  is surjective and that  $\ker(\mathcal{F} \rightarrow \mathcal{F}'') = \text{im}(\mathcal{F}' \rightarrow \mathcal{F})$ .

**Exercise 24.20.** Show that a sequence of sheaves

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}' \rightarrow 0$$

is exact if and only if the corresponding sequence of abelian groups

$$0 \rightarrow \mathcal{F}'_P \rightarrow \mathcal{F}_P \rightarrow \mathcal{F}'_P \rightarrow 0$$

given on stalks is exact for every point  $P \in X$ .

Given a continuous map  $f : X \rightarrow Y$  of topological spaces, we can define several useful operations determined by  $f$  for sheaves on  $X$  and  $Y$ . Here is one such operation.

**Definition 24.21.** If  $\mathcal{F}$  is a sheaf on  $X$ , we define the **direct image presheaf**  $f_*\mathcal{F}$  on  $Y$  to be given by  $f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}(V))$  for any open set  $V$  in  $Y$ .

**Exercise 24.22.** Show that the direct image presheaf of a sheaf is automatically a sheaf.

**Exercise 24.23.** Let  $f : X \rightarrow Y$  be a morphism of quasi-projective varieties, and let  $\mathcal{O}_X$  and  $\mathcal{O}_Y$  be the sheaves of regular functions on  $X$  and  $Y$  respectively. What do you expect the natural induced morphism of sheaves  $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$  to look like? Compute this morphism for some of your favorite such examples  $f$ .

## 25. SHEAVES OF MODULES

We begin by recalling some definitions and exercises from earlier sections on modules to remind you about some of the central constructions.

**Definition 25.1.** Let  $R$  be a domain, then the **fraction field** of  $R$  is the set

$$Q(R) = \{(f, g) | f \in R, g \in R^*\} / \sim, \quad (f, g) \sim (h, k), \text{ iff } fk = gh.$$

Addition is defined by  $(f, g) + (h, k) = (fk + gh, gk)$  and multiplication is defined by  $(f, g)(h, k) = (fh, gk)$ . It will be convenient to denote the equivalence class of  $(f, g)$  simply by  $f/g$ .

We saw that we can generalize this construction, allowing us to “invert” specified elements of any ring  $A$  provided they together form a multiplicatively closed set, a process called **localization** that is essential to many basic constructions in algebraic geometry.

**Definition 25.2.** Let  $A$  be a commutative ring with identity and let  $S \subseteq A$  such that it contains the identity and is **multiplicatively closed**, meaning that  $a, b \in S \implies ab \in S$ . The **localization of  $A$  at  $S$**  (or the **ring of fraction of  $A$  with respect to  $S$** ) is the set

$$S^{-1}A = \{(f, g) | f \in A, g \in S\} / \sim$$

where the equivalence relation  $\sim$  is defined by

$$(f, g) \sim (h, k) \iff \exists x \in S \text{ such that } x(fk - gh) = 0.$$

Addition is defined by  $(f, g) + (h, k) = (fk + gh, gk)$  and multiplication is defined by  $(f, g)(h, k) = (fh, gk)$ . It will be convenient to denote the equivalence class of  $(f, g)$  simply by  $f/g$ .

If you haven't yet done some of the exercises about localization in the earlier sections, we recall them here.

**Exercise 25.3.** Prove that  $\sim$  is indeed an equivalence relation.

**Exercise 25.4.** There is a natural map  $\pi : A \rightarrow S^{-1}A$  sending  $a \mapsto a/1$ . Prove that  $S^{-1}A$  together with this map satisfies the following universal property: given any morphism  $u : A \rightarrow B$  of commutative rings (sending  $1 \mapsto 1$ ) where  $u(d)$  is a unit  $\forall d \in S$ , there exists a unique morphism  $v : S^{-1}A \rightarrow B$  such that  $u = v \circ \pi$ .

**Exercise 25.5.** Prove that  $\pi$  in the previous exercise is one-to-one if and only if  $S$  contains no zero-divisors.

**Exercise 25.6.** Prove that  $S^{-1}A = 0$  if and only if  $0 \in S$ .

If  $R$  is a domain, then choosing  $S = R \setminus \{0\}$  produces the fraction field  $Q(R)$ .

**Exercise 25.7.** Given any element  $f \in A$ , we define  $A_f = S_f^{-1}A$  where  $S_f = \{f^n | n \geq 0\}$  is the set of nonnegative powers of  $f$ . Under what condition is  $A_f = 0$ ? Prove that  $A_f = R[x]/(xf - 1)$ .

Let  $P \subseteq A$  be a prime ideal. Recall that  $S = A \setminus P$  contains the identity and is multiplicatively closed. We call the ring  $S^{-1}A$  the **localization of  $A$  at the prime ideal  $P$** . We now recall the definition of an  $A$ -module.

**Definition 25.8.** Let  $A$  be a ring,  $(M, +)$  an abelian group and  $A \times M \rightarrow M$  an operation (called **scalar multiplication** and denoted by  $(a, m) \rightarrow a \cdot m$ ), then  $M$  is a (left)  **$A$ -module** if the following hold for all  $a, a' \in A$  and  $m, m' \in M$ :

- (1)  $a \cdot (m + m') = a \cdot m + a \cdot m'$ ,
- (2)  $(a + a') \cdot m = a \cdot m + a' \cdot m$ ,
- (3)  $(aa') \cdot m = a \cdot (a' \cdot m)$ , and
- (4)  $1 \cdot m = m$ .

Remind yourself the obvious basic definitions in the theory of modules. For example, an  $A$ -submodule of an  $A$ -module  $M$  is simply a subgroup of  $M$  that is itself closed under scalar multiplication. Morphisms of  $A$ -modules are group homomorphisms that respect the  $A$ -module structure.

**Exercise 25.9.** Go back and review the following notions in the theory of modules:

- (1) Quotient of an  $A$ -module by an  $A$ -submodule,
- (2) kernel of a morphism of  $A$ -modules,
- (3) cokernel of a morphism of  $A$ -modules,
- (4) image of a morphism of  $A$ -modules,

**Exercise 25.10.** Let  $n$  be a positive integer and let  $A$  be a commutative ring with identity. Show that the cartesian product  $A^n$  is an  $A$ -module under componentwise vector addition and componentwise scalar multiplication. We call this the **free  $A$ -module of rank  $n$** . We say an  $A$ -module  $M$  is a **free module of rank  $n$**  if it is isomorphic to  $A^n$  as  $A$ -modules for some positive integer  $n$ .

**Exercise 25.11.** Let  $I \subset A$  be an ideal, show that  $I$  and  $A/I$  are  $A$ -modules and that we have a short exact sequence of  $A$ -modules

$$0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0.$$

**Definition 25.12.** Let  $A$  be a commutative ring with identity and let  $S \subseteq A$  be a multiplicatively closed subset containing the identity. Let  $M$  be an  $A$ -module. The **localization**

of  $M$  at  $S$  (or the **module of fraction of  $M$  with respect to  $S$** ) is the set

$$S^{-1}M = \{(f, m) | f \in S, m \in M\} / \sim$$

where the equivalence relation  $\sim$  is defined by

$$(f, m) \sim (g, n) \iff \exists x \in S \text{ such that } x(fn - gm) = 0.$$

This set becomes a  $S^{-1}A$  module under the following operations: addition is defined by  $(f, m) + (g, n) = (gm + fn, fg)$  and scalar multiplication is defined by  $(g/h)(f, m) = (gm, hf)$ . It will be convenient to denote the equivalence class of  $(d, m)$  simply by  $m/d$ .

**Exercise 25.13.** Given an element  $f \in A$ , what does the  $A_f$ -module  $M_f = S_f^{-1}M$  look like?

**Exercise 25.14.** Let  $M$  be an  $A$ -module and  $P$  a prime ideal of  $A$ . If  $A_P = (A \setminus P)^{-1}A$  is the localization of  $A$  at  $P$ , then show that  $M_P = (A \setminus P)^{-1}M$  is an  $A_P$ -module.

We now introduce sheaves of  $\mathcal{O}_X$ -modules. These form the building blocks for many of the central algebraic objects of study in algebraic geometry.

**Definition 25.15.** Let  $X$  be a quasi-projective variety and  $\mathcal{F}$  be a sheaf on  $X$ , then  $\mathcal{F}$  is a **sheaf of  $\mathcal{O}_X$ -modules** (or just  **$\mathcal{O}_X$ -module** for short) if for any open subset  $U \subset X$ , the group  $\mathcal{F}(U)$  is a  $\mathcal{O}_X(U)$ -module and the module structures are compatible with the restriction morphisms (so that if  $V \subset U$  is an open subset,  $s \in \mathcal{F}(U)$  and  $t \in \mathcal{O}_X(U)$ , then  $(t \cdot s)|_V = t|_V \cdot s|_V$ ).

In particular if  $X$  is an affine variety with coordinate ring  $A = A(X)$ , then  $\Gamma(X, \mathcal{O}_X) = \mathcal{O}_X(X) = A$  and so  $M := \mathcal{F}(X)$  is an  $A$ -module. It turns out that there is a natural way of inverting this procedure.

**Definition 25.16.** Let  $X$  be an affine variety with ring of regular functions  $A = A(X)$  and  $M$  an  $A$ -module, then there is a natural sheaf  $\widetilde{M}$  on  $X$  associated to  $M$  such that for any point  $P \in X$ , the stalk  $\widetilde{M}_P$  of  $\widetilde{M}$  at  $P \in X$  is  $\widetilde{M}_P = M_{\mathfrak{m}_P}$  the localization of  $M$  at the maximal ideal  $\mathfrak{m}_P$  of  $P \in X$ . For any open subset  $U \subset X$ , this is defined by  $\widetilde{M}(U)$  is the set of functions  $s : U \rightarrow \prod_{P \in U} M_{\mathfrak{m}_P}$  such that for every  $P \in U$ ,  $s(P) \in M_{\mathfrak{m}_P}$ , and such that for any  $P \in U$  there is an open subset  $V \subset U$  and elements  $m \in M$  and  $a \in A$  such that for any  $Q \in V$  we have  $s(Q) = m/a \in M_{\mathfrak{m}_Q}$ .

**Exercise 25.17.** Check that  $\widetilde{M}$  is a sheaf of  $\mathcal{O}_X$ -modules such that the stalks  $\widetilde{M}_P$  of  $\widetilde{M}$  at  $P \in X$  is  $\widetilde{M}_P = M_{\mathfrak{m}_P}$  the localization of  $M$  at the maximal ideal  $\mathfrak{m}_P$  of  $P \in X$ .

**Proposition 25.18.** With the notation above, if  $f \in A$  and  $X_f$  is the corresponding principal open subset, then  $\Gamma(X_f, \widetilde{M}) = M_f$  and in particular  $\Gamma(X, \widetilde{M}) = M$ .

**Exercise 25.19.** Prove Proposition 25.18

**Exercise 25.20.** Define the following notions for sheaves of  $\mathcal{O}_X$ -modules:

- (1) submodule of an  $\mathcal{O}_X$ -modules,
- (2) morphisms of  $\mathcal{O}_X$ -modules,
- (3) quotients of an  $\mathcal{O}_X$ -modules by an submodule,
- (4) kernel of a morphism of  $\mathcal{O}_X$ -modules,
- (5) image of a morphism of  $\mathcal{O}_X$ -modules,

**Exercise 25.21.** Let  $X$  be a quasi-projective variety. How can we view  $\bigoplus_{i=1}^n \mathcal{O}_X$  as an  $\mathcal{O}_X$ -module?

**Definition 25.22.** A sheaf of  $\mathcal{O}_X$ -modules  $\mathcal{F}$  is said to be **free of rank  $n$**  if  $\mathcal{F}$  is isomorphic to  $\bigoplus_{i=1}^n \mathcal{O}_X$  for some positive integer  $n$  as sheaves of  $\mathcal{O}_X$ -modules. There is a much less restrictive but related condition: a sheaf of  $\mathcal{O}_X$ -modules  $\mathcal{F}$  is said to be **locally free** if there exists an open covering  $\bigcup_i U_i$  of  $X$  on which, for every open set  $U_i$  in the covering, the restriction  $\mathcal{F}|_{U_i}$  is a free  $\mathcal{O}_X|_{U_i}$ -module.

**Exercise 25.23.** Show that if  $X$  is connected and  $\mathcal{F}$  is a locally free  $\mathcal{O}_X$ -module on  $X$ , then  $\mathcal{F}$  has the same rank everywhere.

**Exercise 25.24.** Define the tensor product  $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$  of two  $\mathcal{O}_X$ -modules to be the sheaf associated to the presheaf  $U \mapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$  (see Section ??). We often write  $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$  as  $\mathcal{F} \otimes \mathcal{G}$  and  $s \otimes t$  as  $st$  without confusion.

**Exercise 25.25.** Prove that the tensor product of two locally free  $\mathcal{O}_X$ -modules of rank  $n$  is again a locally free  $\mathcal{O}_X$ -module of rank  $n$ .

**Exercise 25.26.** Let  $\mathcal{F}$  and  $\mathcal{G}$  be locally free rank 1  $\mathcal{O}_X$ -modules on  $X$ . Show that the sheaf of homomorphisms  $\mathcal{H}om(\mathcal{F}, \mathcal{O}_X)$  (defined by  $U \mapsto \text{Hom}_{\mathcal{O}_X|U}(\mathcal{F}|_U, \mathcal{O}_X|_U)$ ) satisfies  $\mathcal{F} \otimes \mathcal{H}om(\mathcal{F}, \mathcal{O}_X) \cong \mathcal{O}_X$ .

**Definition 25.27.** A sheaf  $\mathcal{F}$  on a quasi-projective variety  $X$  is **quasi-coherent** if there is a cover  $X = \bigcup_i U_i$  of affine open subsets such that  $\mathcal{F}|_{U_i} = \widetilde{M_i}$  where  $M_i$  is the  $\mathcal{O}_X(U_i) = A(U_i)$  module  $\Gamma(U_i, \widetilde{M})$ . If moreover each  $M_i$  is a finitely generated  $A_i$  module, we say that  $\mathcal{F}$  is **coherent**.

In what follows we will be interested in coherent sheaves. Typical examples of coherent sheaves are the structure sheaf  $\mathcal{O}_X$  and any ideal sheaf of  $\mathcal{O}_X$  modules (i.e. any subsheaf of  $\mathcal{O}_X$ -modules  $\mathcal{I} \subset \mathcal{O}_X$ ).

**Exercise 25.28.** Prove that any ideal sheaf is coherent.

**Exercise 25.29.** Find an example of a sheaf which is not quasi-coherent.

**Exercise 25.30.** Prove that if  $X$  is an affine variety and  $\mathcal{F}$  is a quasi-coherent  $\mathcal{O}_X$ -module on  $X$ , then  $X$  can be covered by a finite collection of principle open subsets  $X_{f_i}$  such that for each  $i$ ,  $\mathcal{F}|_{X_{f_i}}$  is isomorphic to  $\widetilde{M_i}$  for some  $S_{f_i}^{-1}\Gamma(X, \mathcal{O}_X)$ -module  $M_i$  (where  $S_{f_i} = \{f_i^n | n \geq 0\}$  as in Exercise 25.7).

**Lemma 25.31.** Let  $X$  be an affine variety, let  $f \in \Gamma(X, \mathcal{O}_X)$ , let  $X_f$  be the corresponding principle open subset of  $X$ , and let  $\mathcal{F}$  be a quasi-coherent sheaf on  $X$ . If  $s$  is a global section of  $\mathcal{F}$  and  $s|_{X_f} = 0$ , then there is some positive integer  $n$  making  $f^n s = 0$  everywhere on  $X$ .

*Proof.* By Exercise 25.30, there is a finite cover of principal open subsets  $X_{f_i}$  over each of which  $\mathcal{F}$  restricts to  $\widetilde{M_i}$  for some  $S_{f_i}^{-1}\Gamma(X, \mathcal{O}_X)$ -module  $M_i$ . Let  $s_i$  be the restriction of  $s$  to each of these subsets, meaning  $s_i \in \Gamma(X_{f_i}, \widetilde{M_i}) = M_i$  for each  $i$ . Consider now the restriction of  $s_i$  to  $X_f \cap X_{f_i} = X_{ff_i}$ . By Proposition 25.18,  $s_i|_{X_{ff_i}}$  is an element of the localized module  $(M_i)_f$ , so looks like an element  $m_i/f^n$  for some  $m_i \in M_i$  and some non-negative integer  $n$ . Since  $s_i|_{X_{ff_i}} = 0$  by assumption,  $f^n s_i = 0$  in  $M_i$ . Since we are only using a finite collection of principal open subsets  $X_{f_i}$  in our cover, we can take the maximum integer  $n$  appearing as  $i$  ranges among these subsets.  $\square$

**Exercise 25.32.** Under the same assumptions as the above lemma, prove that given any section  $t \in \mathcal{F}(X_f)$  there is some positive integer  $n$  making  $f^n t$  extend to a global section of  $\mathcal{F}$  on  $X$ .

Consider a short exact sequence of sheaves

$$0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0.$$

Evaluating this over  $X$  (i.e. taking  $\Gamma(X, \dots)$ ) we obtain a sequence

$$0 \rightarrow \Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'') \rightarrow 0.$$

**Theorem 25.33.** The above sequence is always exact on the left (so that  $\Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F})$  is injective and  $\ker(\Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'')) = \text{im}(\Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F}))$ ) and if  $X$  is affine and  $\mathcal{F}'$  is quasi-coherent, then it is exact on the right (so that  $\Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'')$  is surjective).

*Proof.* (c.f. [Hartshorne, Proposition 5.6]) Let  $\phi : \mathcal{F}' \rightarrow \mathcal{F}$  be the first map and  $\psi : \mathcal{F} \rightarrow \mathcal{F}''$  be the second map in the exact sequence of sheaves  $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}'' \rightarrow 0$  given. We have that  $\phi$  is an injection,  $\psi$  is a surjection, and  $\text{Ker}(\psi) = \text{Im}(\phi)$ . We now consider the maps

$$\phi(X) : \Gamma(X, \mathcal{F}') \rightarrow \Gamma(X, \mathcal{F})$$

and

$$\psi(X) : \Gamma(X, \mathcal{F}) \rightarrow \Gamma(X, \mathcal{F}'').$$

Since  $\phi$  is an injection, we have automatically that  $\phi(X)$  is injective by definition. Furthermore, from Exercise 24.20, we know that for every point  $P \in X$ , the induced sequence on stalks  $0 \rightarrow \mathcal{F}'_P \rightarrow \mathcal{F}_P \rightarrow \mathcal{F}''_P \rightarrow 0$  is exact. We leave it as an exercise to show (using simply the exactness of this sequence on stalks, the definition of a stalk, and the definition of a sheaf) that  $\text{Ker}(\psi(U)) = \text{Im}(\phi(U))$  for any open set  $U \subset X$ . We are left to show that whenever  $X$  is affine and  $\mathcal{F}'$  is quasi-coherent, then  $\psi(X)$  is surjective.

Let  $s \in \Gamma(X, \mathcal{F}'')$  be any global section of  $\mathcal{F}''$ . Since  $\psi$  is surjective on sheaves, given any point  $P \in X$  there is a principal open set  $X_f$  containing  $P$  such that  $s|_{X_f}$  has a preimage  $t \in \mathcal{F}(X_f)$  (why?). We will first show that there exists an integer  $n$  so that the element  $f^n s \in \Gamma(X, \mathcal{F}'')$  has a some preimage in  $\Gamma(X, \mathcal{F})$ .

To this end, let

$$X = \bigcup_{i=1}^N X_{g_i}$$

be a finite covering of  $X$  using principle open subsets  $X_{g_i}$  such that on each  $X_{g_i}$ , the restriction of  $s$  has a preimage  $t_i \in \mathcal{F}(X_{g_i})$  (remind yourself why such a finite cover exists). Notice now that on each open set  $X_f \cap X_{g_i} = X_{fg_i}$  we have two elements  $t|_{X_{fg_i}}$  and  $t_i$  each mapping to  $s$  through the map  $\psi(X_{fg_i})$ . Then  $t - t_i$  is in the kernel of  $\psi(X_{fg_i})$ , so must be in the image of  $\phi(X_{fg_i})$ . Use this fact, the Lemma above with the fact that  $\mathcal{F}'$  is quasi-coherent, the gluing property of sheaves, and some ingenuity, to finish showing that the integer  $n$  desired above exists.

We now show the desired surjectivity of  $\psi(X)$ . Cover  $X$  with a finite cover  $\bigcup_{j=1}^r X_{f_j}$  of principal open sets  $X_{f_j}$  such that  $s|_{X_{f_j}}$  has a preimage in  $\mathcal{F}(X_{f_j})$ . We know that for each  $j$ , we can find an integer  $n_j$  such so that the element  $f_j^{n_j} s \in \Gamma(X, \mathcal{F}'')$  has a some preimage in  $\Gamma(X, \mathcal{F})$ . Let  $N = \max_j \{n_j\}$ . Then the element  $f_j^N s \in \Gamma(X, \mathcal{F}'')$  also has a some preimage

$t_j \in \Gamma(X, \mathcal{F})$  for each  $j$ . Since  $X = \bigcup_{j=1}^r X_{f_j}$  is a covering, we must have  $A = (f_1, \dots, f_r)$ , so in particular we can write  $1 = \sum_{j=1}^r a_j f_j$ . Convince yourself that  $t = \sum_{j=1}^r a_j t_j$  is the preimage we are looking for.  $\square$

## 26. BEZOUT'S THEOREM

Let  $C \subset \mathbb{C}^2$  be a curve in the affine plane, given by a polynomial  $f(x, y) \in \mathbb{C}[x, y]$ . We can write the polynomial  $f$  as a finite sum of *forms* of increasing degree, i.e.

$$f(x, y) = \sum_{i=m}^d f_i(x, y)$$

where each term  $f_i(x, y)$  is a sum only of monomials of degree  $i$ . Obviously, the *degree* of the polynomial  $f(x, y)$  is the nonnegative integer  $d$ . We define the *multiplicity of  $f$  at  $\theta$*  to be the nonnegative integer  $m$ . Given any point  $p = (a, b) \in \mathbb{C}^2$ , we can use a simple affine change of coordinates  $T(u, v) = (u + a, v + b)$  to define a notion of multiplicity for  $f$  at the point  $p$ .

**Definition 26.1.** We define the multiplicity of  $f$  at  $p = (a, b)$ , denoted  $m_p f$ , to be the multiplicity of  $f(x + a, y + b)$  at  $\theta$ .

**Exercise 26.2.** Show that we could equivalently define  $m_p f$  to be the degree of the first nonzero term appearing in the Taylor expansion of  $f$  centered at  $p$ .

**Exercise 26.3.** Show that  $m_p f = 1$  if and only if  $p$  is a nonsingular point of the curve  $C$ .

**Exercise 26.4.** Show that  $m_p f$  is always less than or equal to the degree of  $f$ .

**Exercise 26.5.** Let  $G : \mathbb{C}^2 \rightarrow \mathbb{C}^2$  be a map defined coordinatewise by polynomials that sends  $p \in \mathbb{C}^2$  to  $Q \in \mathbb{C}^2$ . Show that  $m_Q(f \circ G)$  is always greater than or equal to  $m_p f$ . What conditions are necessary for equality to be attained? Are these conditions sufficient?

We quickly recall the construction of the *local ring* at a point  $P$ .

**Definition 26.6.** Let  $X \subset \mathbb{C}^n$  be an affine variety and let  $p \in X$ . Define the local ring of  $X$  at  $p$ , denoted  $\mathcal{O}_p$ , to be the subring of the ring of rational function on  $X$  that are defined at  $p$ . Setting  $\mathfrak{m}_p$  for the maximal ideal at  $p$ , it is not hard to see that the set  $S = \mathcal{O}(X) \setminus \mathfrak{m}_p$  is a multiplicative set and  $\mathcal{O}_p = S^{-1}\mathcal{O}(X)$ .

**Exercise 26.7.** Characterize the value of a function  $f \in \mathcal{O}_p$  at  $p$  using the quotient map  $\mathcal{O}_p \rightarrow \mathcal{O}_p/\mathfrak{m}_p$ .

**Exercise 26.8.** Prove that  $\mathfrak{m}_p$  is the unique maximal ideal in  $\mathcal{O}_p$ .

**Theorem 26.9.** Let  $C \subset \mathbb{C}^2$  be an irreducible curve defined by  $f \in \mathbb{C}[x, y]$ . Then for any positive integer  $n$  sufficiently large,

$$m_P f = \dim_{\mathbb{C}} (\mathfrak{m}_p^n / \mathfrak{m}_p^{n+1}).$$

*Proof.* (c.f. [Fulton, Section 3.2]) First, notice that the sequence

$$0 \rightarrow \mathfrak{m}_p^n / \mathfrak{m}_p^{n+1} \rightarrow \mathcal{O}_p / \mathfrak{m}_p^{n+1} \rightarrow \mathcal{O}_p / \mathfrak{m}_p^n \rightarrow 0$$

is a short exact sequence. It is therefore enough to show that the dimension (as a complex vector space) of  $\mathcal{O}_p / \mathfrak{m}_p^n$  looks like  $n \cdot m_p f + s$  for every  $n$  and some constant  $s$  (why?). Using an



affine change of coordinates, we can assume that  $p$  is the origin, and so  $\mathfrak{m}^n = (x, y)^n \mathcal{O}_p$  and  $\mathcal{O}_p/\mathfrak{m}^n \cong \mathbb{C}[x, y]/((x, y)^n, f)$ . Notice that for any polynomial  $g \in (x, y)^{n-m_p f}$ ,  $fg \in (x, y)^n$ . Use this to show that there is a short exact sequence

$$0 \rightarrow \mathbb{C}[x, y]/(x, y)^{n-m_p f} \rightarrow \mathbb{C}[x, y]/(x, y)^n \rightarrow \mathbb{C}[x, y]/((x, y)^n, f) \rightarrow 0.$$

Compute the desired dimension of the third term of this sequence to be

$$n \cdot m_p f - \frac{m_p f(m_p f - 1)}{2}$$

for all  $n \geq m_p f$ . □

We say curves  $C, D \subset \mathbb{C}^2$  *intersect properly* at  $p \in \mathbb{C}^2$  if they have no common components passing through  $p$ . For two such curves, we now formalize an important multiplicity associated to a point  $p \in C \cap D$  in the intersection.

**Definition 26.10** (Definition-Theorem). *Let  $C, D \subset \mathbb{C}^2$  be curves defined respectively by polynomials  $f, g \in \mathbb{C}[x, y]$  and let  $p \in \mathbb{C}^2$  be a point, and denote  $\mathcal{O} = \mathcal{O}_{\mathbb{C}^2}$ . There exists a unique intersection multiplicity  $I_p(f, g)$  satisfying Fulton's [Fulton] seven axioms:*

- (1)  $I_p(f, g) \in \mathbb{Z}_{\geq 0}$  when  $C$  and  $D$  intersect properly at  $p$ .  $I_p(f, g) = \infty$  otherwise;
- (2)  $I_p(f, g) = 0$  if and only if  $p \notin C \cap D$ ;
- (3) for any affine change of coordinates  $T : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ ,  $I_p(f, g) = I_{T^{-1}(p)}(f \circ T, g \circ T)$ ;
- (4)  $I_p(f, g) = I_p(g, f)$ ;
- (5)  $I_p(f, g) \geq m_p f \cdot m_p g$  with equality if and only if  $C$  and  $D$  have no tangent lines at  $p$  in common;
- (6) If  $f = \prod_i f_i^{r_i}$  and  $g = \prod_j g_j^{s_j}$  then  $I_p(f, g) = \sum_{i,j} r_i s_j I_p(f_i, g_j)$ ;
- (7)  $I_p(f, g) = I_p(f, g + af)$  for any  $a \in \mathbb{C}[x, y]$ ;

given by

$$I_p(f, g) = \dim_{\mathbb{C}} (\mathcal{O}_p/(f, g))$$

*Proof. Proof of uniqueness:* Assuming it exists, there is an inductive procedure for calculating the value of  $I_p(f, g)$  using the axioms above. Using (1), (2), and (3), we may assume it is finite and nonzero, and that  $p$  is the origin. Assume that  $I_p(f, g) = k > 0$  and that any intersection multiplicity values less than  $k$  can always be calculated. Let  $r$  and  $s$  be the respective nonnegative degrees of the single variable polynomials  $f(x, 0)$  and  $g(x, 0)$ , assuming  $r \leq s$  without loss of generality.

If  $r = 0$ , then  $f(x, y) = yh(x, y)$  for some  $h \in \mathbb{C}[x, y]$ . By (6),

$$I_p(f, g) = I_p(y, g) + I_p(h, g).$$

In this case, use the axioms to show that  $I_p(y, g)$  is equal to the multiplicity of  $x$  in  $g(x, 0)$ , which must be positive since  $g(p) = 0$  by assumption. Thus  $I_p(h, g) < k$  and we have computed  $I_p(f, g)$  by induction. If  $r \geq 0$  we can reduce to the case  $r = 0$  in a finite number of steps using (7) and (4). We leave this as an exercise.

**Proof of existence:** Show that the definition

$$I_p(f, g) = \dim_{\mathbb{C}} (\mathcal{O}_p/(f, g))$$

satisfies the axioms. This is some difficult algebra. Try as many axioms as you can, and then read through to the proof of Bézout's theorem below, where we provide another good definition for  $I_p(f, g)$  which should be easier to show satisfies these axioms. □

**Exercise 26.11.** Show that any two parallel lines in  $\mathbb{C}^2$  do not intersect. Homogenize the equations for those lines, and count the number of resulting intersections in  $\mathbb{P}^2$ .

**Exercise 26.12.** How many points of intersection are there between the line  $\mathbb{V}(x - a)$  and the curve  $\mathbb{V}(x^2 + y^2 - 1)$  in  $\mathbb{C}^2$ ? What happens at  $a = 1$ ? What happens when you homogenize and study the intersection in  $\mathbb{P}^2$ ?

**Theorem 26.13** (Bézout's Theorem). Let  $C, D \subset \mathbb{P}^2$  be projective plane curves defined by homogeneous polynomials  $f, g \in \mathbb{C}[x, y, z]$  of degrees  $m$  and  $n$  respectively. Assume  $C$  and  $D$  share no common components. Then

$$\sum_p I_p(f, g) = mn.$$

The value  $I_p(f, g)$  always makes sense since it is defined locally. We can interpret  $f$  and  $g$  as dehomogenized polynomials in two variables in affine coordinates, using one of the three basic affine open sets of the projective plane containing  $p$ .

**Definition 26.14.** Let  $R$  be a ring, and let  $f = \sum_{i=0}^n a_i x^i$  and  $g = \sum_{j=0}^m b_j x^j$  be polynomials in a polynomial ring  $R[x]$  of one variable. The resultant  $\text{Res}(f, g; x)$  is defined to be the determinant of the Sylvester matrix which, by example in the case  $m = 5$  and  $n = 3$ , is the determinant of the  $(m + n) \times (m + n)$  matrix:

$$\begin{vmatrix} a_5 & a_4 & a_3 & a_2 & a_1 & a_0 & 0 & 0 \\ 0 & a_5 & a_4 & a_3 & a_2 & a_1 & a_0 & 0 \\ 0 & 0 & a_5 & a_4 & a_3 & a_2 & a_1 & a_0 \\ b_3 & b_2 & b_1 & b_0 & 0 & 0 & 0 & 0 \\ 0 & b_3 & b_2 & b_1 & b_0 & 0 & 0 & 0 \\ 0 & 0 & b_3 & b_2 & b_1 & b_0 & 0 & 0 \\ 0 & 0 & 0 & b_3 & b_2 & b_1 & b_0 & 0 \\ 0 & 0 & 0 & 0 & b_3 & b_2 & b_1 & b_0 \end{vmatrix}$$

**Exercise 26.15.** Find the resultant of  $x - a$  and  $x - b$  for  $a, b \in \mathbb{C}$ . Show that  $\text{Res}(f, g; x) = 0$  if and only if  $a = b$ .

**Exercise 26.16.** Find the resultant of  $x - a$  and  $(x - b)(x - c)$  for  $a, b, c \in \mathbb{C}$ . Show that  $\text{Res}(f, g; x) = 0$  if and only if  $a = b$  or  $a = c$ .

**Exercise 26.17.** If  $f, g \in \mathbb{C}[x_1, \dots, x_n]$ , show that  $\text{Res}(f, g; x_i) \in \mathbb{C}[x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n]$ .

**Exercise 26.18.** Let  $A$  be the Sylvester matrix of  $f$  and  $g$ . Assume  $f(a) = g(a) = 0$  for some  $a$ . Show that the vector  $\mathbf{x} = (a^{m+n-1}, a^{m+n-2}, \dots, a^1, a^0)^T$  satisfies  $A\mathbf{x} = 0$ . Conclude that  $\text{Res}(f, g; x) = 0$ .

**Exercise 26.19.** Assume  $f(a) = g(a) = 0$  for some  $a$ . Prove that  $\frac{f}{g} = \frac{p}{q}$  for some polynomials  $p$  and  $q$ , each of which is exactly one degree lower. Use this to show that  $f$  and  $g$  share a root if and only if the resultant is zero.

**Exercise 26.20.** Let  $f, g \in \mathbb{C}[x, y, z]$  be homogeneous polynomials of degrees  $m$  and  $n$ , respectively, with  $f(0, 0, 1) \neq 0$  and  $g(0, 0, 1) \neq 0$ . Show their resultant  $\text{Res}(f, g; z)$  (thinking of  $x$  and  $y$  as constants) is homogeneous of degree  $mn$  in  $x$  and  $y$ .

**Exercise 26.21.** Let  $h \in \mathbb{C}[x, y]$  be a nonzero homogeneous polynomial. Prove that  $h$  can be written in the form

$$h(x, y) = c \prod_{i=1}^t (s_i x - r_i y)^{m_i}$$

where  $c$  is a nonzero constant and  $[r_i : s_i]$  are distinct points of  $\mathbb{P}^1$ . Show that

$$\mathbb{V}(h) = \{[r_1 : s_1], \dots, [r_t : s_t]\} \subset \mathbb{P}^1.$$

*Proof. Proof of Bézout's Theorem:* Using an affine transformation of  $\mathbb{C}^3$ , choose coordinates of  $\mathbb{P}^2$  so that  $[0 : 0 : 1]$  is not in  $C$  or  $D$  and, furthermore, is not colinear with any two other points of  $C \cap D$ . Let  $p \in C \cap D$  be described using projective coordinates  $[u : v : w]$ . Prove that another good definition for  $I_p(f, g)$  is the exponent of  $(vx - uy)$  in the factorization of  $\text{Res}(f, g; z)$  given in the above exercise (show it satisfies the seven axioms). Using this and the previous two exercises, you should be able to deduce Bézout's Theorem.  $\square$

## 27. RIEMANN-ROCH THEOREM FOR CURVES

Let  $C$  be a smooth, projective curve over  $\mathbb{C}$  of topological genus  $g$ .

**Definition 27.1.** A divisor  $D$  on  $C$  is a finite linear combination of points  $p_i \in C$ ,

$$D = \sum_i n_i [p_i],$$

with integer coefficients  $n_i \in \mathbb{Z}$ . Under the assumption that only a finite number of coefficients may be non-zero, we may also write

$$D = \sum_{p \in C} n_p [p]$$

for convenience. We say that  $D$  is an effective divisor (and write  $D \geq 0$ ) if all the  $n_i \geq 0$ . We say the degree of  $D$  is the integer  $\deg D = \sum_i n_i$ . For two divisors  $D = \sum_{p \in C} n_p [p]$  and  $D' = \sum_{p \in C} m_p [p]$ , we write  $D \geq D'$  if and only if  $n_p \geq m_p$  for all  $p \in C$ . Given a rational function  $f$  on  $C$ , we can always write

$$\text{div}(f) = \sum_{p \in C} \text{ord}_p(f) [p]$$

where

$$\text{ord}_p(f) = \begin{cases} a & \text{if } f \text{ has a zero of multiplicity } a \text{ at } p \\ -a & \text{if } f \text{ has a pole of order } a \text{ at } p \\ 0 & \text{otherwise.} \end{cases}$$

**Exercise 27.2.** Show that every rational function  $f$  on  $C$  has only a finite number of zeros and poles, so that  $\text{div}(f)$  is a well-defined divisor.

**Exercise 27.3.** Show that the set of divisors on  $C$  form a free abelian group.

**Exercise 27.4.** Show that  $\text{div}(fg) = \text{div}(f) + \text{div}(g)$ . Show that  $\text{div}(f^{-1}) = -\text{div}(f)$ .

**Exercise 27.5.** Show that for any rational function  $f$  on  $C$ ,  $\deg \text{div}(f) = 0$

**Exercise 27.6.** Let  $f$  be a nonzero rational function on  $C$ . Show that  $\text{div}(f)$  is effective if and only if  $f$  is a constant function.

**Exercise 27.7.** Let  $f$  and  $g$  be nonzero rational functions on  $C$ . Show that  $\text{div}(f) = \text{div}(g)$  if and only if  $f = kg$  for some nonzero constant  $k \in \mathbb{C}$ .

**Definition 27.8.** Let  $D = \sum_{p \in C} n_p [p]$  be a divisor on  $C$  and consider the sheaf of regular functions  $\mathcal{O}_C$  on  $C$ . Define  $\mathcal{O}_C(D)$  to be the sheaf of functions on  $C$  whose zeros and poles are bounded locally by  $D$ , meaning that on an open set  $U \subset C$ ,

$$\mathcal{O}_C(D)(U) = \{f \in \mathbb{C}(U) \mid \text{ord}_P(f) \geq n_p \text{ for all } p \in U\}$$

**Exercise 27.9.** Prove that  $\mathcal{O}_C(D)$  is always a locally free sheaf of rank 1 (i.e. the sheaf of sections of a line bundle).

**Exercise 27.10.** Show that the set  $L(D) := \Gamma(C, \mathcal{O}_C(D))$  may be characterized as the set of all global rational function on  $C$  such that  $\text{div}(f) + D$  is effective (or  $f$  is zero). Convince yourself of the following facts:

- (1) If  $D \geq D'$  then  $L(D) \supseteq L(D')$  and  $\dim_{\mathbb{C}}(L(D)/L(D')) \leq \deg(D - D')$ .
- (2)  $L(0) = \mathbb{C}$  and  $L(D) = 0$  whenever  $\deg D < 0$ .

**Exercise 27.11.** Show that  $L(D)$  is always a finite-dimensional complex vector space over  $\mathbb{C}$ . Denote  $l(D) := \dim_{\mathbb{C}} L(D)$ .

Our goal is to better understand the vector spaces  $L(D)$ , their dimension, and how they compare for different choices of  $D$ . One fact to notice is the following.

**Exercise 27.12.** Show that if  $D$  and  $D'$  are divisors on  $C$  with  $D - D' = \text{div}(f)$  for some rational function  $f$ , then multiplication by  $f$  induces an isomorphism between  $L(D)$  and  $L(D')$ .

We would like to work in a sense modulo this isomorphism, motivating the following important definition.

**Definition 27.13.** Two divisors  $D$  and  $D'$  on  $C$  are said to be linearly equivalent if  $D - D' = \text{div}(f)$  for some rational function  $f$ , in which case we write  $D \sim D'$ .

**Exercise 27.14.** Show that:

- (1) linear equivalence is indeed an equivalence relation on the group of divisors on  $C$ ;
- (2)  $D \sim 0$  if and only if  $D = \text{div}(f)$  for some  $f$ ;
- (3) if  $D \sim D'$  then  $\deg D = \deg D'$ ;
- (4) if  $D \sim D'$  and  $E \sim E'$  then  $D + E \sim D' + E'$ .

Essentially, the fundamental collection of objects we want to study is the set of divisors on  $C$  modulo rational equivalence.

**Exercise 27.15.** Denote the set of divisors on  $C$  modulo rational equivalence by  $\text{Div}(C)/\sim$ . Show it is a group.

**Exercise 27.16.** Show that if  $D \sim D'$  then  $\mathcal{O}_C(D) \cong \mathcal{O}_C(D')$  as sheaves. Denote by  $\text{Pic}(C)$  the set of isomorphism classes of locally free rank one sheaves on  $C$ . Show that the mapping  $D \mapsto \mathcal{O}_C(D)$  induces a one-to-one correspondence between  $\text{Div}(C)/\sim$  and  $\text{Pic}(C)$ .

**Exercise 27.17.** *Prove that  $\text{Pic}(C)$  is a group under tensor product. Is the correspondence in the previous question a group homomorphism? This uses the exercises 25.24, 25.25, and 25.26.*

One important locally free rank one sheaf on  $C$  is the sheaf of sections of the cotangent bundle, the cotangent sheaf  $\Omega_C$ . Since  $C$  has dimension one, the cotangent sheaf is its own top exterior power, so we will denote it by  $\Omega_C = \omega_C$ . We denote the divisor corresponding to  $\omega_C$  under the correspondence between  $\text{Div}(C)/\sim$  and  $\text{Pic}(C)$  by  $K_C$ , the *canonical divisor*. We are now ready to state the Riemann-Roch Theorem, which relates the dimension of  $L(D)$  to the degree of  $D$  and genus  $g$  of  $C$  using  $K_C$ .

**Theorem 27.18** (Riemann-Roch Theorem).

$$l(D) - l(K_C - D) = \deg D + 1 - g$$

We first note a couple of facts about the sheaves  $\mathcal{O}_C$  provided to us by a famous and useful theorem called Serre duality. For proofs of these facts, we refer the reader to [Hartshorne, Chapter 4]

**Theorem 27.19.**

$$(1) \ H^1(C, \mathcal{O}_C) \simeq H^0(C, \omega_C), \text{ and}$$

$$(2) \ H^1(C, \mathcal{O}_C(D)) \simeq H^0(C, \omega_C \otimes \mathcal{H}om(\mathcal{O}_C(D), \mathcal{O}_C)).$$

Notice, in particular, that since the topological genus  $g$  of our curve  $C$  can be defined as the dimension of  $H^0(C, \omega_C)$ , this theorem tells us that it is also the dimension of  $H^1(C, \mathcal{O}_C)$ . Notice also that the second isomorphism tells us that the left hand side of the Riemann-Roch Theorem is nothing more than  $\chi(\mathcal{O}_C(D))$ .

*Proof. Proof of Riemann-Roch Theorem* We need to show that

$$\chi(\mathcal{O}_C(D)) = \deg D + 1 - g$$

where

$$\chi(\mathcal{O}_C(D)) = \dim H^0(C, \mathcal{O}_C(D)) - \dim H^1(C, \mathcal{O}_C(D)).$$

As a quick exercise, show that this is true in the case  $D = 0$ . Now, let  $D$  be any divisor and  $p \in C$  be any point. To finish the proof, it is enough to show that the formula is true for  $D$  if and only if it is true for  $D + [p]$  (why?).

Consider the *skyscraper sheaf*  $\mathbb{C}(p)$  at  $p$ , that is, the sheaf given by

$$U \mapsto \begin{cases} \mathbb{C} & \text{if } p \in U \\ 0 & \text{otherwise.} \end{cases}$$

Prove that the following sequence is exact:

$$0 \rightarrow \mathcal{O}_C(-[p]) \rightarrow \mathcal{O}_C \rightarrow \mathbb{C}(p) \rightarrow 0.$$

If you tensor the entire sequence with  $\mathcal{O}_C(D + [p])$ , what is the result? Prove that it is the short exact sequence

$$0 \rightarrow \mathcal{O}_C(D) \rightarrow \mathcal{O}_C(D + [p]) \rightarrow \mathbb{C}(p) \rightarrow 0.$$

By Lemma ??, this gives

$$\begin{aligned}\chi(\mathcal{O}_C(D + [p])) &= \chi(\mathcal{O}_C(D)) + \chi(\mathbb{C}(p)) \\ \chi(\mathcal{O}_C(D + [p])) &= \chi(\mathcal{O}_C(D)) + 1\end{aligned}$$

but  $\deg(D + [p]) = \deg D + 1$ , so the formula is true for  $D + [p]$  if and only if it is true for  $D$ .  $\square$

## 28. TENSOR PRODUCTS

**Definition 28.1.** Let  $R$  be a ring and  $M, N$  be  $R$  modules, then the **tensor product**  $M \otimes_R N$  is the module generated by

$$\{m \otimes n | m \in M, n \in N\} / \sim$$

where  $\sim$  is the equivalence relation induced by

$$(am + a'm') \otimes (bn + bn') = ab(m \otimes n) + a'b(m' \otimes n) + ab'(m \otimes n') + a'b'(m' \otimes n')$$

for all  $m, m' \in M, n, n' \in N$  and  $a, a', b, b' \in R$ . (In other words  $M \otimes_R N$  is the quotient of the free module  $M \times N$  by the relation  $\sim$ ).

**Exercise 28.2.** Show that the above equivalence relation is equivalent to requiring that

- (1)  $(m, n) + (m', n) \sim (m + m', n)$  for  $m, m' \in M$  and  $n \in N$ ,
- (2)  $(m, n) + (m, n') \sim (m, n + n')$  for  $m \in M$  and  $n, n' \in N$ ,
- (3)  $r(m, n) \sim (rm, n) \sim (m, rn)$  for  $r \in R, m \in M$  and  $n \in N$ .

We have a natural bilinear map

$$\pi : M \times N \rightarrow M \otimes_R N, \quad \pi(m \times n) = m \otimes n$$

by which we mean a map of sets such that

$$\pi((am + a'm'), (bn + bn')) = ab\pi(m, n) + a'b\pi(m', n) + ab'\pi(m, n') + a'b'\pi(m', n').$$

**Exercise 28.3.** Show that the set of  $R$  module homomorphisms  $\phi : M \otimes_R N \rightarrow Q$  is in one to one correspondence with the set of bilinear maps  $\psi : M \times_R N \rightarrow Q$ .

**Exercise 28.4.** We say that an element  $v \in M \otimes_R N$  is a *simple tensor* if  $v = m \otimes n$  for some  $m \in M$  and  $n \in N$ . Show that not all tensors are simple.

**Exercise 28.5.** Show that  $M \otimes_R R \cong R \otimes_R M$  and  $M \otimes_R N \cong N \otimes_R M$ .

**Exercise 28.6.** Show that  $(\bigoplus_{i=1}^n M_i) \otimes (\bigoplus_{j=1}^n N_j) \cong \bigoplus_{i,j} M_i \otimes N_j$ .

**Exercise 28.7.** Show that if  $M, N, P$  are  $R$ -modules, then  $(M \otimes_R N) \otimes_R P \cong M \otimes_R (N \otimes_R P)$ .

**Exercise 28.8.** If  $\phi : M \rightarrow M'$  and  $\psi : N \rightarrow N'$  are homomorphisms of  $R$ -modules, then there is a natural induced homomorphism  $\Phi : M \otimes N \rightarrow M' \otimes N'$  such that the induced maps  $M \rightarrow M \otimes N \rightarrow M' \otimes N' \rightarrow M'$  and  $N \rightarrow M \otimes N \rightarrow M' \otimes N' \rightarrow N'$  coincide with  $\phi$  and  $\psi$ .

**Exercise 28.9.** Show that the tensor product is right exact so that if

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

is a short exact sequence of  $R$  modules and  $N$  is an  $R$  module, then

$$M' \otimes N \rightarrow M \otimes N \rightarrow M'' \otimes N \rightarrow 0$$

is exact.

**Exercise 28.10.** If  $R, S$  are rings,  $\phi : R \rightarrow S$  is a homomorphism and  $M$  is an  $S$  module, then  $M \otimes_S R$  is an  $R$  module.

**Exercise 28.11.** Let  $R$  be a ring,  $S$  a multiplicative subset of  $R$  and  $M$  an  $R$  module. Show that  $S^{-1}R \otimes_R M \cong S^{-1}M$ .

**Definition 28.12.** We say that an  $R$  module  $M$  is **flat** if given any injective homomorphism  $N' \rightarrow N$  of  $R$ -modules, then the induced homomorphism  $M \otimes_R N' \rightarrow M \otimes_R N$  is also injective.

**Exercise 28.13.** Show that  $M$  is a flat  $R$ -module if and only if the tensor product by  $M$  is exact so that if

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$$

is a short exact sequence of  $R$  modules and  $N$  is an  $R$  module, then

$$0 \rightarrow M \otimes N' \rightarrow M \otimes N \rightarrow M \otimes N'' \rightarrow 0$$

is a short exact sequence.

**Lemma 28.14.** If  $S$  is a multiplicative subset of a ring  $R$ , then  $S^{-1}R$  is flat.

*Proof.* Let  $\phi : M' \rightarrow M$  be an injective homomorphism, then we must show that  $\phi : S^{-1}M' \rightarrow S^{-1}M$  is injective (see Exercise 28.11). Suppose that  $\phi(m'/s) = 0$ , then  $\phi(m')t = 0$  for some  $t \in S$ . But then  $\phi(tm') = 0$  and so  $tm' = 0$  (as  $\phi$  is injective). But then  $m'/s = 0$  in  $S^{-1}M'$  as required.  $\square$

## 29. FIBER PRODUCTS

We will next introduce the notion of fibered products. Unluckily, in order to define this notion we need to consider a more general setting.

**Definition 29.1.** A **ringed space**  $(X, \mathcal{O}_X)$  is a topological space  $X$  together with a sheaf of rings  $\mathcal{O}_X$  on  $X$ . A **morphism of ringed spaces**  $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  is a continuous map  $f : X \rightarrow Y$  and a homomorphism  $f^\# : \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$  of sheaves of rings on  $Y$ . We say that the ringed space  $(X, \mathcal{O}_X)$  is a **locally ringed space** if each stalk  $\mathcal{O}_{X,P}$  is a local ring and a morphism of ringed spaces  $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$  is a **morphism of locally ringed spaces** if for any  $P \in X$ , the map  $f_P^\# : \mathcal{O}_{Y,f(P)} \rightarrow \mathcal{O}_{X,P}$  is a local homomorphism of local rings.

**Exercise 29.2.** Explain how the map  $f_P^\# : \mathcal{O}_{Y,f(P)} \rightarrow \mathcal{O}_{X,P}$  is defined and show that it is a local homomorphism of local rings (i.e.  $(f^\#)^{-1}\mathfrak{m}_P = \mathfrak{m}_{f(P)}$  where  $\mathfrak{m}_P$  denotes the maximal ideal of  $\mathcal{O}_{X,P}$  and  $\mathfrak{m}_{f(P)}$  denotes the maximal ideal of  $\mathcal{O}_{Y,f(P)}$ ).

**Definition 29.3.** let  $A$  be a ring, then the corresponding **affine scheme**  $(\text{Spec}(A), \mathcal{O})$  is defined as follows:

- (1) The **spectrum** of  $A$  is the set  $\text{Spec}(A)$  given by the union of all prime ideals of  $A$ . The topology on  $\text{Spec}(A)$  is defined by letting the closed sets be of the form  $\mathbb{V}(a) = \{P \supset a\}$  be the union of all prime ideals containing an ideal  $a \subset A$ .
- (2) The sheaf of rings  $\mathcal{O}$  on  $\text{Spec}(A)$  is defined by

$$\mathcal{O}(U) = \{s : U \rightarrow \prod_{P \in U} A_P\}$$

where  $U \subset \operatorname{Spec}(A)$  is an open subset and  $s$  are functions such that  $s(P) \in A_P$  for any  $P \in U$  and for any  $P \in U$  there exists a neighborhood  $P \in V \subset U$  such that  $s(Q) = a/f \in A_Q$  for all  $Q \in V$  where  $a, f \in A$  and  $f(Q) \neq 0$  for all  $Q \in V$ .

**Exercise 29.4.** Show that the sets  $\mathbb{V}(a)$  define a topology on  $\operatorname{Spec}(A)$ .

**Exercise 29.5.** Let  $A = \mathbb{C}[x]$ , then show that the points of  $\operatorname{Spec}(A)$  correspond to the maximal ideals  $\mathfrak{m}_a = (x - a) \subset \mathbb{C}[x]$  and the zero ideal. Show that the zero ideal is the only non-closed point. What is its closure.

**Exercise 29.6.** What are the points (and their closures) of  $X = \operatorname{Spec}(\mathbb{C}[x, y])$ .

**Exercise 29.7.** Show that any affine scheme is a locally ringed space.

**Exercise 29.8.** Show that if  $X$  and  $Y$  are affine schemes corresponding to  $\mathbb{C}$ -algebras  $A, B$ , then there is a natural bijection between the  $\mathbb{C}$  algebra homomorphisms  $\phi : B \rightarrow A$  and morphisms of locally ringed spaces  $f : X \rightarrow Y$ .

**Exercise 29.9.** Let  $(X, \mathcal{O}_X)$  be a scheme. Show that there is a natural morphism  $(X, \mathcal{O}_X) \rightarrow \operatorname{Spec}(\mathbb{Z})$ .

**Exercise 29.10.** Let  $a \subset A$  be an ideal,  $X = \operatorname{Spec}(A)$  and  $Y = \operatorname{Spec}(A/a)$ . Show that the surjective homomorphism  $a : A \rightarrow A/a$  induces a closed immersion  $Y \rightarrow X$  (i.e. a morphism of schemes which is a homeomorphism on the topological space  $Y$  and such that the induced map of sheaves on  $X$ ,  $f^\# : \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$  is surjective).

Note that different ideals  $a, a' \subset A$  may define the same topological space but have different associated scheme structures. For example  $(x^n) \subset \mathbb{C}[x]$  defines a scheme whose topological space is just the origin in  $\mathbb{C}$ , but has scheme structure given by  $\mathcal{O} = \mathbb{C}[x]/(x^n)$ . We think of the corresponding scheme as the  $n$ -th infinitesimal neighborhood of the origin.

For another example, consider the ideals  $(x, y), (x^2, y^2), (x^2, y^2, xy) \subset \mathbb{C}[x, y]$  defining different scheme structures on  $(0, 0) \subset \mathbb{C}^2$ .

Finally consider  $(x^2, xy) \subset \mathbb{C}[x, y]$ . The topological space corresponding to this scheme is given by the  $y$ -axis, however there is a non-reduced structure at the origin corresponding to a non-zero tangent direction parallel to  $x$ -axis.

**Definition 29.11.** A **scheme**  $(X, \mathcal{O}_X)$  is a locally ringed space such that  $X = \cup U_i$  where  $(U_i, \mathcal{O}_X|_{U_i})$  are affine schemes.

**Definition 29.12.** If  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$  are schemes and  $U \subset X$  and  $V \subset Y$  are open subsets such that  $(U, \mathcal{O}_X|_U)$  and  $(V, \mathcal{O}_Y|_V)$  are isomorphic, then there exists a scheme  $(Z, \mathcal{O}_Z)$  and open subsets  $Z_i$  such that  $(Z_1, \mathcal{O}_Z|_{Z_1}) \cong (U, \mathcal{O}_X|_U)$  and  $(Z_2, \mathcal{O}_Z|_{Z_2}) \cong (V, \mathcal{O}_Y|_V)$ . we say that  $(Z, \mathcal{O}_Z)$  is obtained by **glueing the schemes**  $(X, \mathcal{O}_X)$  and  $(Y, \mathcal{O}_Y)$  along  $(U, \mathcal{O}_X|_U) \cong (V, \mathcal{O}_Y|_V)$ .

**Exercise 29.13.** Let  $(X, \mathcal{O}_X) = (Y, \mathcal{O}_Y) = \operatorname{Spec}(\mathbb{C}[x])$  and  $U \cong V = \operatorname{Spec}(\mathbb{C}[x, x^{-1}])$ . Show that the scheme  $Z$  obtained by glueing  $(X, \mathcal{O}_X), (Y, \mathcal{O}_Y)$  along  $U \cong V$  is isomorphic to the affine line with the origin replaced by a double point.

**Definition 29.14.** Let  $f : X \rightarrow S$  and  $g : Y \rightarrow S$  be morphisms of affine varieties so that  $A(X) = A$  and  $A(Y) = B$  are rings and  $f, g$  correspond to homomorphisms  $\phi : R \rightarrow A$  and  $\psi : R \rightarrow B$ . Define the **fiber product**  $X \times_S Y$  to be the affine variety corresponding to the ring  $A \otimes_R B$  and morphisms  $\alpha : X \times_S Y \rightarrow X$  and  $\beta : X \times_S Y \rightarrow Y$  determined by



the homomorphisms  $A \rightarrow A \otimes_R B$  and  $B \rightarrow A \otimes_R B$  which send  $a \rightarrow a \otimes 1$  and  $b \rightarrow 1 \otimes b$  respectively.

**Lemma 29.15.** *Let  $Z$  be an affine variety and  $a : Z \rightarrow X$ ,  $b : Z \rightarrow Y$  be  $S$ -morphisms (i.e. morphisms such that the  $f \circ a = g \circ b$ ), then there exists a unique morphism  $c : Z \rightarrow X \times_S Y$  such that  $\alpha \circ c = a$  and  $\beta \circ c = b$ .*

*Proof.* Let  $T = A(Z)$ , then the morphisms  $a, b$  correspond to homomorphisms  $A \rightarrow T$  and  $A \otimes_R B \rightarrow B$ . These homomorphisms are compatible with the  $S$ -module structure  $T \rightarrow A$  and  $T \rightarrow B$  and hence we obtain a natural homomorphism  $T \rightarrow A \otimes_R B$ .  $\square$

One can define the fiber product of two morphisms of quasi-projective varieties  $f : X \rightarrow S$  and  $g : Y \rightarrow S$  as follows.

**Definition 29.16.** *The fiber product of  $X, Y$  over  $S$  is a quasi-projective variety  $Z$  with morphisms  $\alpha : Z \rightarrow X$  and  $\beta : Z \rightarrow Y$  (compatible with the structure maps  $X \rightarrow S$  and  $Y \rightarrow S$  so that  $f \circ \alpha = g \circ \beta$ ) such that for any quasi-projective variety  $W$  with morphisms  $a : W \rightarrow X$  and  $b : W \rightarrow Y$  (compatible with the structure maps  $X \rightarrow S$  and  $Y \rightarrow S$  so that  $f \circ a = g \circ b$ ), then there exists a unique morphism  $c : W \rightarrow Z$  such that  $\alpha \circ c = a$  and  $\beta \circ c = b$ . We denote the fiber product by  $z = X \times_S Y$ .*

One can use the above lemma to show the following.

**Theorem 29.17.** *Let  $f : X \rightarrow S$  and  $g : Y \rightarrow S$  be morphisms of quasi-projective varieties, then the fiber product  $X \times_S Y$  exists.*

*Proof.* [Hartshorne, II.3.3].  $\square$

We can use fibered products to give a definition of fibers of a morphism.

**Definition 29.18.** *let  $f : X \rightarrow Y$  be a morphism of schemes and  $y \in Y$  a point (corresponding to a morphism  $\text{Spec}(\mathbb{C}) \rightarrow Y$ ), then the fiber of  $f$  over  $y$  is  $X_y := X \times_Y y$ .*

We make the following observation. Even if we assume that  $X$  is a quasi-projective variety, it may happen that the fiber  $X_y$  is not a quasi-projective variety. Consider for example (cf. [Hartshorne, III.3.3.1])  $X = \text{Spec}(\mathbb{C}[x, y, t]/(ty - x^2))$  and  $Y = \text{Spec}(\mathbb{C}[t])$ . Let  $f : X \rightarrow Y$  be the morphism induced by the inclusion  $\mathbb{C}[t] \rightarrow \mathbb{C}[x, y, t]/(ty - x^2)$ , then for any  $a \in \mathbb{C}$ , the fiber  $X_a$  is determined by  $X = \text{Spec}(\mathbb{C}[x, y]/(ay - x^2))$ . For  $a \neq 0$ , this is a plane curve of degree 2 in  $\mathbb{C}^2$ , but for  $a = 0$ , we obtain a double line.

### 30. MONOMIAL ORDERS

Recall that a monomial in  $\mathbb{C}[x_1, \dots, x_n]$  is simply an expression of the form  $x_1^{a_1} \cdots x_n^{a_n} = x^a$  where  $a = (a_1, \dots, a_n) \in \mathbb{Z}_{\geq 0}^n$ . A **monomial ordering**  $>$  on  $\mathbb{C}[x_1, \dots, x_n]$  is a relation on the set of all monomials such that

- (1)  $\geq$  is a **total order**, i.e. for any monomials  $x^a, x^b$  and  $x^c$  ( $a, b, c \in \mathbb{Z}_{\geq 0}^n$ )
  - if  $x^a < x^b$  and  $x^b < x^c$ , then  $x^a < x^c$ , and
  - for any two monomials  $x^a$  and  $x^b$  then one and only one of the following is true:  $x^a < x^b$ ,  $x^b < x^a$  or  $x^a = x^b$ .
- (2) for any monomials  $x^a, x^b$  and  $x^c$  ( $a, b, c \in \mathbb{Z}_{\geq 0}^n$ ), if  $x^a < x^b$ , then  $x^{a+c} < x^{b+c}$  (the order respect multiplication), and

- (3) (**Well ordering**) any decreasing sequence of elements  $x^{a_1} > x^{a_2} > \dots$  is finite (or equivalently, for any non-empty subset of monomials  $\{x^{a_i}\}_{i \in I}$  there is a smallest element say  $x^{a_0}$ ).

Note that we can equivalently talk about the corresponding order on  $\mathbb{Z}_{\geq 0}^n$ . We will switch from one to the other often without any further comment. It is easy to see that the ordering on  $\mathbb{C}[x]$  given by degree:

$$x^a > x^b \quad \text{if and only if} \quad a > b$$

is a monomial order. However, the relation  $\mathbb{C}[x, y]$  given by degree

$$x^a y^b > x^c y^d \quad \text{if and only if} \quad a + b > c + d$$

is not a monomial ordering. It is however possible to give many monomial orderings on  $\mathbb{C}[x_1, \dots, x_n]$ . For example we have

**Definition 30.1. Lexicographic order:** Given  $a, b \in \mathbb{Z}_{\geq 0}^n$ , we say  $a >_{\text{lex}} b$  if the first non-zero entry in  $a - b$  is  $\geq 0$ . Thus

$$x_1^3 x_2 >_{\text{lex}} x_1^2 x_2^7, \quad x_1^2 x_2^3 x_3 >_{\text{lex}} x_1^2 x_2^3 x_4^2.$$

Notice that this is just the usual alphabetical order, eg.  $a = x_1, b = x_2$  etc then

$$\text{artic} > \text{apt}, \quad \text{bbb} > \text{burp}.$$

**Graded lexicographic order:** Given  $a, b \in \mathbb{Z}_{\geq 0}^n$ , we say  $a >_{\text{grlex}} b$  if either  $|a| > |b|$  or  $|a| = |b|$  and  $a >_{\text{lex}} b$ . Thus

$$x_1^3 x_2 >_{\text{grlex}} x_1^2 x_2^7, \quad x_1^2 x_2^3 x_4^2 >_{\text{grlex}} x_1^2 x_2^3 x_3, \quad .$$

**Exercise 30.2.** Show that  $>_{\text{grlex}}$  and  $>_{\text{lex}}$  are monomial orders.

**Exercise 30.3.** Show that the order defined by  $x^a y^b > x^c y^d$  if and only if  $a + b\sqrt{2} > c + d\sqrt{2}$  is a monomial order.

The notion of ordering is extremely useful when we wish to run any recursive procedure. For example the division algorithm and the Euclidean algorithm. Let's begin by recalling these in the standard setting.

**Definition 30.4. Division algorithm:** Let  $a, b \in \mathbb{N}$  then there exist  $q, r \in \mathbb{N}$  such that  $a = bq + r$  where  $0 \leq r < b$ .

Recall that the **greatest common divisor** (or GCD) of two integers  $a, b \in \mathbb{N}$  is an integer  $c = \gcd(a, b)$  such that

- (1)  $c$  divides  $a$  and  $b$  (written  $c|a, c|b$ ),
- (2) for any integer  $d$  dividing  $a$  and  $b$ , then  $d$  divides  $c$ .

**Exercise 30.5.** Show that  $(a, b) = (c)$  as ideals in  $\mathbb{Z}$  where  $c = \gcd(a, b)$ .

**Exercise 30.6.** Show that  $c = \gcd(a, b)$  is the smallest integer in  $(a, b) \cap \mathbb{N}$ .

**Definition 30.7. Euclidean algorithm:** Let  $a \leq b \in \mathbb{N}$  then let  $a = bq + r$  where  $q, r \in \mathbb{Z}_{\geq 0}$  and  $0 \leq r < b$ . Let  $a = a_0, b = b_0$  and  $r = r_0$ . We then proceed inductively as follows. Assume that  $a_i, b_i, r_i$  have been defined for  $0 \leq i \leq n$ . If  $r_n = 0$ , then  $b_n = \gcd(a, b)$  otherwise let  $a_{n+1} = b_n, b_{n+1} = r_n$  and write  $a_{n+1} = b_{n+1}q_{n+1} + r_{n+1}$  where  $0 \leq r_{n+1} < b_{n+1}$ . Since  $r_i > r_{i+1} > \dots \geq 0$  the procedure terminates after finitely many steps.

**Exercise 30.8.** Use the Euclidean Algorithm to find the GCD of 66 and 42.

Once we have fixed a monomial ordering  $>$  we can also perform long division on polynomials. For example you are familiar with long division of polynomials in one variable: Given  $f, g \in \mathbb{C}[x]$  there exist  $q, r \in \mathbb{C}[x]$  such that

$$f = gq + r, \quad \deg r < \deg g.$$

here of course we use the monomial ordering induced by degree.

**Exercise 30.9.** Divide  $x^4 + 3x^2$  by  $x^2 + x + 1$ .

We can also use long division to find the GCD of two polynomials in one variable.

**Exercise 30.10.** Show that  $\gcd(x^4 - 1, x^3 - 3x + x^2 - 3) = x^2 + 1$ .

**Exercise 30.11.** Show that if  $f, g \in \mathbb{C}[x]$ , then  $(f, g) = (h)$  where  $h = \gcd(f, g)$ . thus  $h$  is any non-zero polynomial of minimal degree in  $(f, g)$

Turning now to polynomials in several variables, let's use  $>_{\text{lex}}$  to divide  $x^3y + xy^3$  by  $x + y^2$ . We have  $LT(x^3y + xy^3) = x^3y$  and  $LT(x + y^2) = x$ . Now  $x^3y = (x^2y) \cdot x$  and so we have

$$x^3y + xy^3 = (x^2y) \cdot (x + y) + xy^3 - x^2y^2$$

Since  $LT(xy^3 - x^2y^2) = x^2y^2 = xy^2LT(x + y)$ , we next write

$$xy^3 - x^2y^2 = -xy^2(x + y) + 2xy^3 = (x + y)(xy^2 + 2y^3) - 2y^4.$$

Putting all of this together we obtain

$$x^3y + xy^3 = (x + y)(x^2y + xy^2 + 2y^3) - 2y^4.$$

Therefore the remainder is  $2y^4$  and note that  $x + y > y^4$ . Unluckily, it is not always possible to assume that if we divide  $f$  by  $g$  the remainder will satisfy  $g > r$ . Consider in fact

$$f = x^2 + y^2, \quad g = xy$$

wrt  $>_{\text{lex}}$ . Since  $xy$  does not divide any term in  $f$ , it follows that  $r = f$  but  $g < r$  (not  $r < g$ ). Therefore when dividing  $f$  by  $g$  we can only require that the leading term of  $g$  does not divide any term of  $r$ :

**Definition 30.12.** If  $>$  is a monomial order on  $\mathbb{C}[x_1, \dots, x_n]$ , then for any  $f, g \in \mathbb{C}[x_1, \dots, x_n]$  there exist  $q, r \in \mathbb{C}[x_1, \dots, x_n]$  such that

$$f = gq + r,$$

where  $LT(g)$  does not divide any monomial in  $r$ . Similarly if we divide  $f$  by  $g_1, \dots, g_t$  then we write

$$f = \sum g_i q_i + r, \quad q_i \in \mathbb{C}[x_1, \dots, x_n]$$

where  $LT(g_i)$  does not divide any monomial in  $r$  (for  $1 \leq i \leq n$ ).

**Exercise 30.13.** Use  $>_{\text{grlex}}$  to divide  $x^3y + xy^3$  by  $x + y^2$ . Notice that you get a different answer.

## 31. MONOMIAL IDEALS

**Definition 31.1.** A **monomial ideal**  $I \subset \mathbb{C}[x_1, \dots, x_n]$  is an ideal generated by monomials.

For example  $(x^2, y^3)$  is a monomial ideal, but  $(x^2 + y^3)$  is not a monomial ideal. Notice however that  $(x^2 + y^3, x^2 - y^3)$  is a monomial ideal since  $(x^2 + y^3, x^2 - y^3) = (x^2, y^3)$ .

**Exercise 31.2.** Show that  $I \subset \mathbb{C}[x_1, \dots, x_n]$  is a monomial ideal if and only if for any  $f \in I$  then all the monomials of  $f$  are in  $I$ . It follows that two monomial ideals  $I, J \subset \mathbb{C}[x_1, \dots, x_n]$  are equal if and only if they contain the same monomials.

Next we will reprove the Hilbert's basis theorem in the context of monomial ideals.

**Theorem 31.3** (Dickson's Lemma). Let  $A \subset \mathbb{Z}_{\geq 0}^n$  and  $I = (\{x^\alpha\}_{\alpha \in A}) \subset \mathbb{C}[x_1, \dots, x_n]$  the corresponding ideal, then there exists a finite subset  $B \subset A$  such that  $I = (\{x^\alpha\}_{\alpha \in B}) \subset \mathbb{C}[x_1, \dots, x_n]$

*Proof.* We proceed by induction on  $n$ . If  $n = 1$ , then it suffices to consider the smallest non-zero element  $a \in A$ . We claim that  $I = (x^a)$ . Consider in fact  $\square$

We will now show how to associate a monomial ideal to any ideal  $I$ .

**Definition 31.4.** Let  $I \subset \mathbb{C}[x_1, \dots, x_n]$  be an ideal, then let  $LT(I) = (\{LT(f)\}_{f \in I})$ .

Note that  $LT(I)$  is a monomial ideal. Note that it may happen that  $I = (f_1, \dots, f_r)$  but  $LT(I) \neq (LT(f_1), \dots, LT(f_r))$ .

**Exercise 31.5.** Let  $f = x$  and  $g = x + y$ , and consider  $>_{\text{lex}}$ . Show that  $LT(f, g) \neq (LT(f), LT(g))$ .

We will now use Dickson's Lemma to reprove the Hilbert's basis theorem:

**Theorem 31.6** (Hilbert's basis theorem). Let  $I \subset \mathbb{C}[x_1, \dots, x_n]$  be an ideal, then there exist  $f_1, \dots, f_r \in I$  such that  $I = (f_1, \dots, f_r)$ .

*Proof.* [CLS, Theorem 4].  $\square$

**Definition 31.7.** If  $f_1, \dots, f_r \in I$  are such that  $(LT(f_1), \dots, LT(f_r)) = LT(I)$  then we say that  $f_1, \dots, f_r$  is a **Groebner basis** of  $I$ .

It is an immediate consequence of Dickson's Lemma that every ideal  $I \subset \mathbb{C}[x_1, \dots, x_n]$  has a Groebner basis. We also observe that a Groebner basis is a basis of  $I$  so that  $(f_1, \dots, f_r) = I$ . To see this, pick  $f \in I$  and divide it by  $f_1, \dots, f_r$  to get  $f = \sum q_i f_i + r$  where  $LT(r)$  is not divisible by  $LT(f_i)$ . However  $r \in I$  so that  $LT(r) \in LT(I)$ . But then ...

**Lemma 31.8.** Let  $G = (g_1, \dots, g_r)$  be a Groebner basis for an ideal  $I \subset \mathbb{C}[x_1, \dots, x_n]$  and  $f \in \mathbb{C}[x_1, \dots, x_n]$ , then there exists a unique polynomial  $r \in \mathbb{C}[x_1, \dots, x_n]$  such that  $r \in f + I$ , and no term of  $r$  is divisible by  $LT(g_i)$ .

*Proof.* The existence of such a polynomial  $r$  was observed above: simply divide  $f$  by  $g_1, \dots, g_r$  to get  $f = \sum a_i g_i + r$  with  $a_1, \dots, a_r \in \mathbb{C}[x_1, \dots, x_n]$ .

To see uniqueness, suppose that we have two such expressions  $r \in f + I$  and  $r' \in f + I$  such that no term of  $r, r'$  is divisible by  $LT(g_i)$ , then  $r - r' \in I$  and so some term of  $r - r'$  is divisible by  $LT(g_i)$ . This is impossible and so  $r - r' = 0$ .  $\square$

**Definition 31.9.**  $\{g_1, \dots, g_r\}$  is a **minimal Groebner basis** of an ideal  $I \subset \mathbb{C}[x_1, \dots, x_n]$  if it is a Groebner basis such that for any  $1 \leq i \leq r$  we have  $LC(g_i) = 1$  and

$$LT(g_i) \notin (LT(g_1), \dots, LT(g_{i-1}), LT(g_{i+1}), \dots, LT(g_r)).$$

**Definition 31.10.**  $\{g_1, \dots, g_r\}$  is a **minimal Groebner basis** of an ideal  $I \subset \mathbb{C}[x_1, \dots, x_n]$  if it is a Groebner basis such that for any  $1 \leq i \leq r$  we have

$$LT(g_i) \notin (LT(g_1), \dots, LT(g_{i-1}), LT(g_{i+1}), \dots, LT(g_r)).$$

**Definition 31.11.**  $\{g_1, \dots, g_r\}$  is a **reduced Groebner basis** of an ideal  $I \subset \mathbb{C}[x_1, \dots, x_n]$  if it is a Groebner basis such that for any  $1 \leq i \leq r$  we have  $LC(g_i) = 1$  every monomial of  $g_i$  is not in  $(LT(g_1), \dots, LT(g_{i-1}), LT(g_{i+1}), \dots, LT(g_r))$ .

**Proposition 31.12.** *Reduced Groebner basis always exist and are unique.*

*Proof.* □

## 32. SOLUTIONS AND HINTS FOR THE EXERCISES

- (1) Exercise 1.24. Suppose that  $\mathbb{C}^n = X \cup Y$  where  $X, Y$  are proper closed subsets. We may pick polynomials  $p, q \in \mathbb{C}[x_1, \dots, x_n]$  such that  $p \in \mathbb{I}(X)$ ,  $q \in \mathbb{I}(Y)$ . But then  $pq \in \mathbb{I}(X \cup Y) = \mathbb{I}(\mathbb{C}^n)$ , but this is impossible as any non-constant polynomial has a (infinitely many if  $n > 1$ ) solution.  $\square$
- (2) Exercise 1.40. Suppose that  $X$  is compact in the Euclidean topology. Suppose that there exists a sequence  $x_k$  such that no subsequence has a limit in  $X$ . we may assume that all  $x_k$  are distinct and we pick open neighborhoods  $U_i$  such that  $U_i \cap \{x_j | j \in \mathbb{N}\} = x_i$ . Let  $U = X \setminus \{x_i | i \in \mathbb{N}\}$ .  $U$  is open in  $X$  since no subsequence has a limit in  $X$  (check). Then  $U \cup_{i \in \mathbb{N}} U_i$  is an open cover of  $X$ . Since  $X$  is compact, it admits a finite open subcover, but this is clearly impossible.
- (3) Exercise 2.11. Hint: Consider the projection  $\mathbb{V}(xy = 1) \rightarrow \mathbb{C}$ .
- (4) Exercise 3.31. Hint: Consider the inclusion  $\mathbb{Z} \hookrightarrow \mathbb{Q}$ .
- (5) Exercise 9.7.
- (6) Exercise 29.8. See [Hartshorne, II.2.3].
- (7) Exercise 12.18. Hint: Use the diagonal trick.
- (8) Exercise 12.25. Hint: Use the  $d$ -uple embedding.

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